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Design and Evaluation of Multi-Agent Architectures for Stock Price Prediction: A Vietnam Case Study

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ABSTRACT Traditional stock price prediction methods struggle in emerging markets characterized by retail investor dominance, high volatility, and sentiment-driven trading. We address this gap by developing a multi-agent framework tailored to Vietnam's unique market structure, where 85-90% of participants are retail investors operating under a T+2.5 settlement cycle. Our system integrates six specialized agents that analyze technical indicators, fundamental metrics, risk profiles, news sentiment, macroeconomic signals, and market microstructure through three coordination mechanisms: Hierarchical, Round Robin, and Ensemble Voting. Empirical evaluation reveals strong horizon-dependent architectural effectiveness. For medium-term (14-day) forecasting, Ensemble Voting consistently achieves superior performance across multiple sectors, effectively reconciling heterogeneous technical and fundamental signals. Short-term (3-day) predictions exhibit stock-specific preferences, with Hierarchical architecture excelling in highly volatile sectors while Ensemble Voting proves optimal for stable growth-oriented stocks. Long-term (60-day) forecasting shows sector-dependent architectural advantages: Hierarchical coordination outperforms in highly regulated industries such as banking, while Round Robin demonstrates superiority in cyclical sectors like industrial construction. Across all horizons, the Multi-Agent framework significantly outperforms traditional LSTM baselines and single-agent approaches, offering a robust and interpretable decision-support tool for Vietnam's financial ecosystem.

INDEX TERMS Architectures, Large Language Models, Multi-agent, Price Prediction, Vietnam Stock

I. INTRODUCTION

The rapid growth of digital technology in equity markets and the increasing availability of real-time market data have created new opportunities for applying artificial intelligence (AI) in trading and investment analysis. In Vietnamâ€™s stock market, this transformation is especially relevant. By the end of 2024, the market had experienced substantial growth in securities trading accounts, with the vast majority owned by domestic investors. Retail investors accounted for more than 80% of total trading value [1]. These figures demonstrate that Vietnamâ€™s market is heavily driven by retail investors and exhibits high volatility, structural inefficiencies, and an evolving regulatory framework.

Traditional methods of equity-market analysis, such as technical analysis based on price patterns and fundamental analysis rooted in corporate disclosures, often struggle to capture the complex, non-linear relationships found in emerging markets. For instance, empirical evidence from Vietnam has revealed sluggish information dissemination, intraday trading frictions, and pronounced influences from investor sentiment and speculative behavior [2]. These characteristics underscore the limitations of conventional modeling approaches and reinforce the necessity for more adaptive and robust analytical frameworks.

Integrating deep learning with multi-agent systems presents a promising approach to addressing these challenges. Deep learning models can automatically extract informative representations from large and complex datasets, thereby uncovering latent patterns and improving predictive accuracy. Simultaneously, multi-agent system designs enable the simulation of diverse market participants and their interactions, offering a more realistic representation of market dynamics and investor behavior.

To effectively deploy AI-based strategies within Vietnam's stock market, careful customization is essential to align with its specific operational and behavioral characteristics. However, the application of these advanced systems to Vietnam's unique market context remains underexplored. This gap motivates three critical research questions that guide this study:

- **RQ1:** Under the constraints imposed by the $T + 2.5$ settlement cycle, which coordination architecture (*Hierarchical*, *Round Robin*, or *Ensemble Voting*) is most effective across short-term, medium-term, and long-term forecasting horizons?
- **RQ2:** How does the choice of the underlying Large Language Model (LLM), with a particular focus on *Gemini 2.0 Flash* compared against alternative LLMs, affect the predictive stability and accuracy of the proposed agent-based system?
- **RQ3:** Does the proposed *Multi-Agent* system empirically outperform traditional *Single-Agent* baselines and standard deep learning models in an emerging market context?

Accordingly, we propose a tailored simulation framework that combines specialized agent modules and architectural designs to model Vietnamese stock market dynamics. The focus is on testing trading strategies, liquidity and volatility responses, and inter-agent communication under different coordination architectures. Through this approach, the research contributes both to the academic understanding of agent-to-agent interactions in emerging markets and to the practical design of decision-support tools for Vietnamâ€™s financial ecosystem.

II. LITERATURE REVIEW

A. STOCK MARKET DATA

Data from global stock exchanges have long been recognized as essential for academic research and institutional decision-making. Major trading platforms such as the NYSE, NASDAQ, LSE, and TSE operate at substantial scale, with combined market capitalizations exceeding USD 40â€“50 trillion and daily intraday trading volumes frequently surpassing hundreds of billions [3]. These markets provide millisecond-level order book updates, comprehensive limit order details (Level 3), and standardized high-frequency data streams suitable for automated trading systems and microsecond-level execution studies [4]. Such infrastructure has enabled detailed research on market microstructure, latency-based arbitrage strategies, and systemic risk propagation in high-frequency environments.

In comparison, Vietnam's stock market, comprising HOSE, HNX, and UPCoM, remains structurally smaller and data-constrained. The total market capitalization of listed companies in 2024 was approximately USD 250â€“300 billion [5], with average daily trading volumes ranging between USD 800 million and USD 1.2 billion, significantly lower than those of regional peers such as Thailand and Malaysia [6]. Moreover, available data are limited to daily high, low, opening, and closing prices, order flow summaries, and, more recently, limited order book visibility, without microsecond timestamps or complete order depth information [7].

A key behavioral divergence arises from the dominance of retail investors, who account for more than 85% of total trading volume in Vietnam [8]. In contrast to institutionally driven markets, where prices typically respond to news and earnings reports, price movements in Vietnam are more heavily influenced by government policy announcements, informal communication channels, and sentiment-based speculation [9]. While global markets increasingly incorporate AI-curated news and alternative data into trading systems [10], the Vietnamese market remains predominantly influenced by local social media sentiment and expectations regarding major government policy shifts.

B. AI IN STOCK PRICE PREDICTION

Over the past decades, AI-based methods have been widely applied to stock price prediction. Classical machine learning models such as SVMs, decision trees, random forests, and ANNs can capture non-linear relationships in financial data and often outperform traditional statistical approaches [11]–[13]. Deep learning models such as recurrent neural networks and long short-term memory networks further improved temporal analysis of financial time series [14], [15].

More recently, research has shifted toward *Multi-Agent Systems* (MAS), in which agents interact in simulated markets. Unlike traditional models that treat prices as isolated data streams, MAS capture dynamic relationships among traders, brokers, and regulators, enabling the study of phenomena such as herd behavior, bubbles, and market irregularities [16], [17]. At a methodological level, coordination among distributed agents is strongly connected to consensus theory in networked multi-agent systems, which provides formal guarantees for information aggregation and agreement under dynamic interaction graphs [18], [19]. Combining reinforcement learning with agent-based simulations also deepens the analysis of trading strategies, market stability, and inter-agent communication [20], [21].

While the majority of MAS-based financial research has focused on mature markets such as the U.S. and China, empirical applications to emerging markets remain notably scarce. In the case of Vietnam, the AI-driven research currently available focuses mainly on price prediction trends over time via machine learning or deep learning techniques [22], [23], without considering how actions driven by individual investors, governmental policy shifts, or sudden shifts in market sentiment can shape market dynamics. This highlights a significant gap: it remains unproven whether Multi-Agent frameworks truly outperform traditional single-agent models (RQ3) in such volatile environments. Furthermore, while MAS is evolving, the specific impact of modern Large Language Models as the cognitive core of these agents, and how different models such as Gemini or GPT influence system performance (RQ2), have not been rigorously quantified in financial contexts.

Overall, this shift from single-model learning to multi-agent simulation reflects a growing recognition of stock markets as complex ecosystems shaped by interactions among diverse actors. This approach provides a useful means to examine regulatory effects and investor behavior in experimental settings, which is particularly relevant for developing economies such as Vietnam.

1) Machine Learning for Stock Price Prediction

The utilization of machine learning methodologies has been a long-standing practice in stock price prediction, particularly for capturing non-linear trends within financial time series data. An illustrative example is seen in Kim [11], where support vector machines (SVM) were employed to predict stock market fluctuations, achieving a level of effectiveness that surpassed that of conventional statistical modeling techniques. Similarly, Huang et al. [12] utilized the capabilities of SVM to forecast the directional movement of stock indices, whereas Patel et al. [13] employed a hybrid approach combining artificial neural networks (ANN), random forests, and SVM to achieve heightened predictive precision. Progressing into more contemporary research, deep learning techniques have been increasingly adopted, notably long short-term memory (LSTM) models, which possess the inherent capacity to discern temporal interrelationships within price sequences [14], [15].

However, these pure data-driven models often fail to capture the complex interactions and behavioral dynamics of market participants. To address these challenges, researchers have proposed moving toward *Multi-Agent Systems* (MAS). Multi-Agent Systems, in contrast to traditional machine learning, explicitly represent multiple agents, such as investors, brokers, and regulators, who interact under market rules and adapt their strategies over time. This enables the analysis of complex, emergent behaviors beyond prediction accuracy, offering a more comprehensive understanding of market stability, systemic risks, and the significance of agent communication [16], [17], [20], [21]. From this perspective, Multi-Agent Systems do not merely complement traditional machine learning, but extend it by embedding predictive intelligence into an interactive and behaviorally realistic market environment, a setup particularly well-suited to Vietnamâ€™s retail-dominated and sentiment-driven stock market structure.

2) Multi-Agent for Stock Price Prediction

Initial applications of Multi-Agent Systems in finance were primarily aimed at replicating market behavior rather than predicting prices. Groundbreaking research by LeBaron [16] and Tesfatsion [20] demonstrated that interacting agents can recreate typical market phenomena, including high short-term volatility and speculative bubbles, which standard statistical analysis and single-model learning methods fail to adequately explain. Subsequent research extended MAS into domains where agents could learn and improve their trading strategies through reinforcement learning. For example, Chen et al. [17] demonstrated how emergent market behaviors can arise from agent interactions, and Wang et al. [21] developed investment portfolios that outperformed those based on fixed, rule-based strategies.

Recent advancements in financial modeling have witnessed a paradigm shift toward hybrid learning-based architectures. In these sophisticated environments, automated agents do not merely function in isolation; they engage in complex, asynchronous communication within artificial market settings to simulate trading dynamics. Consequently, the research frontier has moved beyond merely validating the feasibility of MAS. The critical imperative now lies in identifying the optimal coordination topology. Specifically, we aim to determine which structural framework, whether the stratified logic of Hierarchical models, the cyclical consensus of Round Robin, or the democratic aggregation of Ensemble methods, is best equipped to navigate the idiosyncratic volatility of the Vietnamese stock market (RQ1). This framing is also consistent with consensus-oriented coordination principles in distributed multi-agent control [18], [19]. Addressing this question is vital for balancing the conflicting signals of short-term retail speculation against long-term fundamental trends.

III. METHODOLOGY

A. ARCHITECTURAL OVERVIEW

The proposed Multi-Agent System (MAS) represents a sophisticated computational framework designed to address the unique challenges of stock price prediction in Vietnam's emerging market environment. The system integrates six specialized autonomous agents, each equipped with domain-specific analytical capabilities, operating within three distinct coordination architectures to deliver robust and interpretable financial forecasts. At its core, the system employs a foundation layer built upon deep learning architectures (specifically **Long Short-Term Memory (LSTM)** or **Transformer** networks) to capture temporal dependencies in historical price data, enhanced through the coordinated intelligence of six specialized agents: **PricePredictor** (technical analysis), **InvestmentExpert** (fundamental evaluation), **RiskExpert** (risk assessment), **TickerNews** (company-specific sentiment), **MarketNews** (macroeconomic analysis), and **StockInfo** (market microstructure).

The cognitive intelligence of the system is powered by **Google Gemini 2.0 Flash** as the core Large Language Model (LLM), selected for its exceptional scalability, low-latency execution, and expansive context window capabilities essential for processing complex financial narratives and multi-dimensional market data. The system's architectural flexibility is demonstrated through three coordination architectures, namely **Hierarchical, Round Robin, and Ensemble Voting**, each specifically calibrated to handle Vietnam's market characteristics, including the T+2.5 settlement period, 85-90% retail investor dominance, and pronounced sentiment-driven volatility, delivering predictions across short-term (3 days), medium-term (14 days), and long-term (60 days) investment horizons.

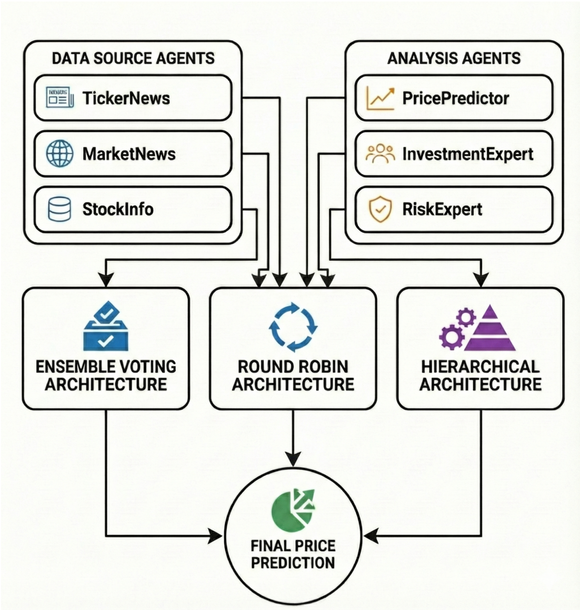

FIGURE 1: System overview diagram

B. THE SPECIALIZED AGENT TEAM

1) Agents Overview

- **PricePredictor Agent:** Using 730 days of OHLCV data from *vnstock* [24], an LSTM [25] or Transformer [26] first produces baseline price forecasts, which are then calibrated by RSI [27], MACD [28], and Bollinger Bands [29], yielding predicted prices at 3-, 14-, and 60-day horizons together with a confidence score and a BUY/HOLD/SELL signal. **Input:** Up to 730 days of OHLCV data retrieved via the *vnstock* library [24] (VCI/KBS sources) with a 60-day lookback period (using the past 60 trading days to forecast future prices) and an 80/20 train–test split. **Output:** Predicted prices at 3-, 14-, and 60-day horizons; a confidence score; a BUY/HOLD/SELL signal; and a trend label (bullish/neutral/bearish).
  1: **Input:** stock symbol; 730-day OHLCV history from *vnstock*
  2: Split data into 80% training and 20% testing sets
  3: Train LSTM or Transformer on training set
  4: Roll out model autoregressively to obtain baseline forecasts for 3-, 14-, and 60-day horizons
  5: Compute RSI, MACD, and Bollinger Bands from historical close prices
  6: Determine a calibration direction (upward or downward) based on RSI overbought/oversold zone, MACD crossover signal, and price position within Bollinger Bands
  7: Apply calibration to baseline forecasts, with stronger weight on shorter horizons; cap total adjustment at five percent
  8: **Output:** calibrated price forecasts; confidence score; BUY/HOLD/SELL signal
- **InvestmentExpert Agent:** Using fundamental data retrieved via *vnstock* [24], this agent scores a stock across three dimensions â€" financial metrics (P/E, P/B, EPS, dividend yield), technical position (historical price range, volume ratio, Beta), and valuation attractiveness (forward P/E trend, dividend payout ratio) â€" and maps the composite score to a BUY/HOLD/SELL recommendation with a calibrated confidence level.
  Input: Stock symbol; financial ratios (P/E, P/B, EPS, dividend yield) and 730-day OHLCV data from *vnstock* [24].
  Output: Investment recommendation (BUY/HOLD/SELL); composite score (0–100); confidence level; per-dimension analysis breakdown.
  1: **Input:** stock symbol; financial ratios and OHLCV history from *vnstock*
  2: Fetch P/E, P/B, EPS, dividend yield, 52-week high/low, volume, and Beta
  3: Score each metric independently on a 0–100 scale based on predefined thresholds
  4: Combine scores with higher priority on financial health, followed equally by technical position and valuation
  5: Map composite score to recommendation: high scores yield BUY, low scores yield SELL, mid-range yields HOLD
  6: **Output:** recommendation; composite score; confidence level
- **RiskExpert Agent:** Using historical price data from *vnstock* [24], this agent computes key risk metrics â€" annualized volatility, maximum drawdown, Beta against the VN-Index, Value-at-Risk, and Sharpe ratio â€" and adjusts the resulting risk level according to the investor's risk tolerance profile, producing position sizing and stop-loss recommendations.
  Input: Stock symbol; investor risk profile (conservative / balanced / aggressive); 730-day OHLCV history from *vnstock* [24].
  Output: Risk level (LOW / MEDIUM / HIGH); key risk metrics; recommended position size and stop-loss level.
  1: **Input:** stock symbol; investor risk profile; 730-day OHLCV history from *vnstock*
  2: Compute annualized volatility, maximum drawdown, Beta vs VN-Index, Value-at-Risk, and Sharpe ratio
  3: Classify risk level based on volatility thresholds (low, medium, or high)
  4: Adjust risk level upward or downward according to the investor's risk tolerance
  5: Derive position size limit and stop-loss percentage from the adjusted risk profile
  6: **Output:** risk level; risk metrics; position sizing and

stop-loss guidance

- **TickerNews Agent:** This agent collects company-specific news articles by crawling prominent Vietnamese financial news portals â€" primarily CafeF (`cafef.vn`), VietStock (`vietstock.vn`), VnExpress (`vnexpress.net`), and VnEconomy (`vneconomy.vn`) â€" filtered strictly to articles published before the *milestone_date* to prevent look-ahead bias. Each article is scored using a bilingual (Vietnamese and English) keyword lexicon, and detected events are classified against a predefined event impact matrix (e.g., earnings surprise, merger, legal issue). A recency-weighted aggregation then produces a final sentiment label and impact score. The aggregated news context is subsequently passed to the LLMs for deeper semantic reasoning, enabling it to interpret nuanced narratives, assess event severity, and generate a structured sentiment summary beyond what keyword matching alone can capture.
  Input: Stock symbol; news articles crawled from Vietnamese financial portals (CafeF, VietStock, VnExpress, and VnEconomy), filtered by *milestone_date*.
  Output: Sentiment label (POS / NEG / NEU); impact score; confidence level; list of detected key events.
- **MarketNews Agent:** This agent collects broad market and macroeconomic news (interest rates, CPI, GDP) from multiple prominent Vietnamese financial portals â€" including CafeF (`cafef.vn`), VietStock (`vietstock.vn`), VnExpress (`vnexpress.net`), and VnEconomy (`vneconomy.vn`) â€" via a unified news engine. The combined corpus is passed to the LLMs for deeper semantic analysis and then classified into a macro regime (RISK-ON / RISK-OFF / NEUTRAL) using bilingual keyword sets covering monetary policy (e.g., SBV rate decisions), inflation signals, and market-wide fund flows. Sector rotation is detected by tallying keyword mentions across seven predefined sectors (banking, real estate, technology, industrials, consumer, energy, healthcare), and a short-term market outlook is generated accordingly.
  Input: Market-wide and macroeconomic news articles from Vietnamese portals (CafeF, VietStock, VnExpress, VnEconomy), filtered by *milestone_date*.
  Output: Macro regime (RISK-ON / RISK-OFF / NEUTRAL); market pressure score; macro factor signals (interest rate, inflation, monetary policy stance); sector rotation summary; short-term market outlook.
- **StockInfo Agent:** Using real-time and historical market microstructure data from *vnstock* [24], this agent performs a five-dimensional composite liquidity analysis covering volume stability, bid-ask spreads, order book depth, capital turnover, and time-to-execute estimates. It calculates a composite liquidity feasibility score and maps it to specific execution advice (AGGRESSIVE, PASSIVE, or WAIT) tailored to manage risk under Vietnam's T+2.5 settlement cycle. Additionally, it monitors for abnormal volume spikes that could signal unusual market activity.
  Input: Stock symbol; historical price and volume data from *vnstock* [24]; target order size.
  Output: Composite feasibility (liquidity) score; execution advice; microstructure indicators (volume ratios, bid-ask imbalance, turnover ratio, and spread).
  1: **Input:** stock symbol; historical price and volume data; target order size
  2: Compute five distinct liquidity dimensions: volume stability, bid-ask spread, order book depth, turnover ratio, and estimated execution timeframe
  3: Detect abnormal volume spikes by comparing recent volume activity against historical averages
  4: Aggregate the individual dimension scores using a weighted combination to derive a composite feasibility score
  5: Map the composite score to an execution advice level suited for the Vietnamese T+2.5 settlement cycle
  6: **Output:** composite feasibility score; execution advice; key microstructure metrics

2) LLMs Overview

Google Gemini 2.0 Flash is selected as the core LLM due to its high scalability, low-latency cloud execution, and exceptionally large context window, making it well-suited for production-grade AI pipelines. To provide a broader comparative perspective, we additionally include **OpenAI GPT-4o** and **Llama-3.1-8B**. GPT-4o is recognized for its strong reasoning quality and balanced multimodal performance, whereas Llama-3.1 offers significant advantages for privacy-focused and offline deployments. The operational characteristics of these models are summarized in Table 1.

TABLE 1: Operational Characteristics and Execution Behavior of Leading Large Language Models.

Criterion	Gemini 2.0 Flash	GPT-4o	Llama-3.1-8B
Deployment	Cloud API	Cloud API	Local / Hosted
Latency	Low	Medium	Low / Variable
Throughput	Very High	High	Infra-dependent
Offline Capability	None	None	Yes
Context Window	Up to 1M tokens	Up to 128k tokens	Model-dependent
Cost Model	SaaS (usage-based)	SaaS (usage-based)	CapEx / SaaS
Optimized for	Cloud scale	Reasoning quality	Privacy / Control

C. COORDINATION ARCHITECTURES

1) Hierarchical Architecture (Design 1)

a: Operational Mechanism

The hierarchical architecture adopts a Master–Subordinate design where a centralized *Master Agent* orchestrates the entire workflow. The Master Agent distributes a shared context packet X_t to all six specialized agents, executes them in parallel, and then aggregates their outputs into a final consolidated forecast.

Mathematically, the Master Agent fuses the individual predictions through a weighted combination:

$$\hat{P}_{\text{final}}(t) = \sum_{i=1}^{6} w_i \cdot Y_t^{(i)} \tag{1}$$

where $Y_t^{(i)}$ represents the decision or forecast output of agent i, and w_i is the dynamic attention weight assigned to that specific domain.

All agents operate independently and run in parallel. Communication follows an asynchronous send–receive pattern: the Master Agent dispatches inputs, waits for returned outputs, and continues aggregation only after validation succeeds. This improves efficiency while preserving modular expertise and robustness to partial failures.

After receiving all agent outputs, the Master Agent executes (1) schema/range validation, (2) strict consistency checks across directional signals and risk flags, (3) conflict resolution via reliability-aware down-weighting, and (4) a weighted Bayesian-style aggregation that interprets the dynamic weights as posterior model probabilities. The final unified prediction is produced together with an overall confidence score and a decision record, ensuring traceability of how heterogeneous signals jointly determine the market view.

This hierarchical coordination provides a computationally efficient yet explainable fusion layer: specialization improves each agentâ€™s local signal quality, while centralized aggregation reduces cross-domain inconsistencies and stabilizes the final decision under market regime changes [16], [20].

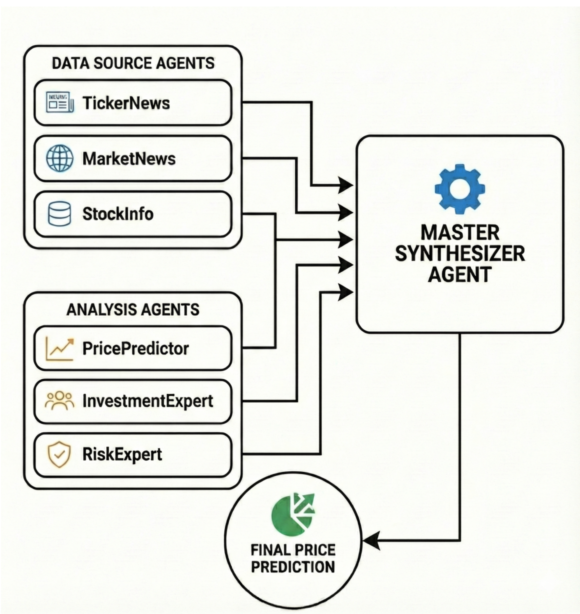

FIGURE 2: Hierarchical architecture where a Master Agent coordinates six specialized agents; agents process domain-specific data in parallel, and the Master Agent aggregates outputs for final prediction.

b: Workflow

The hierarchical architecture operates through a structured sequence of stages that makes the parallelism and aggregation mechanism explicit:

1) **Data Acquisition and Preprocessing:** The system collects and preprocesses multi-source financial data, including technical indicators, fundamental ratios, and market sentiment signals.
2) **Deep Learning Foundation Modeling:** A deep learning foundation model (either LSTM or Transformer) generates a baseline forecast $\{P_{\text{base}}(t,h)\}_{h\in\{3,14,60\}}$, which is included in the shared context. *Clarification:* The foundation model serves as a base module shared by all specialized agents; it is not treated as an independent agent in the hierarchical coordination layer.
3) **Parallel Agent Processing:** Six specialized agents are executed in parallel. Each agent receives the same context packet X_t (ticker, horizon, indicators, and the shared foundation baseline) and returns its structured output $Y_t^{(i)}$.
4) **Validation, Conflict Resolution, and Aggregation:** The Master Agent validates each output package, detects inconsistencies (e.g., directional conflict between technical/fundamental/sentiment signals or risk flags incompatible with the forecast stance), and resolves them by reliability-aware reweighting. In this process, all non-price agents (*InvestmentExpert* via `target_bias`, *RiskExpert* via `risk_level`/`risk_flags`, *TickerNews* via `sentiment`/`impact_score`, *MarketNews* via `macro_regime`/`market_pressure`, and *StockInfo* via `feasibility_score`) are treated as auxiliary context signals that modulate confidence and consistency checks during the final price fusion, rather than being interpreted as independent price forecasts. The master then applies weighted fusion of the expert-adjusted forecasts using dynamic weights to obtain a unified prediction and an overall confidence score.
5) **Final Output Generation:** The system produces the unified result, including final price forecasts, confidence levels, and investment recommendations.

Input: context packet X_t and recent performance cache for dynamic weights.
Compute: baseline forecasts $\{P_{\text{base}}(t,h)\}_{h\in\{3,14,60\}}$.
Dispatch: send X_t (including P_{base}) to all specialized agents in parallel.
Collect: receive $Y_t^{(i)}$ from each agent $i = 1, \ldots, 6$.
Validate: check schema validity, confidence range, and missing-field handling.
Detect conflicts:
  compare directional signals and risk flags across agents; mark conflict indicators.
Resolve conflicts:
  compute dynamic weights $w_i(t)$ from recent reliability; down-weight inconsistent experts.

Fuse: aggregate expert-adjusted forecasts to obtain $\hat{y}_{\text{final}}(t)$ and $\sigma^2_{\text{final}}(t)$.
Output: final prediction, overall confidence, and decision notes for backtesting.

2) Round Robin Architecture (Design 2)

a: Operational Mechanism

The Round Robin architecture implements a sequential refinement process. Instead of processing in parallel, the specialized agents form a chain. Each agent receives the prediction state from the previous agent, refines it with its local domain expertise, and passes the updated prediction to the next agent in the sequence.

Mathematically, let S_k represent the prediction state at step $k \in \{1, \ldots, 6\}$, initialized with the baseline price forecast $S_0 = P_{\text{baseline}}$. The refinement step for each specialized agent A_k is formalized as:

$$S_k = A_k(S_{k-1}, X_t) \tag{2}$$

where the final output is $\hat{y}_{\text{final}} = S_6$, representing the cumulative multi-perspective reasoning of all six specialized agents.

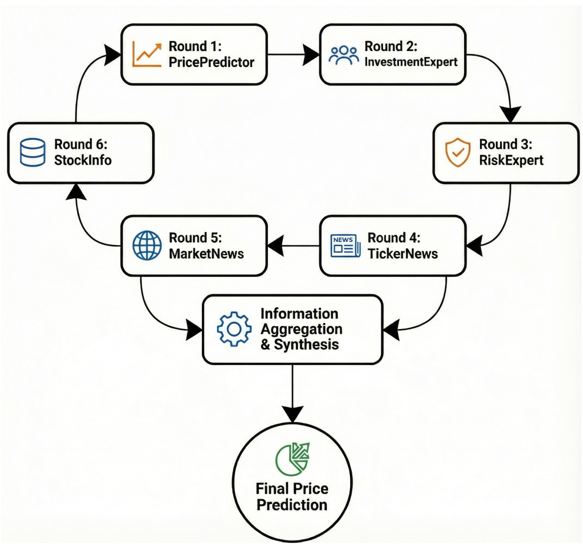

FIGURE 3: Round Robin architecture with six specialized agents operating in a sequential refinement cycle.

b: Workflow

The Round Robin architecture operates through an organized sequence of distinct stages:

1) **Initialization.** The approach begins with the *Deep Learning Foundation Layer* (LSTM or Transformer), which constructs the initial predictive state S_0 by capturing temporal dependencies in the historical data.
2) **Sequential Agent Flow.** The predictive state is propagated through six specialized agents in a fixed chronological order: Round 1 $\rightarrow$ Round 2 $\rightarrow$ Round 3 $\rightarrow$ Round 4 $\rightarrow$ Round 5 $\rightarrow$ Round 6.
3) **Incremental Enhancement.** Within this Round Robin structure, each of the six agents contributes a unique perspective: *PricePredictor* (technical indicators), *InvestmentExpert* (fundamental ratios), *RiskExpert* (risk profile), *TickerNews* (company sentiment), *MarketNews* (macro signals), and *StockInfo* (market microstructure).
4) **Decision Traceability.** All intermediate states S_k and analytical updates are logged to preserve a transparent audit trail.
5) **Cumulative Refinement Output.** Upon completion of the six-round iteration, the framework consolidates the results into a final state S_6 representing the cumulative learning and reasoning of the entire chain.

Input: baseline price forecast S_0; context data X_t
for each specialized agent $k = 1, \ldots, 6$ in chronological sequence **do**
    Pass current prediction state S_{k-1} and context X_t to agent A_k
    Execute agent A_k to refine the state: $S_k = A_k(S_{k-1}, X_t)$
end for
Output: final refined price forecast S_6 and unified investment advice

3) Ensemble Voting Architecture (Design 3)

a: Operational Mechanism

The Ensemble Voting Architecture employs an intelligent voting system in which six specialized agents operate concurrently and independently. Each agent conducts focused analysis on different data domains, ensuring that a wide range of predictive perspectives are considered. Subsequently, the system employs a statistical inference engine to combine these findings. This engine utilizes dynamic weighting based on recent performance, execution time, and confidence scores. This versatile strategy enhances robustness and reduces bias, facilitating more reliable and interpretable collective predictions in financial forecasting [30].

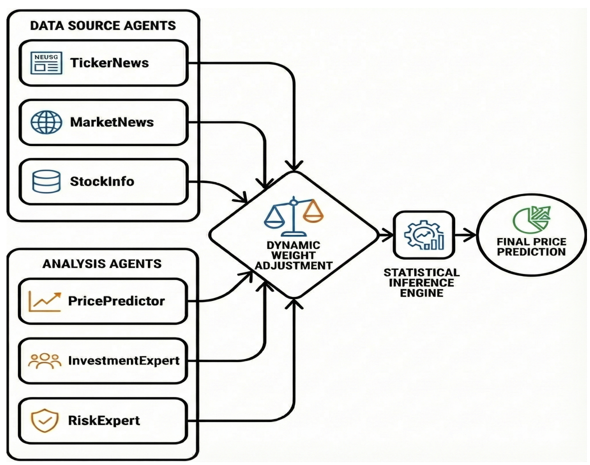

FIGURE 4: Ensemble Voting Architecture with six specialized agents employing an intelligent voting system

b: Workflow

The operational flow of the proposed Ensemble Voting Architecture is summarized as follows:

1) **Data Distribution:** Input data streams are concurrently distributed to all six agents.
2) **Parallel Processing:** Each agent operates independently in parallel, ensuring full task isolation.
3) **Performance Evaluation:** Real-time assessment of individual agent performance is continuously conducted.
4) **Dynamic Weight Adjustment:** Model weights are adaptively updated based on the latest performance metrics.
5) **Statistical Inference Integration:** Final aggregation is performed through statistical inference with uncertainty quantification to enhance predictive reliability.

c: Dynamic Weight Adjustment Process

To maintain adaptability in volatile market regimes, we formalize dynamic weighting over a rolling window of length H. Let $i \in \{1, \ldots, N\}$ denote an agent, and let $E_i(t)$ be its average prediction error (e.g., Mean Absolute Percentage Error) over the rolling window:

$$E_i(t) = \frac{1}{H} \sum_{\tau=t-H+1}^{t} \mathrm{MAPE}_i(\tau) \tag{3}$$

We then convert these errors into normalized dynamic weights $w_i(t)$ via a Softmax function with a sensitivity parameter $\eta > 0$:

$$w_i(t) = \frac{\exp(-\eta E_i(t))}{\sum_{j=1}^{N} \exp(-\eta E_j(t))} \tag{4}$$

where $\sum_{i=1}^{N} w_i(t) = 1$. This formulation ensures that agents with lower historical errors receive higher weights while preserving ensemble stability under regime shifts [30].

d: Statistical Inference Process

The ensemble layer integrates all agent outputs through a simple and robust inference pipeline. Let $\hat{y}_i(t)$ be the forecast of agent i. The aggregated forecast is:

$$\hat{y}_{\mathrm{ens}}(t) = \sum_{i=1}^{N} w_i(t)\hat{y}_i(t) \tag{5}$$

To estimate prediction uncertainty, we compute the weighted variance of the individual forecasts around the ensemble mean:

$$\sigma^2_{\mathrm{ens}}(t) = \sum_{i=1}^{N} w_i(t)\Big(\hat{y}_i(t) - \hat{y}_{\mathrm{ens}}(t)\Big)^2 \tag{6}$$

Finally, the inter-agent consensus score $\mathcal{C}(t)$ is quantified as:

$$\mathcal{C}(t) = 1 - \frac{1}{N(N-1)} \sum_{i \neq j} \frac{|\hat{y}_i(t) - \hat{y}_j(t)|}{|\hat{y}_{\mathrm{ens}}(t)| + \varepsilon} \tag{7}$$

where $\varepsilon > 0$ is a small constant to prevent division by zero. A higher $\mathcal{C}(t)$ indicates stronger agreement among the agents.

In combination, the weights and consensus score act as an adaptive filter that down-weights underperforming agents and preserves stability under abnormal market fluctuations.

1: **Input:** context packet X_t; recent performance cache
2: Run six specialized agents in parallel to obtain individual forecasts $\hat{y}_i(t)$
3: Compute dynamic weights $w_i(t)$ using Eq. (4)
4: Calculate consolidated ensemble forecast $\hat{y}_{\mathrm{ens}}(t)$ using Eq. (5)
5: Estimate ensemble variance $\sigma^2_{\mathrm{ens}}(t)$ and consensus score $\mathcal{C}(t)$ using Eqs. (6)–(7)
6: **Output:** final ensemble forecast $\hat{y}_{\mathrm{ens}}(t)$; prediction uncertainty; consensus score $\mathcal{C}(t)$

IV. EMPIRICAL EVALUATION

This section evaluates the proposed Multi-Agent forecasting framework under Vietnam's $T + 2.5$ settlement cycle constraints, systematically addressing research questions RQ1â€“RQ3 as formulated in the Introduction.

A. RQ1: ARCHITECTURAL EFFECTIVENESS ACROSS HORIZONS

Research Question: Under the constraints imposed by the $T + 2.5$ settlement cycle, which coordination architecture (Hierarchical, Round Robin, or Ensemble Voting) is most effective across short-term, medium-term, and long-term forecasting horizons?

B. RQ2: IMPACT OF FOUNDATION LLMS

Research Question: How does the choice of underlying LLM, with particular focus on Gemini 2.0 Flash compared against alternative LLMs, affect predictive stability and accuracy?

C. RQ3: MULTI-AGENT VS. TRADITIONAL BASELINES

Research Question: Does the proposed Multi-Agent system empirically outperform traditional Single-Agent baselines and standard deep learning models in an emerging market context?

D. DATASET & SETTINGS

A fundamental structural characteristic of the Vietnamese stock market is its **T+2.5 settlement period**. Given that *domestic individual investors overwhelmingly dominate the market, accounting for approximately 85-90% of daily trading value* [31], this configuration strongly incentivizes rapid, short-term speculative trading. This retail-heavy structure, often characterized by significant *herd behavior* rather than fundamental analysis, is identified by analysts as a primary driver of the market's notoriously high price volatility [32], [33].

Inspired by the substantial effects of this behavioral pattern, the present study conducts a sector-specific analysis by selecting representative stocks from four key sectors of the Vietnamese economy: **Banking (VCB)**, **Technology (FPT)**,

Real Estate (DXG), and **Industrial Construction (VCS)**. These sectors are chosen to capture the structural diversity of the Vietnamese stock market in terms of regulatory intensity, growth potential, and sensitivity to macroeconomic conditions.

VCB represents the banking sector due to its large market capitalization, high liquidity, and systemic importance within Vietnamâ€™s financial system, making it highly sensitive to monetary policy and regulatory changes. FPT is selected as a proxy for the technology sector, where price movements are largely driven by innovation, earnings growth, and investor sentiment. DXG reflects the real estate sector, which is characterized by strong cyclicality and elevated volatility arising from credit conditions and macroeconomic fluctuations. Finally, VCS represents the industrial construction sector, whose stock performance is closely linked to infrastructure investment, export demand, and long-term development cycles, typically exhibiting more stable yet trend-driven behavior.

This diversified sectoral selection enables a comprehensive evaluation of forecasting performance across heterogeneous market dynamics.

Historical price data and trading volumes are sourced from both the Ho Chi Minh Stock Exchange (HOSE) and the Hanoi Stock Exchange (HNX) via multiple data providers including VNDirect Securities Corporation's public API, vnstock Python library, and SSI iBoard platform. Daily closing prices, intraday trading volumes, and order book data are collected to support technical analysis. Macroeconomic indicators, monetary policy announcements, and company-specific news are aggregated from official financial news platforms including CafeF, VietStock, SSI Research, and Vndirect Research. Fundamental data such as financial statements, earnings reports, and valuation metrics are obtained from company disclosures and third-party financial databases.

To rigorously evaluate predictive performance across heterogeneous investment horizons, the analysis is structured into three temporal regimes:

- **Short-term (3 days):** aligned with the T+2.5 settlement window, capturing immediate speculative turnover.
- **Medium-term (14 days):** representing a conventional tactical positioning horizon.
- **Long-term (60 days):** proxying strategic buy-and-hold investment behavior.

To ensure robustness and reduce temporal bias, the evaluation at each forecasting horizon (**Short-term**, **Medium-term**, **Long-term**) is conducted across four distinct market intervals:

1) **Late August to early September**: capturing post-summer liquidity shifts;
2) **16/10/2025**: representing a period of heightened trading activity;
3) **21/10/2025**: corresponding to a stabilization phase after short-term volatility;
4) **22/10/2025**: reflecting market behavior during an immediate rebound window.

Assessing performance across these four checkpoints enables a comprehensive examination of model behavior under varying liquidity, volatility, and sentiment conditions, thereby strengthening the validity of comparative results across all three time horizons.

E. EVALUATION METRICS.

To rigorously evaluate the predictive performance of the proposed framework, we utilize three established statistical indicators: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). MAE quantifies the average magnitude of errors in a set of forecasts without considering their direction, providing a linear score that weights all individual differences equally [34]. Conversely, RMSE operates as a quadratic scoring rule; by squaring residuals prior to averaging, it assigns greater weight to larger deviations, thereby increasing sensitivity to outliers [35]. Finally, MAPE measures accuracy as a relative percentage, facilitating the comparison of forecasting performance across datasets with differing scales [36].

1) Diebold-Mariano (DM) Test

To robustly assess whether the differences in predictive accuracy between the proposed models are statistically significant, we employ the Diebold-Mariano (DM) test [37]. The DM test evaluates the null hypothesis H_0 that two competing forecasts have equal predictive accuracy. Given actual observations y_t and two forecasts $\hat{y}_{1t}, \hat{y}_{2t}$, the forecast errors are $e_{it} = y_t - \hat{y}_{it}$ for $i \in \{1, 2\}$. The loss differential is defined as $d_t = g(e_{1t}) - g(e_{2t})$, where $g(\cdot)$ is a loss function (e.g., squared error). The DM test statistic is computed as:

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}_{\bar{d}}}} \tag{8}$$

where $\bar{d}$ is the sample mean of the loss differential and $\hat{V}_{\bar{d}}$ is a consistent estimate of the asymptotic variance of $\sqrt{T}\bar{d}$. A standard normal distribution is used to determine the p-value. A p-value < 0.05 rejects the null hypothesis, indicating a statistically significant difference in model performance.

F. PRACTICAL TRADING SIMULATION (T+2.5 CONSTRAINTS)

To address the practical feasibility concern, we complement forecast-error evaluation with a simplified trading backtest that explicitly accounts for (i) transaction costs, (ii) slippage driven by liquidity conditions, and (iii) the Vietnam settlement cycle $T + 2.5$ [31].

a: Trading policy.

For each forecasting origin t and horizon $h \in \{3, 14, 60\}$, the system outputs a final price forecast and an associated trading stance (BUY/HOLD/SELL) from the forecasting module. We implement a long-only policy for practicality: when the stance is BUY, the system opens/maintains a long position; when it is SELL, it closes the position at the next executable time step;

HOLD leaves the position unchanged. Position sizing uses the risk-oriented output `position_size` from *RiskExpert* when available, otherwise defaults to a fixed fraction of capital.

b: Execution price with costs and slippage.
Let $P_{\text{exec}}(t)$ denote the daily execution reference price. Transaction costs are modeled with a proportional fee rate f (brokerage + statutory charges). Slippage is approximated as a liquidity-dependent perturbation $s(t)$ derived from the feasibility score output of *StockInfo*. The effective buy/sell execution prices are modeled as:

$$P_{\text{buy}}(t) = P_{\text{exec}}(t)\big(1 + f + s(t)\big), \tag{9}$$

$$P_{\text{sell}}(t) = P_{\text{exec}}(t)\big(1 - f - s(t)\big), \tag{10}$$

where $s(t)$ is consistent with execution-cost modeling principles in the optimal execution literature [38].

c: T+2.5 settlement constraint.
Orders submitted at day t are executed at the next trading day, while cash and positions become reusable only after settlement. In the discrete daily setting, we approximate the $T + 2.5$ cycle with a settlement delay of d_s trading days. The simulation maintains a cash-lock mechanism: after a BUY, the corresponding notional plus estimated fees are locked until settlement at $t + d_s$; similarly, a SELL closing action releases cash only after settlement. This constraint prevents unrealistic rapid turnover in the backtest.

d: Backtest metrics.
For each episode (stock, horizon, interval), we compute net trade returns and portfolio-level performance under costs:

$$r_{\text{trade}} = \frac{P_{\text{sell}}(t+h) - P_{\text{buy}}(t)}{P_{\text{buy}}(t)}, \tag{11}$$

$$\text{DD} = \max_{\tau}\left(\max_{u \le \tau} V(u) - V(\tau)\right), \tag{12}$$

where $V(\tau)$ is the simulated portfolio value. Max drawdown (DD) is used as a standard risk measure capturing the largest peak-to-trough decline in portfolio value over the backtest window [39]. We report cumulative net return, max drawdown (DD), and cost-adjusted directional accuracy (the fraction of forecast-driven stances yielding non-negative net returns after frictions).

e: Simulation procedure (pseudo-code).

1: **Input:** historical prices/volumes, feasibility and risk outputs, fee rate f, settlement delay d_s, horizons $h \in \{3, 14, 60\}$.
2: **For each interval and each origin t in the test window:**
3: Run forecasting pipeline to obtain final forecast and stance signal(t).
4: Determine execution day t_{exec} (next trading day) and compute $P_{\text{buy}}(t)$ and $P_{\text{sell}}(t)$.
5: If signal(t)=BUY: open/maintain position; lock cash until $t + d_s$.
6: If signal(t)=SELL: close position; release cash after settlement.
7: Update portfolio value $V(\tau)$ including transaction costs and slippage.
8: **Output:** cumulative net return, max drawdown, and cost-adjusted directional accuracy.

This practical simulation complements MAE/RMSE/MAPE by evaluating whether forecast-driven decisions remain beneficial under Vietnamese settlement constraints and realistic execution frictions.

V. RQ1: ARCHITECTURAL EFFECTIVENESS ACROSS HORIZONS

This section compares the coordination effectiveness of three multi-agent architectures, namely Hierarchical, Round Robin, and Ensemble Voting, across short-term, medium-term, and long-term forecasting horizons in Vietnamâ€™s volatile stock market. The Hierarchical architecture employs centralized aggregation to enhance stability. Round Robin enables sequential refinement but may suffer from error propagation. Ensemble Voting combines parallel predictions to reduce individual model bias through collective decision-making.

Due to the substantial volume of the experimental results, the detailed forecasting performance comparisons for each individual stock in terms of error metrics (MAE, RMSE, MAPE) are provided in the Appendix (see Section , Table 7). Furthermore, the raw forecasting records—including start prices, predicted and actual end prices, and absolute forecasting gaps—are also included in the Appendix (Tables 8 to 13). A systematic, macro-level analysis of these results is presented in the following sections, beginning with an aggregate summary in Table 2.

The macro-level empirical evaluation presented in Table 2 and summarized in Table 3 provides clear evidence to address RQ1 regarding architectural effectiveness. By aggregating the performance metrics across all stocks, we can systematically isolate the coordination strength of each multi-agent architecture independently of individual asset volatility.

First, evaluating the foundational time-series baselines (Table 3), the **LSTM-based Multi-Agent framework** achieves both a lower overall average MAE (**2.9423** vs. **2.9605**) and a lower overall average MAPE (**5.89%** vs. **5.99%**) compared to the Transformer-based equivalent. This demonstrates that the LSTM architecture provides slightly more stable sequence modeling capabilities under Vietnam's volatile market conditions, establishing a robust foundation for multi-agent coordination.

Crucially, when isolating the multi-agent architectures, the **Ensemble Voting architecture** emerges as the unambiguously superior framework. As detailed in Table 3, under the robust LSTM baseline, Ensemble Voting achieves an overall average MAPE of **5.12%**, significantly outperforming both Round Robin (**6.03%**) and Hierarchical (**6.51%**) coordination.

TABLE 2: Aggregated Performance Comparison Across Forecasting Horizons, Architectures, and LLMs.

Horizon	Architecture	LSTM-based Multi-Agent			Transformer-based Multi-Agent		
		MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
Short-term	Hierarchical	2.1117	2.4956	4.03	**1.7922**	**2.2119**	**3.71**
	Round Robin	2.1117	2.4919	3.86	**1.8253**	**2.2561**	**3.51**
	Ensemble	1.7222	2.0703	3.43	**1.5328**	**1.8286**	**3.01**
Medium-term	Hierarchical	3.6433	4.3472	6.89	**3.1861**	**3.9572**	**5.83**
	Round Robin	**2.9714**	**3.6694**	**5.23**	3.2436	4.1267	5.82
	Ensemble	2.7647	**3.4661**	**4.71**	**2.6739**	3.5003	5.06
Long-term	Hierarchical	**3.7917**	**4.6594**	**8.61**	4.1919	5.2447	9.16
	Round Robin	4.3328	5.3242	**8.99**	**4.1294**	**5.1675**	9.09
	Ensemble	**3.0317**	**3.7936**	**7.24**	4.0689	5.0539	8.73

TABLE 3: Summary of Aggregate Performance Metrics Across Models and Coordination Architectures.

Dimension	Configuration	Mean MAE	Mean MAPE (%)
Base Model (Overall)	LSTM-based Multi-Agent	2.9423	**5.89%**
	Transformer-based Multi-Agent	**2.9605**	5.99%
Architecture (under LSTM)	Hierarchical Architecture	3.1822	6.51%
	Round Robin Architecture	3.1386	6.03%
	Ensemble Voting Architecture	**2.5062**	**5.12%**

An analysis of the aggregated outcomes across the specific forecasting horizons (Table 2) yields the following insights regarding architectural behavior:

- **Ensemble Voting Dominance:** The Ensemble architecture consistently maintains the lowest error rates (MAE, RMSE, and MAPE) across almost all combinations of horizons and LLMs. Its parallel prediction and consensus-based voting mechanism effectively reconcile heterogeneous technical and fundamental signals, preventing any single agent's hallucination from dominating the final forecast.
- **Round Robin's Competitive Refinement:** The Round Robin architecture demonstrates competitive capabilities, particularly in the short-term horizon where sequential refinement helps optimize complex signals. However, as the horizon extends to medium and long-term, the risk of error propagation through the sequential chain slightly degrades its aggregate performance compared to the Ensemble approach.
- **Hierarchical Coordination Limitations:** The Hierarchical architecture generally records the highest aggregate errors. While a centralized Master Agent aims to filter out speculative noise, in highly volatile and unpredictable emerging markets, this multi-level supervision introduces a bottleneck that occasionally dilutes nuanced, localized signals detected by specialized lower-level agents.

Furthermore, to rigorously validate the superiority of the proposed configurations, the Diebold-Mariano (DM) test was conducted (Table 4). The test shows that while the Transformer baseline outperforms the LSTM-based framework at the short-term horizon ($p < 0.05$), the LSTM-based framework achieves comparable overall performance across aggregate horizons, showing gradual improvement as the forecasting horizon extends. Crucially, within the LSTM group, the Ensemble Voting architecture demonstrates statistically significant improvements over Hierarchical in medium-term ($p < 0.01$) and long-term forecasting ($p < 0.05$). While the Ensemble Voting architecture achieves lower aggregate MAPE than Round Robin, the DM test shows that their performance difference does not reach statistical significance ($p > 0.1$). No significant architectural difference was observed at the short-term horizon ($p > 0.1$), likely due to the higher noise-to-signal ratio. This provides robust statistical evidence that the Ensemble's lower aggregate MAPE is not a product of stochastic variation, but rather a persistent architectural advantage in resolving heterogeneous signals.

TABLE 4: Diebold-Mariano (DM) Test Statistics for Key Model Comparisons (Loss Function: MSE). Architecture comparisons conducted under Gemini 2.0 Flash + LSTM configuration.

Comparison Pair (Model 1 vs. Model 2)	**Horizon**	**DM Statistic**	***p*-value**
Transformer vs. **LSTM** (Overall)	All	0.5425	0.5875
Transformer vs. **LSTM** (Overall)	Short-term	-2.5135	0.0120**
Transformer vs. **LSTM** (Overall)	Medium-term	0.1422	0.8869
Transformer vs. **LSTM** (Overall)	Long-term	0.7256	0.4681
Hierarchical vs. **Ensemble** (LSTM)	Medium-term	3.6596	0.0003***
Round Robin vs. **Ensemble** (LSTM)	Medium-term	1.2938	0.1957
Hierarchical vs. **Ensemble** (LSTM)	Long-term	2.5474	0.0109**
Round Robin vs. **Ensemble** (LSTM)	Long-term	1.5858	0.1128

Note: ***, **, * indicate significance at 1%, 5%, and 10% levels respectively.
Positive DM stat indicates Model 2 (LSTM / Ensemble) outperforms Model 1.
To address price scale dominance in MSE-based pooling, robustness checks using the MAPE criterion were conducted, yielding consistent rankings.

VI. RQ2: IMPACT OF FOUNDATION LLMS

This section evaluates the impact of the underlying foundation Large Language Model (LLM) on the predictive accuracy and stability of the multi-agent framework. By conducting a horizontal comparison across Table 7, we isolate the performance variations attributable strictly to the choice of the language model under identical coordination architectures and base modeling structures.

To provide a comprehensive quantitative comparison, Table 5 details the average performance metrics (MAE, RMSE, and MAPE) for each foundation LLM family, aggregated across all stock codes and multi-agent coordination architectures for each forecasting horizon.

TABLE 5: **Aggregate Performance Comparison Across Foundation LLM Families (Averaged Over All Stocks and Coordination Architectures).**

Horizon	Metric	Llama-3.1-8B	GPT-4o	Gemini 2.0 Flash
Short-term (3-day)	MAE	**1.6985**	1.9779	1.8715
	RMSE	**2.0819**	2.3849	2.2104
	MAPE (%)	**3.29%**	3.89%	3.60%
Medium-term (14-day)	MAE	3.5615	2.9918	**2.6882**
	RMSE	4.5851	3.7482	**3.2001**
	MAPE (%)	6.14%	**5.30%**	5.32%
Long-term (60-day)	MAE	4.6619	3.9197	**3.1915**
	RMSE	5.9900	4.8493	**3.7824**
	MAPE (%)	9.73%	8.62%	**7.56%**
Overall Average	MAE	3.3073	2.9631	**2.5837**
	RMSE	4.2190	3.6608	**3.0643**
	MAPE (%)	6.39%	5.94%	**5.49%**

A. PERFORMANCE CHARACTERISTICS OF LLM FAMILIES

The empirical results reveal distinct performance patterns for each LLM family, reflecting their architectural scaling, pretraining focus, and instruction-tuning alignment:

- **Google Gemini 2.0 Flash (Strongest Overall Performance):** Gemini 2.0 Flash achieves the lowest overall average MAE (2.5837), RMSE (3.0643), and MAPE (5.49%). Its superiority becomes increasingly pronounced as the forecasting horizon extends, culminating in a dominant performance at the 60-day horizon (7.56% MAPE vs. 8.62% for GPT-4o and 9.73% for Llama). This reflects Gemini's advanced reasoning capabilities and robust long-context window, which are critical for processing multi-source heterogeneous inputs (technical indicators, news sentiment, and fundamental reports) and maintaining predictive coherence over extended horizons.
- **GPT-4o (Strong Competitive Performance):** GPT-4o demonstrates highly competitive performance, particularly in medium-term (14-day) forecasting, where it achieves the lowest average MAPE (5.30%). While slightly trailing Gemini in overall metrics, its robust instruction-following capabilities enable stable coordination within multi-agent frameworks, particularly in reconciling technical signals and sentiment dynamics.
- **Llama-3.1-8B (Short-Term Specialization):** Despite its smaller parameter footprint, Llama-3.1-8B exhibits outstanding performance in short-term forecasting, achieving the lowest MAE (1.6985), RMSE (2.0819), and MAPE (3.29%). This suggests that in short-term horizons where financial time-series are dominated by high-frequency noise and retail investor sentiment, the smaller parameter capacity of Llama-3.1-8B acts as an implicit regularizer, preventing the overfitting that often plagues larger, over-parameterized models.

VII. RQ3: MULTI-AGENT VS. TRADITIONAL BASELINES

A. ANALYSIS OF BASELINES VS. MULTI-AGENT SYSTEMS

The empirical results presented in Table 6 definitively answer RQ3 by demonstrating the overwhelming superiority of multi-agent collaborative frameworks over single-agent baselines and traditional deep learning models.

1. Limitations of Single-Agent Baselines: Both traditional time-series models (**LSTM** and **TRANSFORMER**) and independent LLMs operating as single agents (**LLAMA, GPT, GEMINI Baselines**) struggle to capture the complex, noisy dynamics of the Vietnamese stock market. Across all forecasting horizons, these baseline models exhibit unacceptably high error rates, with MAPE consistently hovering between **14.0% and 15.8%**. This underscores the reality that neither pure quantitative models (lacking contextual reasoning) nor pure LLMs (lacking rigorous numerical optimization) are sufficient for robust financial forecasting in isolation.

2. The Multi-Agent Synergy: When LLMs are integrated with quantitative base models (LSTM/Transformer) within a multi-agent ecosystem, forecasting accuracy improves dramatically. The multi-agent architectures successfully reduce the MAPE from the $\sim$15% baseline down to the **2.5% – 6.0%** range. This drastic reduction confirms that LLMs excel when deployed as high-level orchestrators—synthesizing numeric signals, qualitative news, and market sentiment from specialized sub-agents—rather than acting as solitary predictors.

3. Identifying the Optimal Configuration: Among all tested combinations, the **GEMINI + TRANSFORMER** pairing deployed within an **Ensemble architecture** emerges as the absolute best performer. It achieves unprecedented accuracy across all horizons: **2.92%** (Short-term), **2.34%** (Medium-term), and **5.90%** (Long-term) MAPE. Conversely, while smaller models like Llama-3.1 perform admirably in short-term windows, their performance degrades significantly in long-term scenarios (e.g., Llama+LSTM Round Robin hits 11.0% MAPE in the 60-day horizon), highlighting the necessity of massive parameter capacity (like Gemini's) for sustained long-term logical reasoning.

In conclusion, the proposed Multi-Agent System provides a massive leap in predictive reliability, proving that the future of financial time-series forecasting lies in the structured, debate-driven collaboration between symbolic LLM reasoning and deep neural networks.

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TABLE 6: **Comprehensive Comparison of Single-Agent Baselines vs. Multi-Agent Architectures (RQ3).** Values represent the average error across all evaluated stocks.

Model	**Architecture**	**Short-term**			**Medium-term**			**Long-term**		
		MAE	**MAPE (%)**	**RMSE**	**MAE**	**MAPE (%)**	**RMSE**	**MAE**	**MAPE (%)**	**RMSE**
LSTM	Baseline	8.380	14.626	9.957	8.893	15.326	10.776	7.480	14.321	8.684
TRANSFORMER	Baseline	8.460	13.970	10.367	8.298	15.085	9.537	7.904	15.445	8.983
LLAMA	Baseline	8.329	14.626	9.812	9.085	15.780	10.704	7.956	14.434	9.583
LLAMA + LSTM	Hierarchical	1.620	3.073	2.238	3.859	6.935	6.751	4.332	9.353	6.625
	Round Robin	1.897	3.492	3.060	3.304	5.690	5.788	6.047	11.025	12.819
	Ensemble	1.670	3.258	2.360	3.860	6.034	8.324	3.961	9.473	6.029
LLAMA + TRANSFORMER	Hierarchical	1.551	3.318	2.166	3.077	5.588	5.811	4.673	10.074	7.983
	Round Robin	1.680	3.166	2.573	3.748	6.349	6.579	4.547	9.165	7.437
	Ensemble	1.772	3.385	2.496	3.520	6.231	6.232	4.417	9.305	7.812
GPT	Baseline	8.325	14.475	9.960	8.816	15.554	10.257	8.214	15.352	9.613
GPT + LSTM	Hierarchical	2.754	5.101	4.011	3.081	5.826	4.252	3.069	7.294	4.029
	Round Robin	2.582	4.612	4.034	2.979	5.004	4.574	3.413	7.995	4.336
	Ensemble	1.745	3.683	2.360	2.436	4.076	3.647	2.184	5.627	3.155
GPT + TRANSFORMER	Hierarchical	1.853	3.954	2.436	3.219	5.667	5.367	4.601	9.501	7.084
	Round Robin	1.679	3.293	2.671	3.376	5.838	6.849	4.636	10.331	7.266
	Ensemble	1.254	2.692	1.645	2.859	5.426	5.122	5.616	10.995	10.539
GEMINI	Baseline	8.583	15.170	10.102	7.795	14.618	8.722	8.203	15.326	9.759
GEMINI + LSTM	Hierarchical	1.960	3.923	2.659	3.991	7.889	5.196	3.976	9.204	4.995
	Round Robin	1.856	3.448	2.513	2.632	4.996	3.573	3.611	8.249	5.058
	Ensemble	1.963	4.099	2.575	2.145	4.547	3.103	3.065	7.056	4.036
GEMINI + TRANSFORMER	Hierarchical	1.970	3.886	2.780	3.262	6.263	4.545	3.302	7.884	4.510
	Round Robin	2.116	4.070	3.041	2.608	5.274	3.614	3.208	7.783	4.138
	Ensemble	1.571	2.923	2.203	1.644	3.502	2.344	2.175	5.905	3.057

VIII. DISCUSSION

A. KEY FINDINGS AND IMPLICATIONS

Empirical evaluation reveals strong horizon-dependent architectural effectiveness. Ensemble Voting demonstrates consistent superiority in medium-term forecasting (14-day) across multiple stocks: FPT (5.7%), VCB (4.0%), VCS (1.6%), DXG (10.0%), validating ensemble learning theory [40]. Short-term predictions (3-day) exhibit stock-specific architectural preferences with no single dominant architecture, suggesting idiosyncratic factors require tailored approaches [41]. Long-term predictions (60-day) show mixed preferences: Hierarchical for VCB (8.9%), Ensemble for DXG (19.1%) and FPT (4.6%), Round Robin for VCS (3.4%), reflecting increased uncertainty in extended horizons [42].

Integrating Large Language Models with specialized financial agents enables sophisticated interpretation of unstructured information while maintaining domain-specific expertise [43], [44].

B. THEORETICAL CONTRIBUTIONS

This study provides empirical evidence for horizon-dependent coordination effectiveness, demonstrating that optimal architectural choices must consider temporal prediction horizons [45]. The dynamic weight adjustment mechanism provides a framework for adaptive ensemble learning in non-stationary financial environments [46]. Integrating LLMs with specialized financial agents represents a novel hybrid architecture bridging symbolic AI and connectionist learning paradigms [47], [48]. The modular agent architecture enables transparent attribution of predictions to specific analytical components, supporting regulatory compliance [49].

C. RESEARCH QUESTION ANALYSIS

1) A. RQ1: Horizon-Dependent Architectural Effectiveness

No single coordination architecture is universally optimal; effectiveness is strictly horizon-dependent.

a: Medium-Term (14 Days): Ensemble Dominance.

Ensemble Voting consistently demonstrated superior performance: FPT (5.7%), VCB (4.0%), VCS (1.6%), DXG (10.0%), effectively integrating divergent technical and fundamental signals.

b: Short-Term (3 Days): Stock-Specific Heterogeneity.

Hierarchical architecture proved advantageous in high-volatility environments, with centralized coordination mitigating speculative noise within tight settlement windows.

c: Long-Term (60 Days): Sector-Specific Specialization.

Preferred architecture varied by sector: Hierarchical for regulated banking, Round Robin for cyclical industrial construction.

2) B. RQ2: Impact of Foundation Models (LLMs)

Optimal LLM selection is highly stock-specific and horizon-dependent. GPT-4o demonstrated superior short-term accuracy for DXG, FPT, and VCB. Medium-term forecasting revealed balanced distribution: Gemini 2.0 Flash for FPT and VCS, GPT-4o for DXG, Llama-3.1-8B for VCB. Long-term forecasting showed no single dominant LLM, suggesting practitioners should adopt a *portfolio approach* to LLM selection.

3) C. RQ3: Multi-Agent Systems vs. Traditional Baselines (with a sectoral exception)

Multi-Agent Systems (MAS) significantly outperform LSTM baselines and outperform Single-Agent configurations in most high-volatility contexts, but not in every sector. LSTM models exhibited significant stochastic instability with pronounced amplitude distortion. Single-Agent architectures lacked robust error-correction under high volatility (e.g., DXG and VCS), where MAS acted as *stability anchors* by aggregating heterogeneous perspectives and filtering idiosyncratic noise. However, VCB provides an important exception: in a relatively stable, policy-anchored banking regime, *Single-Agent* achieved the best MAE/MAPE, suggesting that when signal structure is smoother, MAS coordination can introduce additional variance from cross-agent disagreement.

IX. LIMITATIONS AND FUTURE WORK

A. CURRENT LIMITATIONS

Several limitations must be acknowledged when interpreting these results. First, the evaluation scope encompasses only four stocks from the Vietnamese market over a specific temporal period, which may constrain the generalizability of findings to other emerging markets or broader equity universes. While the selected stocks represent key economic sectors, the limited sample size may not capture the full spectrum of market behaviors and sectoral dynamics present in Vietnam's broader equity landscape [50].

Second, the study focuses primarily on price prediction accuracy metrics (MAE, RMSE, MAPE) without incorporating transaction costs, liquidity constraints, or market impact considerations that are crucial for practical trading implementation. Vietnam's $T + 2.5$ settlement period and relatively constrained liquidity compared to developed markets introduce practical constraints not fully reflected in the evaluation framework. Additionally, the impact of bid-ask spreads, market depth variations, and execution timing on actual trading performance remains unexplored.

B. TEMPORAL ISOLATION AND NEWS DATA LEAKAGE PREVENTION

A critical methodological consideration in financial forecasting systems that incorporate news-driven agents is the risk of *temporal data leakage*, wherein future information inadvertently influences historical predictions. In our framework, the TickerNews and MarketNews agents process company-specific and macroeconomic news streams to extract sentiment signals and event-driven impact assessments. To ensure temporal validity and prevent look-ahead bias, we implement strict temporal isolation protocols:

Timestamp-based filtering: All news articles, earnings releases, and policy announcements are timestamped at ingestion. During backtesting, each agent receives only news items with publication timestamps strictly preceding the prediction cutoff time t. For instance, when generating a 14-day forecast at market close on date t, only news published before the close of trading day t is accessible to the agents.

Intraday timing constraints: Vietnamese stock exchanges (HOSE, HNX) operate with specific trading sessions (morning: 9:00–11:30, afternoon: 13:00–15:00). News released during trading hours on day t may already be partially reflected in the closing price of day t. To address this, we apply a conservative cutoff: only news published before the opening of day t is used for predictions targeting day t or later, ensuring that sentiment signals do not incorporate information contemporaneous with or posterior to the target price.

Embargo period for fundamental data: Earnings reports, financial statements, and regulatory filings often exhibit publication delays. We enforce a 24-hour embargo period after official release timestamps before incorporating such data into the InvestmentExpert agent's fundamental analysis, mitigating the risk of using information that market participants could not have acted upon in real time.

Validation through forward-only backtesting: The entire evaluation framework employs a strictly forward-rolling window methodology. At each prediction timestamp t, the system state (including news corpus, technical indicators, and fundamental metrics) is reconstructed using only data available up to $t-1$, with no access to future information. This walk-forward validation ensures that reported performance metrics reflect realistic out-of-sample forecasting conditions.

Despite these safeguards, residual risks remain. First, the precise timing of information dissemination in Vietnam's market is not always deterministic; informal channels (social media, investor forums) may propagate rumors or unofficial news before formal publication, creating ambiguity in timestamp reliability. Second, the LLM-based reasoning layer may inadvertently encode temporal patterns learned during pretraining on datasets that include future information relative to our evaluation period, though this risk is mitigated by the use of structured prompt templates that explicitly condition on historical context only.

Future work should explore more granular timestamp validation, incorporation of intraday tick-level data to better align news timing with price formation, and adversarial testing frameworks to detect potential leakage pathways in multi-agent coordination layers.

C. OPERATIONAL AND ROBUSTNESS LIMITATIONS

The framework's reliance on cloud-based LLM services introduces potential operational risks, including latency variability, service availability concerns, and cost scalability issues that could affect real-time trading applications [51]. Network connectivity disruptions, API rate limitations, and service outages could compromise system reliability in production environments. Furthermore, the cost structure of cloud-based inference may become prohibitive for high-frequency trading applications or for resource-constrained institutional investors.

Third, the framework's performance during extreme market conditions, such as financial crises, high-impact events, or periods of extreme volatility, remains untested. The evaluation period, while including normal market volatility, may not encompass the full range of stress scenarios that could challenge the system's robustness. The coordination mechanisms' behavior during periods of extreme correlation breakdown, liquidity crises, or regulatory interventions requires further investigation to establish operational reliability under adverse conditions [52].

X. CONCLUSION

We investigate the problem of stock price forecasting in Vietnamâ€™s emerging and retail-dominated market through the validation of a specialized *Multi-Agent* forecasting framework. Based on a comprehensive empirical evaluation conducted on four representative stocks spanning multiple sectors (VCB, FPT, DXG, and VCS), the study provides clear and structured answers to its guiding research questions.

First, in addressing **RQ1 (Architectural Effectiveness Across Horizons)**, the results demonstrate that no single coordination architecture is universally optimal across all forecasting horizons. Instead, architectural effectiveness is strongly horizon-dependent. For the medium-term horizon (14 days), *Ensemble Voting* consistently achieves robust performance across multiple stocks: FPT (5.7% MAPE), VCB (4.0% MAPE), VCS (1.6% MAPE), and DXG (10.0% MAPE), effectively reconciling heterogeneous technical, fundamental, and sentiment-driven signals. In contrast, short-term forecasting (3 days), particularly for highly volatile assets, benefits from a *Hierarchical* coordination mechanism, in which centralized decision-making suppresses speculative noise within the $T+2.5$ settlement cycle. For long-term horizons (60 days), the optimal architecture diverges by sector, with *Hierarchical* coordination proving effective for highly regulated industries such as banking (VCB: 8.9% MAPE), *Ensemble Voting* for real estate (DXG: 19.1%) and technology (FPT: 4.6%), and *Round Robin* for industrial construction (VCS: 3.4%).

Second, in response to **RQ2 (Impact of Foundation Models)**, the study establishes that the choice of the underlying Large Language Model (LLM) is highly context-dependent rather than universally applicable. LLM performance varies significantly across forecasting horizons and stock characteristics. For short-term forecasting, GPT-4o demonstrates strong performance for DXG and VCB, while Google Gemini 2.0 Flash proves optimal for FPT and VCS. Medium- and long-term horizons reveal a heterogeneous landscape in which Google Gemini 2.0 Flash, GPT-4o, and Llama-3.1-8B each emerge as optimal under different stock-specific conditions. These findings challenge the assumption of a single dominant LLM for financial forecasting and instead support a portfolio-based approach to LLM selection, in which models are empirically matched to specific assets and forecasting horizons.

Third, in answering **RQ3 (Superiority of Multi-Agent Systems over Traditional Baselines)**, the empirical results support a conditional rather than absolute claim. *Multi-Agent Systems* consistently outperform traditional LSTM baselines and outperform Single-Agent LLMs in most volatile sectors, but VCB is a clear exception where *Single-Agent* records the lowest error (MAE 1.10; MAPE 1.92%). LSTM models exhibit pronounced instability under regime shifts, frequently overshooting during bullish phases and undershooting during bearish corrections, reflecting their limited capacity to incorporate market sentiment and structural changes. Single-Agent LLMs, while effective for relatively stable assets, lack robust error-correction under high volatility. In contrast, Multi-Agent architecturesâ€”particularly Hierarchical and Ensemble systemsâ€—act as stability anchors in sentiment-driven sectors by aggregating diverse perspectives from fundamental, technical, risk, and news-oriented agents. The VCB case therefore serves as a valuable boundary condition: in stable banking dynamics, simpler coordination may preserve cleaner signals than multi-source aggregation.

In summary, this work contributes a *horizon-adaptive Multi-Agent forecasting framework* to the field of financial artificial intelligence. The findings indicate that within emerging markets such as Vietnam, predictive accuracy depends not solely on model complexity but on the strategic alignment between coordination architecture, forecasting horizon, and the underlying cognitive engine. This insight provides practical guidance for both researchers and practitioners seeking robust and adaptive forecasting solutions in volatile, retail-driven financial environments.

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APPENDIX. ADDITIONAL RESULTS

TABLE 7: **Performance Comparison Across Forecasting Horizons, Architectures, and LLMs.** Cell background colors indicate the best-performing architecture for each baseline model (Green = Hierarchical, Blue = Round Robin, Yellow = Ensemble Voting). Bold text highlights the best-performing LLM within each baseline model (horizontal comparison).

Horizon	Stock	Architecture	LSTM									Transformer								
			Llama-3.1-8B			GPT-4o			Gemini 2.0			Llama-3.1-8B			GPT-4o			Gemini 2.0		
			MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Short-term	DXG	Hierarchical	0.55	0.90	**2.3**	**0.50**	**0.62**	**2.3**	0.71	0.94	3.1	**0.82**	**0.93**	**3.8**	1.75	2.08	8.1	1.54	1.83	7.1
		Round Robin	1.13	1.16	5.2	**0.50**	**0.60**	**2.3**	0.79	0.97	3.5	**0.92**	**1.22**	**4.0**	1.04	1.34	4.8	1.09	1.41	5.1
		Ensemble	0.97	1.09	4.3	**0.51**	**0.64**	**2.3**	0.89	1.09	3.9	0.79	0.85	3.5	1.86	1.93	8.4	**0.27**	**0.40**	**1.3**
	FPT	Hierarchical	2.92	3.29	3.0	**2.04**	**2.83**	**2.1**	2.67	2.85	2.8	**2.44**	3.53	**2.5**	3.64	4.08	3.8	2.87	**3.33**	2.9
		Round Robin	3.09	3.52	3.1	2.75	3.23	2.8	**2.16**	**2.50**	**2.3**	**2.83**	3.86	**2.9**	2.86	**3.56**	**2.9**	3.61	3.88	3.7
		Ensemble	2.97	3.68	3.0	2.73	3.20	2.8	**1.38**	**1.58**	**1.4**	3.38	4.45	3.4	**1.57**	**1.83**	**1.7**	2.90	3.36	2.9
	VCB	Hierarchical	1.03	1.72	1.7	**0.58**	**0.73**	**0.9**	1.34	1.55	2.2	1.23	1.51	2.0	**0.89**	**1.01**	**1.4**	0.99	1.36	1.6
		Round Robin	0.48	0.65	0.8	**0.26**	**0.34**	**0.4**	1.09	1.32	1.8	0.82	0.87	1.3	0.66	0.68	1.1	**0.45**	**0.67**	**0.7**
		Ensemble	1.05	1.21	1.7	**0.46**	**0.65**	**0.7**	1.79	1.96	2.9	0.64	0.84	1.0	0.57	0.73	0.9	**0.46**	**0.65**	**0.7**
	VCS	Hierarchical	1.34	1.50	2.9	1.76	1.81	3.8	**0.75**	**0.82**	**1.6**	1.08	**1.23**	2.3	**1.04**	1.25	**2.2**	1.74	1.79	3.7
		Round Robin	1.06	1.15	2.3	1.79	1.85	3.8	**0.72**	**0.81**	**1.6**	0.94	1.21	2.0	**0.57**	**0.65**	**1.2**	1.77	1.83	3.8
		Ensemble	0.81	1.01	1.7	1.31	1.50	2.8	**0.68**	**0.80**	**1.4**	**1.08**	1.25	**2.3**	1.16	**1.24**	2.5	1.80	1.86	3.9
	HPG	Hierarchical	1.39	1.66	5.0	1.61	2.80	5.5	**1.25**	**1.31**	**4.5**	1.80	2.45	6.3	2.37	3.24	8.2	**0.83**	**1.15**	**2.9**
		Round Robin	**1.10**	**1.24**	**3.9**	1.25	1.32	4.4	1.11	1.25	4.0	1.22	1.35	4.4	1.46	2.14	5.0	**0.79**	**0.98**	**2.8**
		Ensemble	**1.12**	1.52	**3.9**	2.02	2.70	7.0	1.25	**1.31**	4.5	1.03	1.15	3.7	**0.53**	**0.71**	**1.9**	0.89	1.17	3.1
	TCB	Hierarchical	1.89	2.59	5.0	**0.96**	**1.10**	**2.6**	1.52	1.76	4.1	1.46	2.06	4.0	1.83	1.99	4.9	**1.39**	**1.51**	**3.7**
		Round Robin	**0.73**	**0.82**	**2.0**	0.90	1.11	2.3	1.42	1.54	3.7	1.23	1.25	3.3	1.85	2.14	5.0	**1.00**	**1.23**	**2.6**
		Ensemble	2.19	2.79	6.0	**0.85**	**1.13**	**2.2**	2.35	2.69	6.0	2.10	2.44	5.7	1.39	1.90	3.8	**1.09**	**1.16**	**2.9**
	VHM	Hierarchical	3.50	4.43	3.1	3.24	**3.79**	2.9	**3.05**	4.63	**2.7**	2.53	**3.27**	2.2	**2.18**	3.38	**1.9**	3.41	4.16	3.0
		Round Robin	7.95	8.96	7.1	**2.65**	**3.68**	**2.4**	3.33	3.95	3.0	5.13	6.24	4.5	4.36	6.30	3.8	**3.58**	**4.47**	**3.2**
		Ensemble	1.74	1.85	1.6	**1.27**	**1.43**	**1.2**	3.19	3.25	2.9	3.72	4.53	3.3	**2.56**	**2.77**	2.2	2.97	3.83	2.6
	MWG	Hierarchical	**1.61**	**2.02**	**1.9**	9.65	9.71	11.6	2.59	2.95	3.2	1.55	2.04	1.9	1.67	1.88	2.0	**1.09**	**1.27**	**1.3**
		Round Robin	**1.31**	**1.51**	**1.5**	8.82	9.26	10.7	2.68	3.45	3.3	**1.04**	**1.25**	**1.2**	1.61	2.05	1.9	1.17	1.31	1.4
		Ensemble	**2.34**	**3.27**	**2.8**	3.53	4.06	4.3	3.93	4.94	4.6	1.50	1.84	1.8	**1.01**	**1.03**	**1.2**	1.13	1.30	1.4
	CMG	Hierarchical	**1.03**	**1.37**	**2.6**	6.02	6.10	15.3	1.75	2.21	4.4	1.29	1.78	3.2	**1.01**	**1.08**	**2.6**	3.45	3.98	8.7
		Round Robin	**1.14**	**1.26**	**3.0**	3.50	4.65	9.0	3.01	3.16	7.6	**0.99**	**1.12**	**2.6**	1.20	1.73	2.9	5.19	5.27	13.2
		Ensemble	**0.89**	**0.99**	**2.2**	1.32	1.62	3.4	0.95	1.36	2.5	1.10	1.32	2.9	**0.40**	**0.46**	**1.0**	3.02	3.49	7.9
	KDH	Hierarchical	**1.35**	**1.49**	**3.9**	2.32	3.05	6.5	4.38	4.59	12.7	1.88	2.40	5.6	**1.55**	2.14	**4.5**	1.62	**1.92**	4.7
		Round Robin	1.95	**2.22**	5.7	**1.60**	2.33	**4.5**	**1.60**	2.33	**4.5**	2.03	2.35	5.9	1.77	2.79	5.2	**1.23**	**1.58**	**3.6**
		Ensemble	2.07	2.73	6.1	2.87	3.22	8.2	**1.74**	**1.81**	**5.0**	**0.96**	**1.19**	**2.8**	1.39	1.96	4.1	1.61	1.84	4.7
	DGC	Hierarchical	**2.00**	**2.13**	**2.1**	3.27	3.87	3.5	2.77	3.33	2.9	**1.34**	**1.81**	**1.4**	3.23	3.28	3.4	4.10	5.55	4.4
		Round Robin	**1.29**	**1.70**	**1.4**	4.91	6.77	5.1	3.80	4.34	4.0	2.09	2.97	2.3	**1.81**	**2.29**	**1.9**	4.50	5.86	4.8
		Ensemble	3.31	4.06	3.6	2.31	3.11	2.5	**2.22**	**2.84**	**2.3**	3.14	3.55	3.4	**1.89**	**2.37**	**2.0**	2.42	2.83	2.6
	MBB	Hierarchical	0.83	1.25	3.3	1.10	1.25	4.3	**0.75**	**0.89**	**2.9**	1.21	1.36	4.8	1.07	1.33	4.3	**0.63**	**0.67**	**2.4**
		Round Robin	1.55	1.63	6.0	2.05	2.41	7.7	**0.55**	**0.72**	**2.2**	**0.91**	1.14	**3.6**	0.97	**1.07**	3.8	1.02	1.16	4.0
		Ensemble	**0.58**	**0.68**	**2.3**	1.75	1.95	6.9	0.66	0.81	2.5	1.81	2.24	7.0	0.73	0.88	2.7	**0.31**	**0.48**	**1.1**
Medium-term	DXG	Hierarchical	**0.53**	**0.57**	**2.5**	0.86	1.03	4.0	2.99	4.53	14.6	2.04	2.23	9.4	1.01	**1.14**	4.7	**0.96**	1.16	**4.6**
		Round Robin	**0.74**	**0.88**	**3.5**	0.88	1.08	4.1	1.84	2.16	8.7	1.02	1.18	4.7	**0.73**	**0.97**	**3.5**	2.29	2.71	10.3
		Ensemble	0.90	1.23	4.0	**0.76**	**0.95**	**3.5**	2.06	3.19	10.0	1.03	1.18	4.7	1.12	1.18	5.3	**0.87**	**1.02**	**4.0**
	FPT	Hierarchical	7.05	**8.01**	6.9	**6.58**	8.31	**6.4**	7.09	8.12	6.9	**4.90**	6.57	**4.8**	4.93	**5.86**	**4.8**	6.58	8.27	6.4
		Round Robin	7.30	9.64	7.1	**6.64**	8.36	**6.5**	7.63	**8.10**	7.5	8.77	11.04	8.6	**6.17**	11.27	**6.0**	6.81	**8.60**	6.6
		Ensemble	6.97	10.31	6.8	6.72	8.36	6.5	**5.81**	**6.97**	**5.7**	7.29	10.98	7.1	6.19	8.63	6.0	**1.42**	**2.23**	**1.4**
	VCB	Hierarchical	**1.09**	**1.27**	**1.7**	1.61	1.80	2.6	2.56	2.68	4.2	2.66	3.17	4.2	1.74	2.33	2.7	**1.65**	**2.05**	**2.6**
		Round Robin	2.57	2.67	4.1	**1.88**	**2.32**	**3.0**	2.49	2.64	4.0	1.71	2.79	2.6	**1.41**	**1.74**	**2.2**	1.78	2.19	2.8
		Ensemble	3.07	3.49	4.9	**2.24**	**2.49**	**3.6**	2.51	2.71	4.0	2.63	2.87	4.3	1.76	**1.99**	2.8	**1.65**	2.04	**2.6**
	VCS	Hierarchical	2.42	2.73	5.1	2.20	2.26	4.6	**0.74**	**1.19**	**1.6**	**1.83**	**2.05**	**3.9**	2.25	2.53	4.7	2.73	2.98	5.7
		Round Robin	2.05	2.47	4.3	1.65	**1.95**	3.4	**1.38**	1.98	**2.8**	1.74	2.18	3.7	1.65	1.98	3.4	**1.17**	**1.39**	**2.4**
		Ensemble	1.54	1.82	3.2	1.85	2.05	3.8	**0.75**	**0.94**	**1.6**	**1.80**	**2.28**	**3.7**	2.93	3.31	6.1	2.05	2.31	4.3
	HPG	Hierarchical	2.25	3.34	7.7	2.62	2.78	9.7	**1.40**	**1.63**	**5.0**	2.23	3.13	7.7	**0.81**	**0.99**	**3.0**	1.33	1.59	4.7
		Round Robin	1.32	1.57	4.7	1.21	**1.36**	4.4	**1.19**	**1.36**	**4.3**	**0.82**	**0.95**	**2.9**	0.92	1.34	3.1	1.23	1.37	4.4
		Ensemble	2.29	2.96	8.3	**0.42**	**0.63**	**1.5**	1.40	1.63	5.0	2.47	3.03	8.6	1.38	1.85	4.8	**1.18**	**1.54**	**4.1**
	TCB	Hierarchical	5.51	6.84	15.7	**1.75**	**2.13**	**5.0**	3.90	4.41	11.1	**1.36**	**1.43**	**3.8**	2.49	2.76	7.0	2.24	3.32	6.3
		Round Robin	2.05	3.04	5.7	2.22	2.76	6.0	**1.68**	**2.36**	**4.8**	3.79	4.64	10.6	3.84	4.71	10.8	**3.50**	**4.13**	**10.0**
		Ensemble	**1.86**	2.51	5.3	2.77	3.34	7.6	1.88	**2.31**	**5.1**	2.40	**3.21**	**6.7**	3.80	4.56	10.8	**2.37**	3.41	**6.7**
	VHM	Hierarchical	13.46	18.43	13.2	**2.70**	**3.41**	**2.7**	8.52	9.52	8.4	12.66	17.06	12.4	13.45	15.53	13.2	**4.87**	**6.70**	**4.8**
		Round Robin	11.82	15.08	11.6	**2.97**	**3.72**	**2.9**	3.66	4.15	3.6	14.38	17.57	14.2	13.36	18.60	13.1	**2.96**	**3.22**	**2.9**
		Ensemble	19.33	25.54	18.9	**3.39**	**4.34**	3.4	3.41	4.47	**3.3**	13.15	16.34	13.0	8.46	13.20	8.4	**0.99**	**1.31**	**1.0**
	MWG	Hierarchical	**2.25**	**2.76**	**2.8**	5.29	5.78	6.5	4.39	4.53	5.4	**1.48**	**1.95**	**1.9**	2.15	3.17	2.7	2.62	3.24	3.2

		Round Robin	**3.37**	5.12	**4.2**	6.52	7.82	8.1	4.03	**4.98**	5.0	2.53	3.70	3.1	**1.62**	**1.90**	**2.0**	2.74	3.34	3.4
		Ensemble	**1.16**	**1.52**	**1.4**	4.94	5.42	6.1	1.93	2.22	2.4	**1.57**	2.30	**2.0**	1.92	**2.28**	2.3	2.42	3.00	3.0
	CMG	Hierarchical	**1.62**	**2.08**	**4.0**	6.74	6.90	16.7	4.08	4.59	10.1	**1.40**	**1.56**	**3.5**	2.78	3.52	6.8	7.52	7.58	18.6
		Round Robin	**1.58**	**1.94**	**3.9**	3.23	4.30	8.1	3.01	3.41	7.4	2.19	3.05	5.4	1.20	1.38	3.0	**1.00**	**1.14**	**2.5**
		Ensemble	1.90	2.80	4.7	**0.45**	**0.75**	**1.1**	1.54	1.63	3.8	1.84	1.96	4.6	**0.89**	**1.15**	**2.2**	2.10	3.66	5.2
	KDH	Hierarchical	4.30	5.01	12.1	**1.35**	**1.72**	**3.9**	5.40	6.10	15.3	**2.29**	3.91	**6.5**	3.21	**3.31**	9.1	3.13	3.93	8.8
		Round Robin	1.92	2.16	5.4	**1.56**	**1.98**	**4.4**	1.61	1.99	4.5	3.63	4.33	10.3	**2.09**	**2.86**	**6.0**	2.87	3.25	8.1
		Ensemble	**1.37**	1.65	**3.9**	1.76	2.10	5.0	1.38	**1.41**	**3.9**	3.09	3.29	8.7	1.89	2.88	5.3	**1.05**	**1.33**	**3.0**
	DGC	Hierarchical	**4.04**	5.86	**4.2**	4.48	**5.48**	4.7	5.09	5.82	5.3	2.52	3.60	2.7	**2.12**	**2.81**	**2.2**	4.38	4.59	4.6
		Round Robin	2.20	2.60	2.3	6.36	7.97	6.6	**1.68**	**2.05**	**1.8**	**2.54**	**2.69**	**2.7**	4.54	5.67	4.8	3.38	3.72	3.5
		Ensemble	4.26	4.98	4.4	3.08	3.55	3.2	**0.59**	**0.79**	**0.6**	2.94	3.76	3.1	**1.76**	**1.97**	**1.9**	2.65	3.05	2.8
	MBB	Hierarchical	1.79	2.13	7.3	**0.78**	**0.83**	**3.2**	1.73	1.92	6.9	1.55	2.36	6.5	1.69	2.63	6.9	**1.14**	**1.45**	**4.6**
		Round Robin	2.74	3.82	11.3	**0.62**	**0.67**	**2.5**	1.37	1.64	5.6	1.86	1.89	7.6	2.99	3.47	12.2	**1.54**	**1.60**	**6.3**
		Ensemble	1.67	1.87	6.7	0.87	1.06	3.5	**0.70**	**0.79**	**2.8**	2.03	3.20	8.4	2.21	2.64	9.1	**0.96**	**1.07**	**4.0**
Long-term	**DXG**	Hierarchical	**3.21**	4.75	**19.0**	3.83	**4.29**	21.8	3.60	4.55	21.2	3.98	5.48	23.1	**1.79**	**2.02**	**9.8**	4.16	4.55	23.7
		Round Robin	**3.57**	**4.20**	**20.7**	4.19	4.63	23.9	4.73	5.42	27.0	**2.01**	**2.55**	**11.0**	3.86	5.42	22.8	4.24	4.68	24.2
		Ensemble	4.06	4.38	23.0	3.86	4.29	22.1	**3.25**	**4.06**	**19.1**	**2.54**	**2.79**	**13.6**	4.19	4.31	23.2	3.91	4.32	22.4
	FPT	Hierarchical	3.61	3.88	3.7	**2.24**	**2.86**	**2.4**	5.50	7.82	5.8	2.75	3.55	2.9	**1.37**	**1.58**	**1.4**	1.94	2.62	2.1
		Round Robin	4.19	4.66	4.3	**1.48**	**1.99**	**1.6**	5.52	7.88	5.9	**1.85**	**2.21**	**2.0**	4.82	5.69	5.0	2.12	2.79	2.2
		Ensemble	3.12	3.72	3.3	**2.11**	**2.68**	**2.2**	4.36	6.03	4.6	3.20	3.64	3.3	3.30	3.46	3.4	**1.51**	**2.19**	**1.6**
	VCB	Hierarchical	6.98	9.38	12.1	5.26	5.95	9.0	**5.21**	**5.81**	**8.9**	**4.60**	7.96	**7.6**	8.75	11.78	14.8	5.05	**5.89**	8.6
		Round Robin	**4.14**	**4.52**	**7.2**	5.09	5.75	8.7	8.83	10.10	15.0	6.53	8.17	11.1	5.07	6.60	8.6	**4.42**	**5.25**	**7.5**
		Ensemble	**2.34**	**3.08**	**4.0**	4.37	5.03	7.4	5.66	5.74	9.7	3.96	4.94	6.9	4.77	7.39	8.0	**3.87**	**4.82**	**6.5**
	VCS	Hierarchical	1.82	2.15	4.1	**1.33**	**1.54**	**3.0**	4.12	4.19	9.2	2.20	3.00	5.0	2.40	2.94	5.3	**1.22**	**1.49**	**2.8**
		Round Robin	1.82	1.91	4.0	**1.07**	**1.29**	**2.4**	1.55	1.74	3.4	**1.58**	2.44	**3.6**	2.18	2.79	5.0	1.75	**2.06**	4.0
		Ensemble	1.47	1.67	3.3	**1.20**	**1.42**	**2.7**	3.79	3.97	8.5	1.31	**1.39**	3.0	1.88	2.62	4.2	**1.27**	1.47	**2.9**
	HPG	Hierarchical	1.50	2.46	5.7	1.51	1.95	5.7	**1.13**	**1.16**	**4.2**	0.76	0.94	2.9	**0.64**	**0.72**	**2.4**	1.32	1.65	5.0
		Round Robin	1.54	1.84	5.8	1.34	1.66	5.1	**0.85**	**0.94**	**3.2**	**1.07**	**1.12**	**4.0**	1.39	2.34	5.3	1.15	1.59	4.3
		Ensemble	**0.22**	**0.25**	**0.8**	0.67	0.88	2.5	1.13	1.16	4.2	0.98	1.42	3.7	**0.81**	1.18	**3.1**	1.00	**1.17**	3.8
	TCB	Hierarchical	4.22	4.88	12.6	**2.37**	**2.72**	**7.0**	5.49	5.92	16.5	9.20	11.02	27.9	7.23	8.09	21.9	**4.11**	**4.66**	**12.3**
		Round Robin	4.38	6.25	13.3	2.98	3.21	8.8	**2.29**	**2.77**	**6.9**	6.84	9.69	21.0	6.69	7.24	20.1	**3.94**	**4.36**	**11.6**
		Ensemble	6.11	6.34	18.1	**0.42**	**0.45**	**1.3**	2.24	3.14	6.9	7.18	7.68	21.5	6.55	6.83	19.6	**4.97**	**6.03**	**15.0**
	VHM	Hierarchical	11.22	13.03	11.4	**5.70**	7.13	**5.8**	5.85	**6.99**	6.0	9.65	12.66	9.9	10.55	10.95	10.6	**8.18**	**9.68**	**8.3**
		Round Robin	24.54	34.21	25.6	6.44	6.72	6.5	**5.76**	**6.11**	**5.9**	11.11	11.58	11.2	8.02	9.22	8.2	**6.25**	**6.66**	**6.3**
		Ensemble	6.89	8.59	7.0	**3.29**	**3.67**	**3.4**	5.19	5.83	5.3	6.29	6.72	6.4	23.87	30.85	24.6	**2.52**	**2.75**	**2.5**
	MWG	Hierarchical	4.22	4.93	5.2	4.37	4.78	5.3	**3.45**	**3.61**	**4.2**	**4.08**	**5.19**	**5.0**	4.99	7.02	6.3	4.69	5.53	5.9
		Round Robin	2.69	3.82	3.4	7.09	7.38	8.7	**2.21**	**3.02**	**2.7**	4.25	5.79	5.3	**1.99**	**2.03**	**2.4**	3.86	4.78	4.8
		Ensemble	4.62	6.30	5.8	**1.14**	**1.76**	**1.4**	2.18	3.27	2.6	2.40	2.92	2.9	5.09	6.46	6.4	**2.33**	**2.54**	**2.8**
	CMG	Hierarchical	**2.66**	**2.85**	**7.4**	3.54	4.36	9.2	4.38	5.11	11.4	**2.45**	**2.67**	**6.9**	2.76	3.38	7.7	3.87	4.38	10.2
		Round Robin	**3.39**	3.77	9.3	4.84	5.05	13.5	3.40	**3.59**	**9.1**	**2.56**	**3.10**	**7.2**	3.49	4.58	9.8	4.71	5.42	13.1
		Ensemble	3.65	4.17	10.3	**1.45**	**1.68**	**3.9**	2.40	2.95	6.7	3.58	4.09	10.1	3.03	4.09	8.6	**2.73**	**3.08**	**7.6**
	KDH	Hierarchical	2.50	4.27	8.2	**2.26**	**2.37**	**6.9**	3.48	3.84	10.7	**1.38**	**1.46**	**4.2**	1.41	1.91	4.5	2.77	2.80	8.4
		Round Robin	**1.17**	**1.30**	**3.6**	1.48	1.90	4.5	1.48	1.90	4.5	**1.59**	**1.70**	**4.8**	2.08	3.67	7.0	2.72	3.95	8.4
		Ensemble	2.36	3.05	7.6	3.17	5.01	10.4	**1.93**	**2.51**	**5.9**	1.46	1.72	4.4	2.22	3.24	7.2	**0.64**	**0.78**	**1.9**
	DGC	Hierarchical	6.80	11.51	9.4	**2.49**	**3.98**	**3.4**	3.85	4.83	5.3	13.38	18.04	18.8	9.46	13.12	13.0	**0.80**	**1.01**	**1.1**
		Round Robin	19.20	25.66	26.9	**3.04**	**4.31**	**4.2**	4.39	5.72	6.1	12.38	17.08	17.2	13.41	18.08	18.9	**2.33**	**2.77**	**2.6**
		Ensemble	8.19	13.18	11.5	3.11	4.33	4.4	**2.77**	**3.81**	**3.8**	17.36	23.06	24.3	8.93	12.72	12.0	**0.63**	**0.80**	**0.8**
	MBB	Hierarchical	3.21	3.69	13.4	1.94	2.25	8.1	**1.65**	**1.95**	**6.9**	1.63	2.31	6.8	3.86	5.08	16.2	**1.53**	**1.68**	**6.3**
		Round Robin	1.93	2.60	8.1	1.92	2.26	8.0	**1.45**	**1.59**	**6.0**	2.79	3.54	11.7	2.61	3.48	11.0	**1.00**	**1.61**	**4.2**
		Ensemble	4.51	6.06	18.9	1.41	1.51	5.9	**0.50**	**0.60**	**2.1**	2.73	4.35	11.5	2.76	3.23	11.6	**0.71**	**0.89**	**2.9**

Note: Bold values indicate best performance within each LLM group. Each stock has three rows: Hierarchical, Round Robin, and Ensemble architectures.

This appendix presents the complete set of granular forecast results that underpin the aggregate performance analysis reported in the main body of the paper. For each combination of foundation LLM (Google Gemini 2.0 Flash, GPT-4o, and Llama-3.1-8B) and base time-series model (LSTM and Transformer), a dedicated table reports the predicted versus actual closing prices for all twelve stocks across three forecasting horizons—short-term (3 days), medium-term (14 days), and long-term (60 days)—under each of the three multi-agent coordination architectures (Hierarchical, Round Robin, and Ensemble). These tables are intended to enable full reproducibility and to facilitate fine-grained, stock-level comparisons that go beyond the aggregated metrics summarized in the main results.

TABLE 8: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (LLM: Google Gemini 2.0 Flash, LSTM Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Hierarchical	31/08	22.8	03/09	22.25	24	↓-1.75	14/09	23.38	24.05	↑+0.67	30/10	21.83	21.2	↑+0.63
		16/10	22.45	19/10	22.1	22.6	↓-0.50	30/10	23.34	21.2	↓-2.14	15/12	24.46	16.3	↓-8.16
		21/10	20.0	24/10	20.51	20.9	↑+0.39	04/11	20.52	20.15	↑+0.37	20/12	20.81	17.8	↓-3.01
		22/10	20.5	25/10	21.1	20.9	↑+0.20	05/11	28.88	20.1	↓-8.78	21/12	20.42	17.8	↑+2.62
	Round Robin	31/08	22.8	03/09	22.25	24	↓-1.75	14/09	21.98	24.05	↓-2.07	30/10	24.47	21.2	↓-3.27
		16/10	22.45	19/10	22.15	22.6	↓-0.45	30/10	22.28	21.2	↑+1.08	15/12	25.63	16.3	↓-9.33
		21/10	20.0	24/10	20.34	20.9	↑+0.56	04/11	20.79	20.15	↑+0.64	20/12	20.96	17.8	↓-3.16
		22/10	20.5	25/10	20.48	20.9	↓-0.42	05/11	23.68	20.1	↓-3.58	21/12	20.96	17.8	↓-3.16
	Ensemble	31/08	22.8	03/09	22.11	24	↓-1.89	14/09	22.98	24.05	↑+1.07	30/10	21.47	21.2	↑+0.27
		16/10	22.45	19/10	21.92	22.6	↓-0.68	30/10	21.75	21.2	↑+0.55	15/12	23.36	16.3	↓-7.06
		21/10	20.0	24/10	20.07	20.9	↑+0.83	04/11	20.52	20.15	↑+0.37	20/12	20.81	17.8	↓-3.01
		22/10	20.5	25/10	21.05	20.9	↑+0.15	05/11	26.35	20.1	↓-6.25	21/12	20.45	17.8	↑+2.65
FPT	Hierarchical	01/09	101.6	04/09	103.6	105	↑+1.40	15/09	106.07	101.9	↑+4.17	31/10	105.57	103.9	↑+1.67
		16/10	89.8	19/10	90.83	88.1	↓-2.73	30/10	89.32	102.7	↓-13.38	15/12	92.13	93.8	↑+1.67
		21/10	93	24/10	95.37	97.7	↑+2.33	04/11	95.8	103.3	↑+7.50	20/12	97.56	93.9	↑+3.66
		22/10	97	25/10	93.5	97.7	↓-4.20	05/11	97.6	100.9	↑+3.30	21/12	108.92	93.9	↓-15.02
	Round Robin	01/09	101.6	04/09	104.42	105	↑+0.58	15/09	105.66	101.9	↑+3.76	31/10	104.27	103.9	↑+0.37
		16/10	89.8	19/10	90.46	88.1	↓-2.36	30/10	91.28	102.7	↑+11.42	15/12	90.7	93.8	↑+3.10
		21/10	93	24/10	96.0	97.7	↑+1.70	04/11	95.25	103.3	↑+8.05	20/12	97.48	93.9	↑+3.58
		22/10	97	25/10	93.68	97.7	↓-4.02	05/11	93.6	100.9	↓-7.30	21/12	108.92	93.9	↓-15.02
	Ensemble	01/09	101.6	04/09	103.6	105	↑+1.40	15/09	103.6	101.9	↑+1.70	31/10	102.77	103.9	↑+1.13
		16/10	89.8	19/10	88.55	88.1	↑+0.45	30/10	91.17	102.7	↑+11.53	15/12	92.24	93.8	↑+1.56
		21/10	93	24/10	95.14	97.7	↑+2.56	04/11	96.26	103.3	↑+7.04	20/12	97.22	93.9	↑+3.32
		22/10	97	25/10	96.59	97.7	↓-1.11	05/11	97.93	100.9	↑+2.97	21/11	105.33	93.9	↓-11.43
VCB	Hierarchical	30/08	68.6	02/09	68.66	68.6	0.06	13/09	68.52	65.8	↑+2.72	29/10	67.6	60.7	↑+6.90
		16/10	62.9	19/10	63.29	61.9	↓-1.39	30/10	64.29	60.6	↓-3.69	15/12	64.47	56.8	↓-7.67
		21/10	59.3	24/10	61.68	59.5	↑+2.18	04/11	62.37	60.1	↑+2.27	20/12	58.5	57.5	↑+1.00
		22/10	59.6	25/10	61.21	59.5	↓-1.71	05/11	62.37	60.8	↑+1.57	21/12	62.76	57.5	↓-5.26
	Round Robin	30/08	68.6	02/09	68.46	68.6	0.14	13/09	68.9	65.8	↓-3.10	29/10	77.81	60.7	↓-17.11
		16/10	62.9	19/10	62.51	61.9	↑+0.61	30/10	64.03	60.6	↓-3.43	15/12	64.57	56.8	↓-7.77
		21/10	59.3	24/10	61.15	59.5	↑+1.65	04/11	62.37	60.1	↑+2.27	20/12	63.0	57.5	↓-5.50
		22/10	59.6	25/10	61.46	59.5	↓-1.96	05/11	61.96	60.8	↑+1.16	21/12	62.45	57.5	↓-4.95
	Ensemble	30/08	68.6	02/09	67.64	68.6	0.96	13/09	68.85	65.8	↓-3.05	29/10	67.9	60.7	↑+7.20
		16/10	62.9	19/10	63.65	61.9	↓-1.75	30/10	64.34	60.6	↓-3.74	15/12	61.94	56.8	↑+5.14
		21/10	59.3	24/10	62.58	59.5	↑+3.08	04/11	62.36	60.1	↑+2.26	20/12	63.01	57.5	↓-5.51
		22/10	59.6	25/10	60.88	59.5	↓-1.38	05/11	61.78	60.8	↑+0.98	21/12	62.29	57.5	↓-4.79
VCS	Hierarchical	30/08	48.8	02/09	47.97	48.8	0.83	13/09	50.2	50	↑+0.20	29/10	50.51	47.6	↓-2.91
		16/10	47.2	19/10	47.86	46.8	↓-1.06	30/10	49.74	47.4	↑+2.34	15/12	42.13	46.5	↑+4.37
		21/10	45.5	24/10	47.12	46.2	↑+0.92	04/11	47.06	47.1	↑+0.04	20/12	47.66	43	↓-4.66
		22/10	45.5	25/10	46.38	46.2	↑+0.18	05/11	47.27	46.9	↑+0.37	21/12	47.56	43	↓-4.56
	Round Robin	30/08	48.8	02/09	48.62	48.8	0.18	13/09	46.2	50	↓-3.80	29/10	50.51	47.6	↓-2.91
		16/10	47.2	19/10	47.55	46.8	↓-0.75	30/10	46.94	47.4	↓-0.46	15/12	45.5	46.5	↑+1.00
		21/10	45.5	24/10	47.43	46.2	↑+1.23	04/11	46.12	47.1	↑+0.98	20/12	44.16	43	↑+1.16
		22/10	45.5	25/10	46.92	46.2	↑+0.72	05/11	46.6	46.9	↑+0.30	21/12	44.15	43	↑+1.15
	Ensemble	30/08	48.8	02/09	48.66	48.8	0.14	13/09	49.2	50	↑+0.80	29/10	49.65	47.6	↓-2.05
		16/10	47.2	19/10	45.48	46.8	↑+1.32	30/10	45.74	47.4	↓-1.66	15/12	41.33	46.5	↑+5.17
		21/10	45.5	24/10	46.95	46.2	↑+0.75	04/11	46.78	47.1	↑+0.32	20/12	47.43	43	↓-4.43
		22/10	45.5	25/10	45.7	46.2	↑+0.50	05/11	46.66	46.9	↑+0.24	21/12	46.53	43	↓-3.53
HPG	Hierarchical	01/09	27.5	04/09	28.29	29.85	↑+1.56	15/09	28.0	30.35	↑+2.35	31/10	28.1	26.7	↓-1.40
		16/10	28.25	19/10	28.6	28	↓-0.60	30/10	29.01	26.9	↓-2.11	15/12	25.11	26.25	↑+1.14
		21/10	26.8	24/10	28.02	26.4	↓-1.62	04/11	27.2	26.75	↓-0.45	20/12	28.03	26.7	↓-1.33
		22/10	26.7	25/10	27.61	26.4	↓-1.21	05/11	27.0	26.3	↓-0.70	21/12	27.33	26.7	0.63
	Round Robin	01/09	27.5	04/09	28.84	29.85	↑+1.01	15/09	28.84	30.35	↑+1.51	31/10	26.41	26.7	↑+0.29
		16/10	28.25	19/10	29.63	28	↓-1.63	30/10	29.01	26.9	↓-2.11	15/12	30.22	26.25	↓-3.97
		21/10	26.8	24/10	26.6	26.4	↑+0.20	04/11	27.2	26.75	↓-0.45	20/12	28.11	26.7	↓-1.41
		22/10	26.7	25/10	28.0	26.4	↓-1.60	05/11	27.0	26.3	↓-0.70	21/12	27.86	26.7	1.16
	Ensemble	01/09	27.5	04/09	25.22	29.85	↓-4.63	15/09	25.24	30.35	↓-5.11	31/10	25.3	26.7	↑+1.40
		16/10	28.25	19/10	25.92	28	↑+2.08	30/10	25.78	26.9	↑+1.12	15/12	25.86	26.25	↑+0.39
		21/10	26.8	24/10	22.63	26.4	↑+3.77	04/11	24.3	26.75	↑+2.45	20/12	23.6	26.7	↑+3.10
		22/10	26.7	25/10	21.75	26.4	↑+4.65	05/11	22.21	26.3	↑+4.09	21/12	21.55	26.7	5.15
	Ensemble	01/09	39.6	04/09	35.28	39.6	4.32	15/09	35.1	39	↑+3.90	31/10	35.42	35.1	↑+0.32
		16/10	41.25	19/10	37.9	40.65	↑+2.75	30/10	37.94	35.7	↑+2.24	15/12	37.97	32	↑+5.97

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TABLE 8 – continued from previous page

Stock	Architecture	Start Date	Start Price	Short-term End Date	Short-term Pred	Short-term Act	Short-term Gap	Medium-term End Date	Medium-term Pred	Medium-term Act	Medium-term Gap	Long-term End Date	Long-term Pred	Long-term Act	Long-term Gap
		21/10	37.65	24/10	34.78	36.1	↑+1.32	04/11	34.67	35	↑+0.33	20/12	34.76	33.6	↑+1.16
		22/10	38	25/10	35.1	36.1	↑+1.00	05/11	35.14	34.1	↑+1.04	21/12	35.1	33.6	↑+1.50
	Hierarchical	01/09	39.6	04/09	40.02	39.6	0.42	15/09	39.82	39	↓-0.82	31/10	40.03	35.1	↓-4.93
		16/10	41.25	19/10	42.0	40.65	↓-1.35	30/10	42.27	35.7	↓-6.57	15/12	41.2	32	↑+9.20
		21/10	37.65	24/10	37.5	36.1	↑+1.40	04/11	38.82	35	↓-3.82	20/12	38.0	33.6	↓-4.40
		22/10	38	25/10	39.01	36.1	↓-2.91	05/11	38.5	34.1	↓-4.40	21/12	37.03	33.6	↑+3.43
	RoundRobin	01/09	39.6	04/09	37.81	39.6	1.79	15/09	38.25	39	↑+0.75	31/10	36.91	35.1	↑+1.81
		16/10	41.25	19/10	42.81	40.65	↓-2.16	30/10	36.07	35.7	↑+0.37	15/12	36.03	32	↑+4.03
		21/10	37.65	24/10	36.73	36.1	↑+0.63	04/11	36.08	35	↑+1.08	20/12	33.59	33.6	↑+0.01
		22/10	38	25/10	37.22	36.1	↑+1.12	05/11	38.62	34.1	↓-4.52	21/12	36.93	33.6	↑+3.33
	Hierarchical	01/09	104.5	04/09	105.5	102.3	↓-3.20	15/09	107.54	104.5	3.04	31/10	106.01	99.2	↓-6.81
		16/10	122	19/10	124.69	116	↓-8.69	30/10	115.5	104	↑+11.50	15/12	105.04	93.9	↑+11.14
		21/10	110.9	24/10	114.33	114.5	↑+0.17	04/11	113.84	100.2	↓-13.64	20/12	102.01	101.5	↑+0.51
		22/10	113	25/10	114.64	114.5	↑+0.14	05/11	105.49	99.6	↑+5.89	21/12	106.45	101.5	↑+4.95
	Round Robin	01/09	104.5	04/09	105.13	102.3	↓-2.83	15/09	107.47	104.5	2.97	31/10	107.83	99.2	↓-8.63
VHM		16/10	122	19/10	122.73	116	↓-6.73	30/10	105.23	104	↑+1.23	15/12	100.56	93.9	↑+6.66
		21/10	110.9	24/10	111.57	114.5	↑+2.93	04/11	104.05	100.2	↑+3.85	20/12	105.05	101.5	↑+3.55
		22/10	113	25/10	113.68	114.5	↑+0.82	05/11	106.21	99.6	↑+6.61	21/12	105.71	101.5	↑+4.21
	Ensemble	01/09	104.5	04/09	105.18	102.3	↓-2.88	15/09	105.31	104.5	0.81	31/10	95.79	99.2	↑+3.41
		16/10	122	19/10	111.7	116	↑+4.30	30/10	112.0	104	↑+8.00	15/12	102.3	93.9	↑+8.40
		21/10	110.9	24/10	111.66	114.5	↑+2.84	04/11	101.29	100.2	↑+1.09	20/12	99.62	101.5	↑+1.88
		22/10	113	25/10	117.23	114.5	↑+2.73	05/11	95.86	99.6	↑+3.74	21/12	94.42	101.5	↑+7.08
	Hierarchical	01/09	78	04/09	73.12	77.5	↑+4.38	15/09	73.42	79.5	↓-6.08	31/10	78.64	82.6	↑+3.96
		16/10	84.5	19/10	87.88	84.5	3.38	30/10	88.18	83.9	↓-4.28	15/12	81.35	77.7	↑+3.65
		21/10	83	24/10	86.4	85.7	↑+0.70	04/11	86.08	81.8	↓-4.28	20/12	81.23	82.9	↑+1.67
		22/10	84.5	25/10	87.6	85.7	↑+1.90	05/11	83.14	80.2	↑+2.94	21/12	87.42	82.9	↓-4.52
	Round Robin	01/09	78	04/09	71.61	77.5	↑+5.89	15/09	71.86	79.5	↓-7.64	31/10	78.62	82.6	↑+3.98
MWG		16/10	84.5	19/10	87.75	84.5	3.25	30/10	77.72	83.9	↑+6.18	15/12	77.81	77.7	↑+0.11
		21/10	83	24/10	85.63	85.7	↑+0.07	04/11	80.45	81.8	↑+1.35	20/12	82.69	82.9	↑+0.21
		22/10	84.5	25/10	87.2	85.7	↑+1.50	05/11	79.27	80.2	↑+0.93	21/12	78.35	82.9	↑+4.55
	Ensemble	01/09	78	04/09	77.31	77.5	↑+0.19	15/09	75.71	79.5	↓-3.79	31/10	76.21	82.6	↓-6.39
		16/10	84.5	19/10	77.04	84.5	7.46	30/10	82.79	83.9	↑+1.11	15/12	77.09	77.7	↑+0.61
		21/10	83	24/10	79.49	85.7	↓-6.21	04/11	80.18	81.8	↑+1.62	20/12	81.82	82.9	↑+1.08
		22/10	84.5	25/10	87.54	85.7	↑+1.84	05/11	81.39	80.2	↑+1.19	21/12	83.53	82.9	↑+0.63
	Hierarchical	01/09	40.3	04/09	39.13	41.85	↓-2.72	15/09	33.28	40.8	↓-7.52	31/10	33.24	41.9	↓-8.66
		16/10	38	19/10	37.52	37.3	↑+0.22	30/10	38.14	41.3	↑+3.16	15/12	33.63	35.1	↑+1.47
		21/10	38.4	24/10	38.12	38.75	↓-0.63	04/11	35.82	39.6	↓-3.78	20/12	31.82	35.2	↑+3.38
		22/10	38.5	25/10	35.33	38.75	↓-3.42	05/11	37.44	39.3	↓-1.86	21/12	31.21	35.2	↑+3.99
	Round Robin	01/09	40.3	04/09	37.27	41.85	↓-4.58	15/09	36.84	40.8	↓-3.96	31/10	36.88	41.9	↓-5.02
CMG		16/10	38	19/10	39.28	37.3	↓-1.98	30/10	36.58	41.3	↓-4.72	15/12	32.52	35.1	↑+2.58
		21/10	38.4	24/10	36.22	38.75	↓-2.53	04/11	36.72	39.6	↓-2.88	20/12	31.29	35.2	↑+3.91
		22/10	38.5	25/10	35.82	38.75	↓-2.93	05/11	38.83	39.3	↑+0.47	21/12	33.11	35.2	↑+2.09
	Ensemble	01/09	40.3	04/09	42.15	41.85	↑+0.30	15/09	42.21	40.8	↑+1.41	31/10	42.88	41.9	↑+0.98
		16/10	38	19/10	39.85	37.3	↓-2.55	30/10	39.25	41.3	↑+2.05	15/12	35.52	35.1	↑+0.42
		21/10	38.4	24/10	38.66	38.75	↑+0.09	04/11	38.87	39.6	↑+0.73	20/12	31.21	35.2	↑+3.99
		22/10	38.5	25/10	39.62	38.75	↑+0.87	05/11	41.26	39.3	↑+1.96	21/12	39.41	35.2	↓-4.21
	Hierarchical	01/09	36.5	04/09	33.18	36.75	↓-3.57	15/09	33.38	34.85	↑+1.47	31/10	33.47	35.85	↑+2.38
		16/10	34.2	19/10	31.31	33.9	↑+2.59	30/10	31.44	35.85	↓-4.41	15/12	31.44	29.6	↑+1.84
		21/10	31.75	24/10	28.48	33.8	↓-5.32	04/11	29.07	35.6	↓-6.53	20/12	28.95	32.65	↓-3.70
		22/10	32.2	25/10	27.76	33.8	↓-6.04	05/11	25.76	34.95	↓-9.19	21/12	26.64	32.65	↓-6.01
	Round Robin	01/09	36.5	04/09	32.25	36.75	↓-4.50	15/09	34.61	34.85	↑+0.24	31/10	35.21	35.85	↑+0.64
KDH		16/10	34.2	19/10	34.89	33.9	↓-0.99	30/10	34.61	35.85	↑+1.24	15/12	29.52	29.6	↑+0.08
		21/10	31.75	24/10	33.45	33.8	↑+0.35	04/11	32.14	35.6	↑+3.46	20/12	29.56	32.65	↓-3.09
		22/10	32.2	25/10	33.24	33.8	↑+0.56	05/11	33.45	34.95	↑+1.50	21/12	30.54	32.65	↓-2.11
	Ensemble	01/09	36.5	04/09	39.25	36.75	↑+2.50	15/09	35.71	34.85	↑+0.86	31/10	34.97	35.85	↑+0.88
		16/10	34.2	19/10	32.39	33.9	↑+1.51	30/10	37.51	35.85	↑+1.66	15/12	28.92	29.6	↑+0.68
		21/10	31.75	24/10	35.55	33.8	↑+1.75	04/11	37.04	35.6	↑+1.44	20/12	34.16	32.65	↑+1.51
		22/10	32.2	25/10	32.59	33.8	↑+1.21	05/11	33.4	34.95	↑+1.55	21/12	27.99	32.65	↓-4.66
	Hierarchical	01/09	98.2	04/09	96.01	99.3	↓-3.29	15/09	94.5	99.5	↓-5.00	31/10	95.0	96	↑+1.00
		16/10	95	19/10	93.52	93.2	↑+0.32	30/10	92.82	93.6	↑+0.78	15/12	92.01	93	↑+0.99
		21/10	89.2	24/10	87.01	92.4	↓-5.39	04/11	86.0	94.6	↓-8.60	20/12	76.0	70.2	↑+5.80
		22/10	91	25/10	90.3	92.4	↓-2.10	05/11	89.02	95	↓-5.98	21/12	77.8	70.2	↑+7.60
	Round Robin	01/09	98.2	04/09	95.27	99.3	↓-4.03	15/09	99.77	99.5	↑+0.27	31/10	95.82	96	↑+0.18
DGC		16/10	95	19/10	92.82	93.2	↑+0.38	30/10	95.12	93.6	↓-1.52	15/12	94.36	93	↑+1.36

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TABLE 8 – continued from previous page

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
		21/10	89.2	24/10	98.37	92.4	↑+5.97	04/11	91.06	94.6	↑+3.54	20/12	78.52	70.2	↑+8.32
		22/10	91	25/10	97.22	92.4	↑+4.82	05/11	96.4	95	↑+1.40	21/12	77.92	70.2	↑+7.72
	Ensemble	01/09	98.2	04/09	94.36	99.3	↓-4.94	15/09	100.97	99.5	↑+1.47	31/10	95.31	96	↑+0.69
		16/10	95	19/10	93.35	93.2	↑+0.15	30/10	93.84	93.6	↑+0.24	15/12	93.6	93	↑+0.60
		21/10	89.2	24/10	94.82	92.4	↑+2.42	04/11	95.11	94.6	↑+0.51	20/12	77.25	70.2	↑+7.05
		22/10	91	25/10	93.78	92.4	↑+1.38	05/11	95.13	95	↑+0.13	21/12	72.95	70.2	↑+2.75
MBB	**Hierarchical**	01/09	27.75	04/09	26.77	28.25	↓-1.48	15/09	29.54	26.85	↓-2.69	31/10	22.21	23.6	↑+1.39
		16/10	27.2	19/10	26.81	27.1	↑+0.29	30/10	21.58	23.95	↑+2.37	15/12	26.95	23.75	↑+3.20
		21/10	25.3	24/10	24.78	24.4	↑+0.38	04/11	25.18	24	↑+1.18	20/12	24.96	24.7	↑+0.26
		22/10	25.45	25/10	23.54	24.4	↑+0.86	05/11	24.58	23.9	↑+0.68	21/12	26.44	24.7	↓-1.74
	Round Robin	01/09	27.75	04/09	28.2	28.25	↑+0.05	15/09	25.51	26.85	↑+1.34	31/10	25.93	23.6	↑+2.33
		16/10	27.2	19/10	27.7	27.1	↓-0.60	30/10	26.78	23.95	↑+2.83	15/12	22.67	23.75	↑+1.08
		21/10	25.3	24/10	24.7	24.4	↑+0.30	04/11	24.41	24	↑+0.41	20/12	24.07	24.7	↑+0.63
		22/10	25.45	25/10	23.13	24.4	↑+1.27	05/11	22.99	23.9	↑+0.91	21/12	22.94	24.7	↑+1.76
	Ensemble	01/09	27.75	04/09	29.65	28.25	↑+1.40	15/09	25.61	26.85	↑+1.24	31/10	24.48	23.6	↑+0.88
		16/10	27.2	19/10	26.97	27.1	↑+0.13	30/10	23.1	23.95	↑+0.85	15/12	23.02	23.75	↑+0.73
		21/10	25.3	24/10	24.97	24.4	↑+0.57	04/11	24.35	24	↑+0.35	20/12	24.73	24.7	↑+0.03
		22/10	25.45	25/10	24.94	24.4	↑+0.54	05/11	23.55	23.9	↑+0.35	21/12	24.34	24.7	↑+0.36

TABLE 9: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (LLM: Google Gemini 2.0 Flash, Transformer Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Hierarchical	31/08	22.8	03/09	23	24	↑+1	14/09	23.31	24.05	↑+0.74	30/10	23.86	21.2	↓-2.66
		16/10	22.45	19/10	21.69	22.6	↓-0.91	30/10	20.94	21.2	↓-0.26	15/12	23.61	16.3	↓-7.31
		21/10	20.0	24/10	19.89	20.9	↓-1.01	04/11	18.13	20.15	↓-2.02	20/12	20.92	17.8	↓-3.12
		22/10	20.5	25/10	17.66	20.9	↓-3.24	05/11	20.94	20.1	↓-0.84	21/12	21.35	17.8	↓+3.55
	Round Robin	31/08	22.8	03/09	23.07	24	↑+0.93	14/09	19.38	24.05	↓-4.67	30/10	24.01	21.2	↓-2.81
		16/10	22.45	19/10	22.77	22.6	↑+0.17	30/10	23.10	21.2	↓-1.90	15/12	23.92	16.3	↓-7.62
		21/10	20.0	24/10	20.17	20.9	↑+0.73	04/11	18.34	20.15	↓+1.81	20/12	20.74	17.8	↓-2.94
		22/10	20.5	25/10	18.35	20.9	↓-2.55	05/11	20.89	20.1	↓-0.79	21/12	21.40	17.8	↓-3.6
	Ensemble	31/08	22.8	03/09	23.97	24	↑+1.03	14/09	23.18	24.05	↑+0.87	30/10	23.58	21.2	↓-2.38
		16/10	22.45	19/10	22.65	22.6	↑+0.05	30/10	22.91	21.2	↓-1.71	15/12	23.33	16.3	↓-7.03
		21/10	20.0	24/10	20.15	20.9	↑+0.75	04/11	20.36	20.15	↑+0.21	20/12	20.66	17.8	↓-2.86
		22/10	20.5	25/10	20.65	20.9	↑+0.25	05/11	20.79	20.1	↓-0.69	21/12	21.19	17.8	↓-3.39
FPT	Hierarchical	01/09	101.6	04/09	101.09	105	↓-3.91	15/09	101.91	101.9	↑+0.09	31/10	103.71	103.9	↑+0.19
		16/10	89.8	19/10	89.47	88.1	↑+0.91	30/10	90.01	102.7	↑+12.69	15/12	91.69	93.8	↑+2.11
		21/10	93	24/10	92.61	97.7	↓-5.09	04/11	93.36	103.3	↑+9.94	20/12	94.59	93.9	↑+0.69
		22/10	97	25/10	96.6	97.7	↓-1.1	05/11	97.22	100.9	↑-3.68	21/12	98.65	93.9	↓-4.75
	Round Robin	01/09	101.6	04/09	101.16	105	↓-3.84	15/09	102.01	101.9	↑+0.01	31/10	104.16	103.9	↑+0.26
		16/10	89.8	19/10	89.35	88.1	↑+1.25	30/10	89.19	102.7	↓-13.51	15/12	90.96	93.8	↑+2.84
		21/10	93	24/10	92.59	97.7	↓-5.11	04/11	93.29	103.3	↑+10.01	20/12	94.52	93.9	↑+0.62
		22/10	97	25/10	93.48	97.7	↓-4.22	05/11	97.28	100.9	↑+3.62	21/12	98.65	93.9	↓-4.75
	Ensemble	01/09	101.6	04/09	101.08	105	↑+3.92	15/09	101.63	101.9	↑+0.27	31/10	103.14	103.9	↑+0.76
		16/10	89.8	19/10	89.37	88.1	↑+1.29	30/10	101.88	102.7	↑+0.82	15/12	93.09	93.8	↑+0.71
		21/10	93	24/10	92.53	97.7	↓-5.17	04/11	103.09	103.3	↑+0.21	20/12	94.23	93.9	↑+0.33
		22/10	97	25/10	96.48	97.7	↓-1.22	05/11	96.52	100.9	↓-4.38	21/11	98.15	93.9	↓-4.25
VCB	Hierarchical	30/08	68.6	02/09	68.37	68.6	0.23	13/09	68.70	65.8	↓+2.9	29/10	69.41	60.7	↓-8.71
		16/10	62.9	19/10	63.19	61.9	↓-1.29	30/10	63.43	60.6	↓-2.83	15/12	64.05	56.8	↓-7.25
		21/10	59.3	24/10	59.56	59.5	↑+0.04	04/11	59.87	60.1	↑+0.23	20/12	60.44	57.5	↓-2.94
		22/10	59.6	25/10	61.88	59.5	↓-2.38	05/11	60.18	60.8	↑+0.62	21/12	58.79	57.5	↑+1.29
	Round Robin	30/08	68.6	02/09	68.44	68.6	0.16	13/09	68.89	65.8	↓-3.10	29/10	69.99	60.7	↓-9.29
		16/10	62.9	19/10	63.19	61.9	↑+1.29	30/10	63.60	60.6	↓-3	15/12	54.31	56.8	↑-2.49
		21/10	59.3	24/10	59.53	59.5	↓-0.03	04/11	59.79	60.1	↓-0.31	20/12	60.37	57.5	↓-2.87
		22/10	59.6	25/10	59.82	59.5	↓-0.32	05/11	60.08	60.8	↑+0.72	21/12	60.55	57.5	↓-3.05
	Ensemble	30/08	68.6	02/09	68.36	68.6	0.24	13/09	68.66	65.8	↓-2.86	29/10	69.33	60.7	↓-8.63
		16/10	62.9	19/10	63.14	61.9	↓-1.24	30/10	63.41	60.6	↓-2.51	15/12	55.87	56.8	↑+0.93
		21/10	59.3	24/10	59.52	59.5	↑+0.02	04/11	59.86	60.1	↑+0.24	20/12	60.25	57.5	↓-2.75
		22/10	59.6	25/10	59.83	59.5	↓-0.33	05/11	60.11	60.8	↑+0.69	21/12	60.69	57.5	↓-3.19
VCS	Hierarchical	30/08	48.8	02/09	47.26	48.8	1.54	13/09	47.72	50	↓-2.28	29/10	48.32	47.6	↑+0.72
		16/10	47.2	19/10	45.67	46.8	↑+1.13	30/10	46.07	47.4	↓-1.33	15/12	46.59	46.5	↑+0.09
		21/10	45.5	24/10	44.06	46.2	↓-2.14	04/11	44.37	47.1	↓-2.73	20/12	44.97	43	↑+1.97
		22/10	45.5	25/10	44.05	46.2	↓-0.15	05/11	42.31	46.9	↓-4.59	21/12	45.11	43	↑+2.11
	Round Robin	30/08	48.8	02/09	47.23	48.8	1.57	13/09	47.58	50	↓-2.42	29/10	48.05	47.6	↑+0.45
		16/10	47.2	19/10	45.71	46.8	↑-1.09	30/10	47.99	47.4	↑+0.59	15/12	47.39	46.5	↓-0.89
		21/10	45.5	24/10	43.98	46.2	↓-2.22	04/11	48.21	47.1	↑+1.11	20/12	45.88	43	↓-2.88
		22/10	45.5	25/10	44.01	46.2	↓-2.19	05/11	46.33	46.9	↑+0.57	21/12	45.77	43	↓-2.77
	Ensemble	30/08	48.8	02/09	47.19	48.8	1.61	13/09	52.54	50	↑+2.54	29/10	48.86	47.6	↓-1.26
		16/10	47.2	19/10	45.64	46.8	↑+1.16	30/10	47.16	47.4	↑+0.24	15/12	46.57	46.5	↑+0.07
		21/10	45.5	24/10	43.99	46.2	↓-2.21	04/11	44.28	47.1	↓-2.82	20/12	44.85	43	↑+1.85
		22/10	45.5	25/10	43.97	46.2	↓-2.23	05/11	44.28	46.9	↓-2.62	21/12	44.9	43	↑+1.9
HPG	Hierarchical	01/09	27.5	04/09	27.67	29.85	↑+2.18	15/09	27.83	30.35	↑+2.52	31/10	28.17	26.7	↓-1.47
		16/10	28.25	19/10	28.44	28	↓-0.44	30/10	28.68	26.9	↓-1.78	15/12	29.12	26.25	↓-2.87
		21/10	26.8	24/10	26.94	26.4	↓-0.54	04/11	27.1	26.75	↓-0.35	20/12	26.42	26.7	↑+0.28
		22/10	26.7	25/10	26.55	26.4	↑+0.15	05/11	26.97	26.3	↓-0.67	21/12	27.37	26.7	0.67
	Round Robin	01/09	27.5	04/09	28.84	29.85	↑+1.01	15/09	28.84	30.35	↑+1.51	31/10	26.41	26.7	↑+0.49
		16/10	28.25	19/10	29.63	28	↓-1.63	30/10	29.01	26.9	↓-2.11	15/12	29.30	26.25	↓+3.05
		21/10	26.8	24/10	26.29	26.4	↑+0.11	04/11	26.06	26.75	↑+0.69	20/12	27.41	26.7	↓-0.71
		22/10	26.7	25/10	26.79	26.4	↓-0.39	05/11	26.89	26.3	↓-0.59	21/12	27.23	26.7	0.53
	Ensemble	01/09	27.5	04/09	27.65	29.85	↑+2.2	15/09	27.8	30.35	↑+2.55	31/10	28.11	26.7	↓-1.41
		16/10	28.25	19/10	28.4	28	↓-0.4	30/10	28.58	26.9	↓-1.68	15/12	27.98	26.25	↑+1.73
		21/10	26.8	24/10	26.93	26.4	↓-0.53	04/11	27.06	26.75	↓-0.31	20/12	26.5	26.7	↑+0.2
		22/10	26.7	25/10	26.83	26.4	↓-0.43	05/11	26.5	26.3	↑+0.2	21/12	27.35	26.7	0.65
	Ensemble	01/09	39.6	04/09	38.78	39.6	0.82	15/09	39.03	39	↑+0.03	31/10	39.43	35.1	↑+4.33
		16/10	41.25	19/10	41.5	40.65	↓-0.85	30/10	41.77	35.7	↓-6.07	15/12	42.14	32	↓-10.14

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TABLE 9 – continued from previous page

Stock	Architecture	Start Date	Start Price	Short-term End Date	Short-term Pred	Short-term Act	Short-term Gap	Medium-term End Date	Medium-term Pred	Medium-term Act	Medium-term Gap	Long-term End Date	Long-term Pred	Long-term Act	Long-term Gap
		21/10	37.65	24/10	37.88	36.1	↓-1.78	04/11	38.08	35	↓-3.08	20/12	38.43	33.6	↓-4.83
		22/10	38	25/10	35.2	36.1	↑+0.9	05/11	34.41	34.1	↑+0.31	21/12	33.02	33.6	↑+0.58
	Hierarchical	01/09	39.6	04/09	38.84	39.6	0.76	15/09	39.09	39	↑+0.09	31/10	39.68	35.1	↓-4.58
		16/10	41.25	19/10	41.52	40.65	↓-0.87	30/10	41.85	35.7	↓-6.15	15/12	38.43	32	↑+6.43
		21/10	37.65	24/10	37.88	36.1	↓-1.78	04/11	35.23	35	↑+0.23	20/12	38.52	33.6	↓-4.92
		22/10	38	25/10	38.25	36.1	↓-2.15	05/11	36.59	34.1	↑+2.49	21/12	34.09	33.6	↑+0.49
	RoundRobin	01/09	39.6	04/09	38.85	39.6	0.75	15/09	39.23	39	↑+0.23	31/10	39.88	35.1	↓-4.78
		16/10	41.25	19/10	41.54	40.65	↓-0.89	30/10	41.97	35.7	↓-6.27	15/12	32.71	32	↑+0.71
		21/10	37.65	24/10	36.27	36.1	↑+0.17	04/11	38.12	35	↓-3.12	20/12	38.6	33.6	↓-5
		22/10	38	25/10	38.27	36.1	↓-2.17	05/11	38.49	34.1	↓-4.39	21/12	38.87	33.6	↓-5.27
	Hierarchical	01/09	104.5	04/09	105.28	102.3	↓-2.98	15/09	106.31	104.5	1.81	31/10	97.97	99.2	↑+1.23
		16/10	122	19/10	123.23	116	↓-7.23	30/10	104.43	104	↑+0.43	15/12	107	93.9	↑+13.1
		21/10	110.9	24/10	111.77	114.5	↑+2.73	04/11	112.53	100.2	↓-12.33	20/12	114.76	101.5	↓-13.26
		22/10	113	25/10	113.8	114.5	↑+0.7	05/11	104.52	99.6	↑+4.92	21/12	106.61	101.5	↑+5.11
	Round Robin	01/09	104.5	04/09	105.54	102.3	↓-3.24	15/09	106.72	104.5	2.22	31/10	108.85	99.2	↓-9.65
VHM		16/10	122	19/10	123.89	116	↓-7.89	30/10	105.79	104	↑+1.79	15/12	100.95	93.9	↑+7.05
		21/10	110.9	24/10	111.92	114.5	↑+2.58	04/11	103	100.2	↑+2.8	20/12	105.22	101.5	↑+3.72
		22/10	113	25/10	113.88	114.5	↑+0.62	05/11	104.64	99.6	↑+5.04	21/12	106.08	101.5	↑+4.58
	Ensemble	01/09	104.5	04/09	101.14	102.3	↑+1.16	15/09	106.03	104.5	1.53	31/10	97.62	99.2	↑+1.58
		16/10	122	19/10	122.91	116	↓-6.91	30/10	103.8	104	↑+0.2	15/12	95.53	93.9	↑+1.63
		21/10	110.9	24/10	111.53	114.5	↑+2.97	04/11	102.32	100.2	↑+2.12	20/12	104.03	101.5	↑+2.53
		22/10	113	25/10	113.67	114.5	↑+0.83	05/11	99.5	99.6	↑+0.1	21/12	105.83	101.5	↑+4.33
	Hierarchical	01/09	78	04/09	78.54	77.5	↓-1.04	15/09	79.14	79.5	↑+0.36	31/10	80.53	82.6	↑+2.07
		16/10	84.5	19/10	85.09	84.5	0.59	30/10	85.93	83.9	↓-2.03	15/12	87.25	77.7	↓-9.55
		21/10	83	24/10	83.53	85.7	↑+2.17	04/11	84.26	81.8	↓-2.46	20/12	85.63	82.9	↓-2.73
		22/10	84.5	25/10	85.15	85.7	↑+0.55	05/11	85.82	80.2	↓-5.62	21/12	87.32	82.9	↓-4.42
	Round Robin	01/09	78	04/09	78.8	77.5	↓-1.3	15/09	79.76	79.5	↑+0.26	31/10	81.67	82.6	↑+0.93
MWG		16/10	84.5	19/10	85.31	84.5	0.81	30/10	86.44	83.9	↓-2.54	15/12	85.75	77.7	↓-8.05
		21/10	83	24/10	83.63	85.7	↑+2.07s	04/11	84.31	81.8	↓-2.51	20/12	81.22	82.9	↑+1.68
		22/10	84.5	25/10	85.2	85.7	↑+0.5	05/11	85.85	80.2	↓-5.65	21/12	87.7	82.9	↑+4.8
	Ensemble	01/09	78	04/09	78.49	77.5	↓-0.99	15/09	79	79.5	↑+0.5	31/10	80.38	82.6	↑+2.22
		16/10	84.5	19/10	85.06	84.5	0.56	30/10	85.64	83.9	↓-1.74	15/12	76.79	77.7	↑+0.91
		21/10	83	24/10	83.48	85.7	↑+2.22	04/11	83.92	81.8	↑+2.12	20/12	85.31	82.9	↓-2.41
		22/10	84.5	25/10	84.95	85.7	↑+0.75	05/11	85.52	80.2	↓-5.32	21/12	86.68	82.9	↓-3.78
	Hierarchical	01/09	40.3	04/09	36.08	41.85	↓-5.77	15/09	34.13	40.8	↓-6.67	31/10	49.31	41.9	↑+7.41
		16/10	38	19/10	34	37.3	↓-3.3	30/10	32.06	41.3	↓-9.24	15/12	37.64	35.1	↑+2.54
		21/10	38.4	24/10	34.38	38.75	↓-4.37	04/11	32.52	39.6	↓-7.08	20/12	37.93	35.2	↑+2.73
		22/10	38.5	25/10	39.12	38.75	↑+0.37	05/11	32.22	39.3	↓-7.08	21/12	38	35.2	↑+2.8
	Round Robin	01/09	40.3	04/09	35.39	41.85	↓-6.46	15/09	41.18	40.8	↑+0.38	31/10	39.31	41.9	↓-2.59
CMG		16/10	38	19/10	33.39	37.3	↑+3.91	30/10	39.46	41.3	↑+1.84	15/12	27.64	35.1	↑+7.46
		21/10	38.4	24/10	33.52	38.75	↓-5.23	04/11	38.88	39.6	↑+0.72	20/12	27.93	35.2	↓-7.27
		22/10	38.5	25/10	33.57	38.75	↓-5.18	05/11	38.22	39.3	↓-1.08	21/12	36.74	35.2	↑+1.54
	Ensemble	01/09	40.3	04/09	41.94	41.85	↑+0.09	15/09	33.51	40.8	↓-7.29	31/10	42.88	41.9	↑+0.98
		16/10	38	19/10	33.97	37.3	↓-3.33	30/10	41.94	41.3	↑+0.64	15/12	36.78	35.1	↑+1.68
		21/10	38.4	24/10	34.36	38.75	↓-4.39	04/11	39.42	39.6	↑+0.18	20/12	39.52	35.2	↓-4.32
		22/10	38.5	25/10	34.47	38.75	↓-4.28	05/11	39.59	39.3	↑+0.29	21/12	39.13	35.2	↓-3.93
	Hierarchical	01/09	36.5	04/09	35.84	36.75	↓-0.91	15/09	33.52	34.85	↑+1.33	31/10	39.2	35.85	↓-3.35
		16/10	34.2	19/10	33.59	33.9	↑+0.31	30/10	31.46	35.85	↓-4.39	15/12	27.36	29.6	↑+2.24
		21/10	31.75	24/10	31.19	33.8	↓-2.61	04/11	29.22	35.6	↓-6.38	20/12	35.21	32.65	↑+2.56
		22/10	32.2	25/10	31.16	33.8	↓-2.64	05/11	34.52	34.95	↑+0.43	21/12	35.56	32.65	↑+2.91
	Round Robin	01/09	36.5	04/09	35.53	36.75	↓-1.22	15/09	32.2	34.85	↑+2.65	31/10	35.88	35.85	↑+0.03
KDH		16/10	34.2	19/10	33.12	33.9	↑+0.78	30/10	30.45	35.85	↓-5.4	15/12	29.12	29.6	↑+0.48
		21/10	31.75	24/10	33.67	33.8	↑+0.13	04/11	37.01	35.6	↑+1.41	20/12	25.4	32.65	↓-7.25
		22/10	32.2	25/10	31	33.8	↓-2.8	05/11	36.98	34.95	↑+2.03	21/12	35.76	32.65	↑-3.11
	Ensemble	01/09	36.5	04/09	35.62	36.75	↑+1.13	15/09	32.59	34.85	↑+2.26	31/10	35.32	35.85	↑+0.53
		16/10	34.2	19/10	33.48	33.9	↑+0.42	30/10	35.96	35.85	↑+0.11	15/12	30.02	29.6	↑+0.42
		21/10	31.75	24/10	31.12	33.8	↓-2.68	04/11	36.9	35.6	↑+1.3	20/12	34.05	32.65	↑+1.4
		22/10	32.2	25/10	31.58	33.8	↓-2.22	05/11	34.41	34.95	↑+0.54	21/12	32.85	32.65	↑+0.2
	Hierarchical	01/09	98.2	04/09	99.9	99.3	↑+0.6	15/09	94.33	99.5	↓-5.17	31/10	95.27	96	↑+0.73
		16/10	95	19/10	88.47	93.2	↑+4.73	30/10	99.7	93.6	↓-6.1	15/12	92.82	93	↑+0.18
		21/10	89.2	24/10	82.43	92.4	↓-9.97	04/11	92.19	94.6	↑+2.41	20/12	68.37	70.2	↑+1.83
		22/10	91	25/10	93.48	92.4	↑+1.08	05/11	91.18	95	↑+3.82	21/12	69.75	70.2	↑+0.45
	Round Robin	01/09	98.2	04/09	98.86	99.3	↑+0.44	15/09	95.27	99.5	↓-4.23	31/10	100.02	96	↓-4.02
DGC		16/10	95	19/10	86.2	93.2	↑+7	30/10	92.82	93.6	↑+0.78	15/12	96.33	93	↓-3.33

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TABLE 9 – continued from previous page

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
		21/10	89.2	24/10	91.14	92.4	↑+1.26	04/11	98.37	94.6	↑+3.77	20/12	68.37	70.2	↑+1.83
		22/10	91	25/10	83.09	92.4	↓-9.31	05/11	99.75	95	↑+4.75	21/12	70.35	70.2	↓-0.15
	Ensemble	01/09	98.2	04/09	99.03	99.3	↑+0.27	15/09	94.86	99.5	↓-4.64	31/10	95.27	96	↑+0.73
		16/10	95	19/10	88.79	93.2	↑+4.41	30/10	91.06	93.6	↑+2.54	15/12	92.82	93	↑+0.18
		21/10	89.2	24/10	94.61	92.4	↑+2.21	04/11	91.57	94.6	↑+3.03	20/12	69.98	70.2	↑+0.22
		22/10	91	25/10	89.62	92.4	↓-2.78	05/11	94.59	95	↑+0.41	21/12	71.6	70.2	↑+1.4
MBB	**Hierarchical**	01/09	27.75	04/09	28.87	28.25	↑+0.62	15/09	28.23	26.85	↓-1.38	31/10	24.22	23.6	↑+0.62
		16/10	27.2	19/10	27.41	27.1	↓-0.31	30/10	23.43	23.95	↑+0.52	15/12	22.19	23.75	↑+1.56
		21/10	25.3	24/10	24.98	24.4	↑+0.58	04/11	24.19	24	↑+0.19	20/12	27.22	24.7	↓-2.52
		22/10	25.45	25/10	23.41	24.4	↑+0.99	05/11	21.42	23.9	↑+2.48	21/12	26.14	24.7	↓-1.44
	Round Robin	01/09	27.75	04/09	29.43	28.25	↑+1.18	15/09	28.42	26.85	↓-1.57	31/10	20.43	23.6	↑+3.17
		16/10	27.2	19/10	27.35	27.1	↓-0.25	30/10	22.94	23.95	↑+1.01	15/12	24.14	23.75	↑+0.39
		21/10	25.3	24/10	25.21	24.4	↑+0.81	04/11	26.13	24	↓-2.13	20/12	24.43	24.7	↑+0.27
		22/10	25.45	25/10	26.22	24.4	↓-1.82	05/11	22.43	23.9	↑+1.47	21/12	24.88	24.7	↑+0.18
	Ensemble	01/09	27.75	04/09	28.35	28.25	↑+0.10	15/09	27.11	26.85	↑+0.26	31/10	24.22	23.6	↑+0.62
		16/10	27.2	19/10	26.15	27.1	↑+0.95	30/10	23.14	23.95	↑+0.81	15/12	23.42	23.75	↑+0.33
		21/10	25.3	24/10	24.3	24.4	↑+0.10	04/11	25.25	24	↑+1.25	20/12	26.31	24.7	↓-1.61
		22/10	25.45	25/10	24.47	24.4	↑+0.07	05/11	22.39	23.9	↑+1.51	21/12	24.98	24.7	↑+0.28

TABLE 10: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (LLM: GPT-4o, LSTM Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Hierarchical	31/08	22.8	03/09	23.01	24	↑+0.99	14/09	23.3	24.05	↑+0.75	30/10	23.78	21.2	↓-2.58
		16/10	22.45	19/10	22.7	22.6	↑+0.10	30/10	22.96	21.2	↓-1.76	15/12	23.39	16.3	↓-7.09
		21/10	20.0	24/10	20.2	20.9	↑+0.70	04/11	20.34	20.15	↑+0.19	20/12	19.91	17.8	↑+2.11
		22/10	20.5	25/10	20.67	20.9	↑+0.23	05/11	20.85	20.1	↓-0.75	21/12	21.32	17.8	↓-3.52
	Round Robin	31/08	22.8	03/09	23.09	24	↑+0.91	14/09	23.36	24.05	↑+0.69	30/10	23.99	21.2	↓-2.79
		16/10	22.45	19/10	22.75	22.6	↑+0.15	30/10	23.07	21.2	↓-1.87	15/12	23.87	16.3	↓-7.57
		21/10	20.0	24/10	20.16	20.9	↑+0.74	04/11	20.33	20.15	↑+0.18	20/12	20.69	17.8	↓-2.89
		22/10	20.5	25/10	20.68	20.9	↑+0.22	05/11	20.9	20.1	↓-0.80	21/12	21.33	17.8	↓-3.53
	Ensemble	31/08	22.8	03/09	22.98	24	↑+1.02	14/09	23.15	24.05	↑+0.90	30/10	23.54	21.2	↓-2.34
		16/10	22.45	19/10	22.65	22.6	↑+0.05	30/10	22.84	21.2	↓-1.64	15/12	23.32	16.3	↓-7.02
		21/10	20.0	24/10	20.16	20.9	↑+0.74	04/11	20.32	20.15	↑+0.17	20/12	20.63	17.8	↓-2.83
		22/10	20.5	25/10	20.65	20.9	↑+0.25	05/11	20.44	20.1	↑+0.34	21/12	21.07	17.8	↓-3.27
FPT	Hierarchical	01/09	101.6	04/09	104.66	105	↑+0.34	15/09	101.96	101.9	↑+0.06	31/10	103.27	103.9	↑+0.63
		16/10	89.8	19/10	89.4	88.1	↑+1.30	30/10	89.93	102.7	↑+12.77	15/12	91.24	93.8	↑+2.56
		21/10	93	24/10	92.33	97.7	↓-5.37	04/11	93.26	103.3	↑+10.04	20/12	94.66	93.9	↑+0.76
		22/10	97	25/10	96.54	97.7	↓-1.16	05/11	97.43	100.9	↑+3.47	21/12	98.91	93.9	↓-5.01
	Round Robin	01/09	101.6	04/09	101.29	105	↓-3.71	15/09	102.08	101.9	↑+0.18	31/10	103.92	103.9	↑+0.02
		16/10	89.8	19/10	89.32	88.1	↑+1.22	30/10	89.95	102.7	↑+12.75	15/12	90.85	93.8	↑+2.95
		21/10	93	24/10	92.67	97.7	↓-5.03	04/11	93.04	103.3	↑+10.26	20/12	94.19	93.9	↑+0.29
		22/10	97	25/10	96.67	97.7	↓-1.03	05/11	97.51	100.9	↑+3.39	21/12	96.55	93.9	↑+2.65
	Ensemble	01/09	101.6	04/09	101.71	105	↑+3.29	15/09	101.6	101.9	↓-0.30	31/10	103.01	103.9	↑+0.89
		16/10	89.8	19/10	89.3	88.1	↑+1.20	30/10	89.98	102.7	↑+12.72	15/12	91.14	93.8	↑+2.66
		21/10	93	24/10	92.49	97.7	↓-5.21	04/11	93.1	103.3	↑+10.20	20/12	94.25	93.9	↑+0.35
		22/10	97	25/10	96.48	97.7	↓-1.22	05/11	97.25	100.9	↑+3.65	21/11	98.46	93.9	↓-4.56
VCB	Hierarchical	30/08	68.6	02/09	69.12	68.6	0.52	13/09	68.64	65.8	↓-2.84	29/10	69.15	60.7	↓-8.45
		16/10	62.9	19/10	63.21	61.9	↓-1.31	30/10	61.92	60.6	↑+1.32	15/12	64.33	56.8	↓-7.53
		21/10	59.3	24/10	59.59	59.5	↑+0.09	04/11	61.77	60.1	↑+1.67	20/12	55.62	57.5	↑+1.88
		22/10	59.6	25/10	59.88	59.5	↓-0.38	05/11	60.2	60.8	↑+0.60	21/12	60.66	57.5	↓-3.16
	Round Robin	30/08	68.6	02/09	68.5	68.6	0.10	13/09	68.83	65.8	↓-3.03	29/10	68.44	60.7	↑+7.74
		16/10	62.9	19/10	62.51	61.9	↑+0.61	30/10	64.03	60.6	↓-3.43	15/12	64.57	56.8	↓-7.77
		21/10	59.3	24/10	59.45	59.5	↑+0.05	04/11	59.68	60.1	↑+0.42	20/12	60.29	57.5	↓-2.79
		22/10	59.6	25/10	59.79	59.5	↓-0.29	05/11	60.17	60.8	↑+0.63	21/12	59.55	57.5	↑+2.05
	Ensemble	30/08	68.6	02/09	68.38	68.6	0.22	13/09	68.67	65.8	↓-2.87	29/10	69.38	60.7	↓-8.68
		16/10	62.9	19/10	63.12	61.9	↓-1.22	30/10	63.35	60.6	↓-2.75	15/12	59.44	56.8	↑+2.64
		21/10	59.3	24/10	59.53	59.5	↑+0.03	04/11	59.76	60.1	↑+0.34	20/12	60.43	57.5	↓-2.93
		22/10	59.6	25/10	59.86	59.5	↓-0.36	05/11	63.78	60.8	↑+2.98	21/12	60.74	57.5	↓-3.24
VCS	Hierarchical	30/08	48.8	02/09	47.25	48.8	1.55	13/09	47.65	50	↓-2.35	29/10	48.33	47.6	↑+0.73
		16/10	47.2	19/10	45.68	46.8	↑+1.12	30/10	46.05	47.4	↓-1.35	15/12	46.9	46.5	↑+0.40
		21/10	45.5	24/10	44.01	46.2	↓-2.19	04/11	44.43	47.1	↓-2.67	20/12	44.98	43	↑+1.98
		22/10	45.5	25/10	44.03	46.2	↓-2.17	05/11	44.46	46.9	↓-2.44	21/12	45.21	43	↑+2.21
	Round Robin	30/08	48.8	02/09	47.18	48.8	1.62	13/09	47.42	50	↓-2.58	29/10	48.32	47.6	↑+0.72
		16/10	47.2	19/10	45.69	46.8	↑+1.11	30/10	46.1	47.4	↓-1.30	15/12	46.56	46.5	↑+0.06
		21/10	45.5	24/10	43.98	46.2	↓-2.22	04/11	47.22	47.1	↑+0.12	20/12	44.71	43	↑+1.71
		22/10	45.5	25/10	43.98	46.2	↓-2.22	05/11	44.28	46.9	↓-2.62	21/12	44.78	43	↑+1.78
	Ensemble	30/08	48.8	02/09	47.19	48.8	1.61	13/09	47.47	50	↓-2.53	29/10	48.18	47.6	↑+0.58
		16/10	47.2	19/10	45.63	46.8	↑+1.17	30/10	46.03	47.4	↓-1.37	15/12	46.79	46.5	↑+0.29
		21/10	45.5	24/10	43.97	46.2	↓-2.23	04/11	44.25	47.1	↓-2.85	20/12	45.02	43	↑+2.02
		22/10	45.5	25/10	45.95	46.2	↑+0.25	05/11	47.55	46.9	↑+0.65	21/12	44.89	43	↑+1.89
HPG	Hierarchical	01/09	27.5	04/09	24.28	29.85	↓-5.57	15/09	28.93	30.35	↑+1.42	31/10	26.25	26.7	↑+0.45
		16/10	28.25	19/10	28.21	28	↑+0.21	30/10	23.15	26.9	↑+3.75	15/12	28.83	26.25	↓-2.58
		21/10	26.8	24/10	26.18	26.4	↑+0.22	04/11	29.95	26.75	↓-3.20	20/12	26.58	26.7	↑+0.12
		22/10	26.7	25/10	26.83	26.4	↓-0.43	05/11	24.18	26.3	↑+2.12	21/12	23.82	26.7	2.88
	Round Robin	01/09	27.5	04/09	27.89	29.85	↑+1.96	15/09	29.37	30.35	↑+0.98	31/10	27.82	26.7	↓-1.12
		16/10	28.25	19/10	29.15	28	↓-1.15	30/10	29.15	26.9	↓-2.25	15/12	29.22	26.25	↓-2.97
		21/10	26.8	24/10	27.12	26.4	↓-0.72	04/11	27.66	26.75	↓-0.91	20/12	27.12	26.7	↓-0.42
		22/10	26.7	25/10	27.55	26.4	↓-1.15	05/11	27.01	26.3	↓-0.71	21/12	27.55	26.7	0.85
	Ensemble	01/09	27.5	04/09	25	29.85	↓-4.85	15/09	30.12	30.35	↑+0.23	31/10	25.07	26.7	↑+1.63
		16/10	28.25	19/10	25.73	28	↑+2.27	30/10	25.67	26.9	↑+1.23	15/12	25.8	26.25	↑+0.45
		21/10	26.8	24/10	26.98	26.4	↓-0.58	04/11	26.87	26.75	↓-0.12	20/12	26.26	26.7	↑+0.44
		22/10	26.7	25/10	26.77	26.4	↓-0.37	05/11	26.22	26.3	↑+0.08	21/12	26.55	26.7	0.15
	Ensemble	01/09	39.6	04/09	37.5	39.6	2.10	15/09	33.8	39	↑+5.20	31/10	34.87	35.1	↑+0.23
		16/10	41.25	19/10	39.97	40.65	↑+0.68	30/10	35.62	35.7	↑+0.08	15/12	32.72	32	↑+0.72

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TABLE 10 – continued from previous page

Stock	Architecture	Start Date	Start Price	Short-term End Date	Short-term Pred	Short-term Act	Short-term Gap	Medium-term End Date	Medium-term Pred	Medium-term Act	Medium-term Gap	Long-term End Date	Long-term Pred	Long-term Act	Long-term Gap
		21/10	37.65	24/10	36.54	36.1	↑+0.44	04/11	32.74	35	↑+2.26	20/12	33.24	33.6	↑+0.36
		22/10	38	25/10	36.27	36.1	↑+0.17	05/11	30.57	34.1	↑+3.53	21/12	33.25	33.6	↑+0.35
	Hierarchical	01/09	39.6	04/09	38.63	39.6	0.97	15/09	38.81	39	↑+0.19	31/10	39.61	35.1	↓-4.51
		16/10	41.25	19/10	40.77	40.65	↑+0.12	30/10	39.06	35.7	↑+3.36	15/12	33	32	↑+1.00
		21/10	37.65	24/10	37.3	36.1	↑+1.20	04/11	36.03	35	↑+1.03	20/12	31.19	33.6	↑+2.41
		22/10	38	25/10	37.67	36.1	↑+1.57	05/11	36.51	34.1	↑+2.41	21/12	32.02	33.6	↑+1.58
	RoundRobin	01/09	39.6	04/09	37.63	39.6	1.97	15/09	34.36	39	↑+4.64	31/10	30.86	35.1	↑+4.24
		16/10	41.25	19/10	40.08	40.65	↑+0.57	30/10	36.06	35.7	↑+0.36	15/12	33	32	↑+1.00
		21/10	37.65	24/10	36.42	36.1	↑+0.32	04/11	32.25	35	↑+2.75	20/12	30.12	33.6	↑+3.48
		22/10	38	25/10	36.86	36.1	↑+0.76	05/11	32.97	34.1	↑+1.13	21/12	30.4	33.6	↑+3.20
	Hierarchical	01/09	104.5	04/09	104.82	102.3	↓-2.52	15/09	106.01	104.5	1.51	31/10	111.13	99.2	↓-11.93
		16/10	122	19/10	122.43	116	↓-6.43	30/10	104.02	104	↑+0.02	15/12	100.87	93.9	↑+6.97
		21/10	110.9	24/10	111.57	114.5	↑+2.93	04/11	104.04	100.2	↑+3.84	20/12	105	101.5	↑+3.50
		22/10	113	25/10	113.43	114.5	↑+1.07	05/11	105.02	99.6	↑+5.42	21/12	101.9	101.5	↑+0.40
	Round Robin	01/09	104.5	04/09	105.13	102.3	↓-2.83	15/09	107.47	104.5	2.97	31/10	107.83	99.2	↓-8.63
VHM		16/10	122	19/10	122.73	116	↓-6.73	30/10	105.47	104	↑+1.47	15/12	101.56	93.9	↑+7.66
		21/10	110.9	24/10	114.27	114.5	↑+0.23	04/11	101.05	100.2	↑+0.85	20/12	105.05	101.5	↑+3.55
		22/10	113	25/10	113.68	114.5	↑+0.82	05/11	106.21	99.6	↑+6.61	21/12	107.41	101.5	↑+5.91
	Ensemble	01/09	104.5	04/09	100.22	102.3	↑+2.08	15/09	105.98	104.5	1.48	31/10	93.6	99.2	↑+5.60
		16/10	122	19/10	116.89	116	↑+0.89	30/10	104.92	104	↑+0.92	15/12	97.6	93.9	↑+3.70
		21/10	110.9	24/10	116.18	114.5	↑+1.68	04/11	103.54	100.2	↑+3.34	20/12	98.72	101.5	↑+2.78
		22/10	113	25/10	114.07	114.5	↑+0.43	05/11	91.78	99.6	↑+7.82	21/12	100.4	101.5	↑+1.10
	Hierarchical	01/09	78	04/09	66.76	77.5	↑+10.74	15/09	70.65	79.5	↓-8.85	31/10	85.03	82.6	↑+2.43
		16/10	84.5	19/10	75.55	84.5	8.95	30/10	79.88	83.9	↑+4.02	15/12	74.98	77.7	↑+2.72
		21/10	83	24/10	74.96	85.7	↓-10.74	04/11	76.07	81.8	↑+5.73	20/12	75.71	82.9	↑+7.19
		22/10	84.5	25/10	77.54	85.7	↓-8.16	05/11	77.66	80.2	↑+2.54	21/12	77.75	82.9	↑+5.15
	Round Robin	01/09	78	04/09	64.18	77.5	↑+13.32	15/09	65.82	79.5	↓-13.68	31/10	73.69	82.6	↓-8.91
MWG		16/10	84.5	19/10	78.11	84.5	6.39	30/10	78.47	83.9	↑+5.43	15/12	68.65	77.7	↑+9.05
		21/10	83	24/10	76.62	85.7	↓-9.08	04/11	76.93	81.8	↑+4.87	20/12	76.64	82.9	↑+6.26
		22/10	84.5	25/10	79.21	85.7	↓-6.49	05/11	78.12	80.2	↑+2.08	21/12	78.76	82.9	↑+4.14
	Ensemble	01/09	78	04/09	71.76	77.5	↑+5.74	15/09	71.8	79.5	↓-7.70	31/10	81.77	82.6	↑+0.83
		16/10	84.5	19/10	87.77	84.5	3.27	30/10	87.86	83.9	↓-3.96	15/12	77.67	77.7	↑+0.03
		21/10	83	24/10	86.12	85.7	↑+0.42	04/11	75.55	81.8	↑+6.25	20/12	82.6	82.9	↑+0.30
		22/10	84.5	25/10	81	85.7	↓-4.70	05/11	78.36	80.2	↑+1.84	21/12	79.5	82.9	↑+3.40
	Hierarchical	01/09	40.3	04/09	34.06	41.85	↓-7.79	15/09	35.52	40.8	↓-5.28	31/10	34.09	41.9	↓-7.81
		16/10	38	19/10	32.06	37.3	↑+5.24	30/10	32.84	41.3	↓-8.46	15/12	34	35.1	↑+1.10
		21/10	38.4	24/10	33.23	38.75	↓-5.52	04/11	31.65	39.6	↓-7.95	20/12	32.58	35.2	↑+2.62
		22/10	38.5	25/10	33.23	38.75	↓-5.52	05/11	34.01	39.3	↓-5.29	21/12	32.59	35.2	↑+2.61
	Round Robin	01/09	40.3	04/09	39.51	41.85	↓-2.34	15/09	39.46	40.8	↓-1.34	31/10	39.43	41.9	↓-2.47
CMG		16/10	38	19/10	39.55	37.3	↓-2.25	30/10	38.89	41.3	↑+2.41	15/12	30.15	35.1	↑+4.95
		21/10	38.4	24/10	38.04	38.75	↓-0.71	04/11	31.54	39.6	↓-8.06	20/12	29.13	35.2	↑+6.07
		22/10	38.5	25/10	30.06	38.75	↓-8.69	05/11	38.18	39.3	↓-1.12	21/12	29.32	35.2	↑+5.88
	Ensemble	01/09	40.3	04/09	40.23	41.85	↓-1.62	15/09	39.31	40.8	↓-1.49	31/10	40.23	41.9	↓-1.67
		16/10	38	19/10	39.99	37.3	↓-2.69	30/10	41.27	41.3	↑+0.03	15/12	34.58	35.1	↑+0.52
		21/10	38.4	24/10	38.53	38.75	↑+0.22	04/11	39.67	39.6	↑+0.07	20/12	37.93	35.2	↑+2.73
		22/10	38.5	25/10	38	38.75	↓-0.75	05/11	39.52	39.3	↑+0.22	21/12	36.07	35.2	↑+0.87
	Hierarchical	01/09	36.5	04/09	31.13	36.75	↓-5.62	15/09	31.78	34.85	↑+3.07	31/10	38.72	35.85	↓-2.87
		16/10	34.2	19/10	34.21	33.9	↓-0.31	30/10	36.82	35.85	↑+0.97	15/12	27.66	29.6	↑+1.94
		21/10	31.75	24/10	32.12	33.8	↑+1.68	04/11	34.42	35.6	↑+1.18	20/12	31.42	32.65	↓-1.23
		22/10	32.2	25/10	32.15	33.8	↓-1.65	05/11	35.15	34.95	↑+0.20	21/12	35.65	32.65	↑+3.00
	Round Robin	01/09	36.5	04/09	32.25	36.75	↓-4.50	15/09	34.88	34.85	↑+0.03	31/10	35.21	35.85	↑+0.64
KDH		16/10	34.2	19/10	34.89	33.9	↓-0.99	30/10	34.61	35.85	↑+1.24	15/12	29.52	29.6	↑+0.08
		21/10	31.75	24/10	33.45	33.8	↑+0.35	04/11	32.14	35.6	↑+3.46	20/12	29.56	32.65	↓-3.09
		22/10	32.2	25/10	33.24	33.8	↑+0.56	05/11	33.45	34.95	↑+1.50	21/12	30.54	32.65	↓-2.11
	Ensemble	01/09	36.5	04/09	31.55	36.75	↓-5.20	15/09	35.13	34.85	↑+0.28	31/10	34.11	35.85	↑+1.74
		16/10	34.2	19/10	32.74	33.9	↑+1.16	30/10	32.61	35.85	↓-3.24	15/12	39.42	29.6	↓-9.82
		21/10	31.75	24/10	31.23	33.8	↓-2.57	04/11	34.48	35.6	↑+1.12	20/12	31.64	32.65	↓-1.01
		22/10	32.2	25/10	31.26	33.8	↓-2.54	05/11	32.54	34.95	↑+2.41	21/12	32.54	32.65	↑+0.11
	Hierarchical	01/09	98.2	04/09	96.52	99.3	↓-2.78	15/09	97.04	99.5	↓-2.46	31/10	96.9	96	↑+0.90
		16/10	95	19/10	93.01	93.2	↑+0.19	30/10	93.22	93.6	↑+0.38	15/12	94.05	93	↑+1.05
		21/10	89.2	24/10	86.68	92.4	↓-5.72	04/11	87.01	94.6	↓-7.59	20/12	78.04	70.2	↑+7.84
		22/10	91	25/10	88	92.4	↓-4.40	05/11	87.5	95	↓-7.50	21/12	70.04	70.2	↑+0.16
	Round Robin	01/09	98.2	04/09	86.46	99.3	↓-12.84	15/09	86.87	99.5	↓-12.63	31/10	96.73	96	↑+0.73
DGC		16/10	95	19/10	94.18	93.2	↑+0.98	30/10	84.17	93.6	↑+9.43	15/12	94.16	93	↑+1.16

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TABLE 10 – continued from previous page

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
		21/10	89.2	24/10	88.88	92.4	↓-3.52	04/11	96.27	94.6	↑+1.67	20/12	78.47	70.2	↑+8.27
		22/10	91	25/10	94.7	92.4	↑+2.30	05/11	93.3	95	↑+1.70	21/12	72.2	70.2	↑+2.00
	Ensemble	01/09	98.2	04/09	99.82	99.3	↑+0.52	15/09	96.05	99.5	↓-3.45	31/10	95.81	96	↑+0.19
		16/10	95	19/10	92.82	93.2	↑+0.38	30/10	98.86	93.6	↓-5.26	15/12	92.59	93	↑+0.41
		21/10	89.2	24/10	95.26	92.4	↑+2.86	04/11	94.94	94.6	↑+0.34	20/12	74.65	70.2	↑+4.45
		22/10	91	25/10	97.89	92.4	↑+5.49	05/11	98.27	95	↑+3.27	21/12	77.61	70.2	↑+7.41
MBB	**Hierarchical**	01/09	27.75	04/09	28.05	28.25	↑+0.20	15/09	27.61	26.85	↑+0.76	31/10	20.91	23.6	↑+2.69
		16/10	27.2	19/10	25.23	27.1	↑+1.87	30/10	24.34	23.95	↑+0.39	15/12	27.05	23.75	↑+3.30
		21/10	25.3	24/10	25.63	24.4	↓-1.23	04/11	23.21	24	↑+0.79	20/12	26.13	24.7	↓-1.43
		22/10	25.45	25/10	23.29	24.4	↑+1.11	05/11	22.72	23.9	↑+1.18	21/12	25.03	24.7	↑+0.33
	Round Robin	01/09	27.75	04/09	24.24	28.25	↓-4.01	15/09	27.51	26.85	↑+0.66	31/10	27.53	23.6	↑+3.93
		16/10	27.2	19/10	25.31	27.1	↑+1.79	30/10	23.01	23.95	↑+0.94	15/12	22.05	23.75	↑+1.70
		21/10	25.3	24/10	23.93	24.4	↑+0.47	04/11	24.66	24	↑+0.66	20/12	23.53	24.7	↑+1.17
		22/10	25.45	25/10	26.33	24.4	↓-1.93	05/11	23.68	23.9	↑+0.22	21/12	25.58	24.7	↓-0.88
	Ensemble	01/09	27.75	04/09	27.93	28.25	↑+0.32	15/09	27.42	26.85	↑+0.57	31/10	21.52	23.6	↑+2.08
		16/10	27.2	19/10	24.92	27.1	↑+2.18	30/10	23.81	23.95	↑+0.14	15/12	25.43	23.75	↑+1.68
		21/10	25.3	24/10	26.93	24.4	↓-2.53	04/11	24.93	24	↑+0.93	20/12	23.42	24.7	↑+1.28
		22/10	25.45	25/10	26.37	24.4	↓-1.97	05/11	25.72	23.9	↓-1.82	21/12	25.31	24.7	↑+0.61

TABLE 11: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (LLM: GPT-4o, Transformer Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Hierarchical	31/08	22.8	03/09	23.03	24	↑+0.97	14/09	23.14	24.05	↑+0.91	30/10	23.91	21.2	↓-2.71
		16/10	22.45	19/10	21.5	22.6	↓-1.10	30/10	22.83	21.2	↓-1.63	15/12	19.03	16.3	↑+2.73
		21/10	20.0	24/10	17.21	20.9	↓-3.69	04/11	19.95	20.15	↓-0.20	20/12	18.69	17.8	↑+0.89
		22/10	20.5	25/10	19.67	20.9	↓-1.23	05/11	18.8	20.1	↑+1.30	21/12	18.61	17.8	↑+0.81
	Round Robin	31/08	22.8	03/09	22.86	24	↑+1.14	14/09	23.87	24.05	↑+0.18	30/10	20.26	21.2	↑+0.94
		16/10	22.45	19/10	22.28	22.6	↓-0.32	30/10	22.79	21.2	↓-1.59	15/12	26.57	16.3	↓-10.27
		21/10	20.0	24/10	18.51	20.9	↓-2.39	04/11	20.22	20.15	↑+0.07	20/12	16.78	17.8	↑+1.02
		22/10	20.5	25/10	20.59	20.9	↑+0.31	05/11	21.19	20.1	↓-1.09	21/12	21.0	17.8	↓-3.20
	Ensemble	31/08	22.8	03/09	22.15	24	↓-1.85	14/09	22.91	24.05	↑+1.14	30/10	25.54	21.2	↓-4.34
		16/10	22.45	19/10	25.15	22.6	↑+2.55	30/10	20.1	21.2	↑+1.10	15/12	21.53	16.3	↑+5.23
		21/10	20.0	24/10	19.84	20.9	↓-1.06	04/11	21.8	20.15	↑+1.65	20/12	20.29	17.8	↓-2.49
		22/10	20.5	25/10	18.93	20.9	↓-1.97	05/11	20.7	20.1	↓-0.60	21/12	22.49	17.8	↓-4.69
FPT	Hierarchical	01/09	101.6	04/09	101.82	105	↑+3.18	15/09	102.4	101.9	↑+0.50	31/10	101.45	103.9	↓-2.45
		16/10	89.8	19/10	94.05	88.1	↓-5.95	30/10	93.77	102.7	↑+8.93	15/12	94.24	93.8	↑+0.44
		21/10	93	24/10	93.21	97.7	↑+4.49	04/11	99.68	103.3	↑+3.62	20/12	92.13	93.9	↓-1.77
		22/10	97	25/10	96.76	97.7	↓-0.94	05/11	94.24	100.9	↓-6.66	21/12	94.7	93.9	↑+0.80
	Round Robin	01/09	101.6	04/09	101.27	105	↓-3.73	15/09	101.08	101.9	↓-0.82	31/10	99.11	103.9	↓-4.79
		16/10	89.8	19/10	87.11	88.1	↑+0.99	30/10	101.65	102.7	↑+1.05	15/12	85.43	93.8	↓-8.37
		21/10	93	24/10	91.76	97.7	↓-5.94	04/11	80.81	103.3	↓-22.49	20/12	93.79	93.9	↑+0.11
		22/10	97	25/10	96.93	97.7	↓-0.77	05/11	100.57	100.9	↑+0.33	21/12	99.93	93.9	↓-6.03
	Ensemble	01/09	101.6	04/09	103.76	105	↑+1.24	15/09	101.74	101.9	↑+0.16	31/10	100.99	103.9	↓-2.91
		16/10	89.8	19/10	91.28	88.1	↓-3.18	30/10	87.0	102.7	↓-15.70	15/12	88.74	93.8	↓-5.06
		21/10	93	24/10	96.74	97.7	↑+0.96	04/11	96.43	103.3	↑+6.87	20/12	90.97	93.9	↓-2.93
		22/10	97	25/10	98.58	97.7	↑+0.88	05/11	102.92	100.9	↑+2.02	21/11	96.2	93.9	↑+2.30
VCB	Hierarchical	30/08	68.6	02/09	68.01	68.6	0.59	13/09	69.85	65.8	↓-4.05	29/10	81.38	60.7	↓-20.68
		16/10	62.9	19/10	60.4	61.9	↑+1.50	30/10	61.01	60.6	↑+0.41	15/12	67.82	56.8	↓-11.02
		21/10	59.3	24/10	58.32	59.5	↓-1.18	04/11	59.87	60.1	↑+0.23	20/12	59.41	57.5	↓-1.91
		22/10	59.6	25/10	59.8	59.5	↓-0.30	05/11	58.53	60.8	↓-2.27	21/12	58.91	57.5	↑+1.41
	Round Robin	30/08	68.6	02/09	68.98	68.6	0.38	13/09	68.15	65.8	↑+2.35	29/10	72.46	60.7	↓-11.76
		16/10	62.9	19/10	61.05	61.9	↑+0.85	30/10	63.09	60.6	↓-2.49	15/12	53.5	56.8	↑+3.30
		21/10	59.3	24/10	58.76	59.5	↓-0.74	04/11	60.61	60.1	↑+0.51	20/12	57.26	57.5	↑+0.24
		22/10	59.6	25/10	58.83	59.5	↑+0.67	05/11	61.07	60.8	↑+0.27	21/12	62.5	57.5	↓-5.00
	Ensemble	30/08	68.6	02/09	69.08	68.6	0.48	13/09	68.53	65.8	↑+2.73	29/10	75.07	60.7	↓-14.37
		16/10	62.9	19/10	63.22	61.9	↓-1.32	30/10	58.67	60.6	↑+1.93	15/12	59.85	56.8	↑+3.05
		21/10	59.3	24/10	59.17	59.5	↓-0.33	04/11	59.86	60.1	↑+0.24	20/12	59.08	57.5	↑+1.58
		22/10	59.6	25/10	59.34	59.5	↑+0.16	05/11	58.66	60.8	↓-2.14	21/12	57.58	57.5	↑+0.08
VCS	Hierarchical	30/08	48.8	02/09	49.07	48.8	0.27	13/09	48.43	50	↓-1.57	29/10	51.44	47.6	↓-3.84
		16/10	47.2	19/10	46.41	46.8	↑+0.39	30/10	48.26	47.4	↑+0.86	15/12	45.44	46.5	↑+1.06
		21/10	45.5	24/10	44.52	46.2	↓-1.68	04/11	44.42	47.1	↓-2.68	20/12	42.58	43	↑+0.42
		22/10	45.5	25/10	44.4	46.2	↓-1.80	05/11	43.0	46.9	↓-3.90	21/12	47.3	43	↓-4.30
	Round Robin	30/08	48.8	02/09	48.31	48.8	0.49	13/09	47.65	50	↓-2.35	29/10	48.89	47.6	↓-1.29
		16/10	47.2	19/10	47.45	46.8	↓-0.65	30/10	46.4	47.4	↓-1.00	15/12	47.56	46.5	↓-1.06
		21/10	45.5	24/10	46.08	46.2	↑+0.12	04/11	46.87	47.1	↑+0.23	20/12	41.81	43	↑+1.19
		22/10	45.5	25/10	45.2	46.2	↓-1.00	05/11	43.89	46.9	↓-3.01	21/12	48.18	43	↓-5.18
	Ensemble	30/08	48.8	02/09	50.26	48.8	1.46	13/09	46.13	50	↓-3.87	29/10	47.75	47.6	↑+0.15
		16/10	47.2	19/10	48.01	46.8	↓-1.21	30/10	47.99	47.4	↑+0.59	15/12	49.06	46.5	↓-2.56
		21/10	45.5	24/10	45.8	46.2	↑+0.40	04/11	42.4	47.1	↓-4.70	20/12	47.57	43	↓-4.57
		22/10	45.5	25/10	44.64	46.2	↓-1.56	05/11	44.36	46.9	↓-2.54	21/12	42.76	43	↑+0.24
HPG	Hierarchical	01/09	27.5	04/09	23.78	29.85	↓-6.07	15/09	29.96	30.35	↑+0.39	31/10	27.05	26.7	↑+0.35
		16/10	28.25	19/10	29.7	28	↓-1.70	30/10	27.68	26.9	↑+0.78	15/12	27.12	26.25	↑+0.87
		21/10	26.8	24/10	26.22	26.4	↑+0.18	04/11	28.49	26.75	↓-1.74	20/12	27.77	26.7	↓-1.07
		22/10	26.7	25/10	24.88	26.4	↑+1.52	05/11	26.63	26.3	↑+0.33	21/12	26.43	26.7	0.27
	Round Robin	01/09	27.5	04/09	25.77	29.85	↓-4.08	15/09	27.79	30.35	↑+2.56	31/10	27.06	26.7	↑+0.36
		16/10	28.25	19/10	29.22	28	↓-1.22	30/10	27.53	26.9	↑+0.63	15/12	21.61	26.25	↑+4.64
		21/10	26.8	24/10	26.27	26.4	↑+0.13	04/11	27.19	26.75	↓-0.44	20/12	27.2	26.7	↓-0.50
		22/10	26.7	25/10	26.0	26.4	↑+0.40	05/11	26.25	26.3	↑+0.05	21/12	26.62	26.7	0.08
	Ensemble	01/09	27.5	04/09	29.26	29.85	↑+0.59	15/09	26.98	30.35	↓-3.37	31/10	26.93	26.7	↑+0.23
		16/10	28.25	19/10	29.28	28	↓-1.28	30/10	26.84	26.9	↑+0.06	15/12	28.54	26.25	↓-2.29
		21/10	26.8	24/10	26.26	26.4	↑+0.14	04/11	28.02	26.75	↓-1.27	20/12	27.16	26.7	↓-0.46
		22/10	26.7	25/10	26.5	26.4	↑+0.10	05/11	25.47	26.3	↑+0.83	21/12	26.42	26.7	0.28
	Ensemble	01/09	39.6	04/09	38.95	39.6	0.65	15/09	38.67	39	↑+0.33	31/10	40.53	35.1	↓-5.43
		16/10	41.25	19/10	41.18	40.65	↑+0.53	30/10	43.02	35.7	↓-7.32	15/12	39.46	32	↑+7.46

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TABLE 11 – continued from previous page

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
		21/10	37.65	24/10	36.85	36.1	↑+0.75	04/11	38.11	35	↓-3.11	20/12	37.72	33.6	↓-4.12
		22/10	38	25/10	39.73	36.1	↓-3.63	05/11	38.55	34.1	↓-4.45	21/12	42.78	33.6	↓-9.18
	Hierarchical	01/09	39.6	04/09	38.7	39.6	0.90	15/09	40.61	39	↓-1.61	31/10	37.34	35.1	↑+2.24
		16/10	41.25	19/10	42.16	40.65	↓-1.51	30/10	40.14	35.7	↑+4.44	15/12	42.99	32	↓-10.99
		21/10	37.65	24/10	34.15	36.1	↑+1.95	04/11	36.45	35	↑+1.45	20/12	38.94	33.6	↓-5.34
		22/10	38	25/10	39.08	36.1	↓-2.98	05/11	36.57	34.1	↑+2.47	21/12	43.97	33.6	↓-10.37
	RoundRobin	01/09	39.6	04/09	37.66	39.6	1.94	15/09	40.64	39	↓-1.64	31/10	40.68	35.1	↓-5.58
		16/10	41.25	19/10	40.49	40.65	↑+0.16	30/10	44.17	35.7	↓-8.47	15/12	43.14	32	↓-11.14
		21/10	37.65	24/10	38.24	36.1	↓-2.14	04/11	37.02	35	↑+2.02	20/12	37.13	33.6	↑+3.53
		22/10	38	25/10	39.26	36.1	↓-3.16	05/11	37.32	34.1	↑+3.22	21/12	40.09	33.6	↓-6.49
	Hierarchical	01/09	104.5	04/09	102.17	102.3	↑+0.13	15/09	103.53	104.5	0.97	31/10	108.73	99.2	↓-9.53
		16/10	122	19/10	122.6	116	↓-6.60	30/10	126.31	104	↓-22.31	15/12	100.29	93.9	↑+6.39
		21/10	110.9	24/10	115.51	114.5	↑+1.01	04/11	115.77	100.2	↓-15.57	20/12	115.81	101.5	↓-14.31
		22/10	113	25/10	113.5	114.5	↑+1.00	05/11	84.63	99.6	↑+14.97	21/12	113.47	101.5	↓-11.97
	Round Robin	01/09	104.5	04/09	103.32	102.3	↑+1.02	15/09	103.59	104.5	0.91	31/10	104.88	99.2	↓-5.68
VHM		16/10	122	19/10	128.13	116	↓-12.13	30/10	135.55	104	↓-31.55	15/12	105.51	93.9	↑+11.61
		21/10	110.9	24/10	113.14	114.5	↑+1.36	04/11	101.53	100.2	↑+1.33	20/12	114.54	101.5	↓-13.04
		22/10	113	25/10	111.56	114.5	↓-2.94	05/11	119.26	99.6	↓-19.66	21/12	103.27	101.5	↑+1.77
	Ensemble	01/09	104.5	04/09	103.06	102.3	↑+0.76	15/09	105.01	104.5	0.51	31/10	105.34	99.2	↓-6.14
		16/10	122	19/10	118.75	116	↑+2.75	30/10	106.66	104	↑+2.66	15/12	148.1	93.9	↓-54.20
		21/10	110.9	24/10	111.05	114.5	↑+3.45	04/11	95.32	100.2	↑+4.88	20/12	108.7	101.5	↑+7.20
		22/10	113	25/10	111.22	114.5	↓-3.28	05/11	125.4	99.6	↓-25.80	21/12	129.43	101.5	↓-27.93
	Hierarchical	01/09	78	04/09	78.24	77.5	↓-0.74	15/09	80.65	79.5	↑+1.15	31/10	77.37	82.6	↓-5.23
		16/10	84.5	19/10	82.77	84.5	1.73	30/10	83.1	83.9	↑+0.80	15/12	90.67	77.7	↓-12.97
		21/10	83	24/10	84.52	85.7	↑+1.18	04/11	82.3	81.8	↑+0.50	20/12	83.66	82.9	↓-0.76
		22/10	84.5	25/10	82.67	85.7	↓-3.03	05/11	86.36	80.2	↓-6.16	21/12	83.9	82.9	↑+1.00
	Round Robin	01/09	78	04/09	76.98	77.5	↑+0.52	15/09	77.93	79.5	↓-1.57	31/10	80.0	82.6	↑+2.60
MWG		16/10	84.5	19/10	83.77	84.5	0.73	30/10	82.21	83.9	↑+1.69	15/12	79.25	77.7	↑+1.55
		21/10	83	24/10	84.26	85.7	↑+1.44	04/11	84.82	81.8	↓-3.02	20/12	84.54	82.9	↓-1.64
		22/10	84.5	25/10	81.96	85.7	↓-3.74	05/11	80.02	80.2	↑+0.18	21/12	85.06	82.9	↓-2.16
	Ensemble	01/09	78	04/09	78.34	77.5	↓-0.84	15/09	77.52	79.5	↓-1.98	31/10	75.39	82.6	↓-7.21
		16/10	84.5	19/10	83.27	84.5	1.23	30/10	87.7	83.9	↓-3.80	15/12	88.21	77.7	↓-10.51
		21/10	83	24/10	84.5	85.7	↑+1.20	04/11	83.31	81.8	↓-1.51	20/12	82.37	82.9	↑+0.53
		22/10	84.5	25/10	84.93	85.7	↑+0.77	05/11	80.6	80.2	↑+0.40	21/12	85.0	82.9	↓-2.10
	Hierarchical	01/09	40.3	04/09	42.26	41.85	↑+0.41	15/09	42.26	40.8	↑+1.46	31/10	40.96	41.9	↑+0.94
		16/10	38	19/10	38.52	37.3	↓-1.22	30/10	35.02	41.3	↓-6.28	15/12	38.16	35.1	↓-3.06
		21/10	38.4	24/10	39.74	38.75	↑+0.99	04/11	36.82	39.6	↓-2.78	20/12	36.41	35.2	↑+1.21
		22/10	38.5	25/10	37.33	38.75	↓-1.42	05/11	38.68	39.3	↑+0.62	21/12	41.03	35.2	↓-5.83
	Round Robin	01/09	40.3	04/09	38.57	41.85	↓-3.28	15/09	38.49	40.8	↓-2.31	31/10	40.92	41.9	↑+0.98
CMG		16/10	38	19/10	37.15	37.3	↑+0.15	30/10	42.26	41.3	↑+0.96	15/12	41.63	35.1	↓-6.53
		21/10	38.4	24/10	37.7	38.75	↓-1.05	04/11	38.51	39.6	↑+1.09	20/12	41.55	35.2	↓-6.35
		22/10	38.5	25/10	38.43	38.75	↓-0.32	05/11	38.87	39.3	↑+0.43	21/12	35.32	35.2	↑+0.12
	Ensemble	01/09	40.3	04/09	41.13	41.85	↑+0.72	15/09	41.01	40.8	↑+0.21	31/10	41.52	41.9	↑+0.38
		16/10	38	19/10	37.62	37.3	↑+0.32	30/10	41.51	41.3	↑+0.21	15/12	34.85	35.1	↑+0.25
		21/10	38.4	24/10	38.66	38.75	↑+0.09	04/11	37.68	39.6	↓-1.92	20/12	40.41	35.2	↓-5.21
		22/10	38.5	25/10	38.29	38.75	↓-0.46	05/11	38.08	39.3	↓-1.22	21/12	41.48	35.2	↓-6.28
	Hierarchical	01/09	36.5	04/09	38.19	36.75	↑+1.44	15/09	37.75	34.85	↓-2.90	31/10	35.8	35.85	↑+0.05
		16/10	34.2	19/10	33.58	33.9	↑+0.32	30/10	32.82	35.85	↓-3.03	15/12	32.31	29.6	↑+2.71
		21/10	31.75	24/10	29.82	33.8	↓-3.98	04/11	33.28	35.6	↑+2.32	20/12	29.97	32.65	↓-2.68
		22/10	32.2	25/10	34.27	33.8	↑+0.47	05/11	30.38	34.95	↓-4.57	21/12	32.44	32.65	↑+0.21
	Round Robin	01/09	36.5	04/09	36.95	36.75	↑+0.20	15/09	37.47	34.85	↓-2.62	31/10	35.63	35.85	↑+0.22
KDH		16/10	34.2	19/10	34.85	33.9	↓-0.95	30/10	36.31	35.85	↑+0.46	15/12	36.91	29.6	↓-7.31
		21/10	31.75	24/10	33.34	33.8	↑+0.46	04/11	35.81	35.6	↑+0.21	20/12	32.33	32.65	↑+0.32
		22/10	32.2	25/10	28.33	33.8	↓-5.47	05/11	29.88	34.95	↓-5.07	21/12	33.14	32.65	↑+0.49
	Ensemble	01/09	36.5	04/09	36.23	36.75	↓-0.52	15/09	34.71	34.85	↑+0.14	31/10	36.65	35.85	↓-0.80
		16/10	34.2	19/10	33.66	33.9	↑+0.24	30/10	36.08	35.85	↑+0.23	15/12	35.87	29.6	↓-6.27
		21/10	31.75	24/10	30.07	33.8	↓-3.73	04/11	30.1	35.6	↓-5.50	20/12	33.92	32.65	↑+1.27
		22/10	32.2	25/10	32.74	33.8	↑+1.06	05/11	33.27	34.95	↑+1.68	21/12	33.2	32.65	↑+0.55
	Hierarchical	01/09	98.2	04/09	96.81	99.3	↓-2.49	15/09	99.87	99.5	↑+0.37	31/10	99.54	96	↓-3.54
		16/10	95	19/10	96.43	93.2	↓-3.23	30/10	96.65	93.6	↓-3.05	15/12	94.19	93	↑+1.19
		21/10	89.2	24/10	96.52	92.4	↑+4.12	04/11	94.94	94.6	↑+0.34	20/12	94.72	70.2	↓-24.52
		22/10	91	25/10	89.31	92.4	↓-3.09	05/11	99.7	95	↑+4.70	21/12	78.79	70.2	↑+8.59
	Round Robin	01/09	98.2	04/09	98.45	99.3	↑+0.85	15/09	97.09	99.5	↓-2.41	31/10	96.95	96	↑+0.95
DGC		16/10	95	19/10	93.94	93.2	↑+0.74	30/10	93.81	93.6	↑+0.21	15/12	94.66	93	↑+1.66

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TABLE 11 – continued from previous page

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
		21/10	89.2	24/10	88.21	92.4	↓-4.19	04/11	87.8	94.6	↓-6.80	20/12	96.51	70.2	↓-26.31
		22/10	91	25/10	93.86	92.4	↑+1.46	05/11	86.25	95	↓-8.75	21/12	94.93	70.2	↓-24.73
	Ensemble	01/09	98.2	04/09	98.99	99.3	↑+0.31	15/09	99.11	99.5	↑+0.39	31/10	98.54	96	↓-2.54
		16/10	95	19/10	96.98	93.2	↓-3.78	30/10	95.2	93.6	↓-1.60	15/12	98.38	93	↓-5.38
		21/10	89.2	24/10	89.64	92.4	↑+2.76	04/11	92.07	94.6	↑+2.53	20/12	94.71	70.2	↓-24.51
		22/10	91	25/10	91.68	92.4	↑+0.72	05/11	92.48	95	↑+2.52	21/12	73.49	70.2	↑+3.29
MBB	**Hierarchical**	01/09	27.75	04/09	28.74	28.25	↑+0.49	15/09	27.92	26.85	↓-1.07	31/10	31.87	23.6	↓-8.27
		16/10	27.2	19/10	27.51	27.1	↓-0.41	30/10	23.49	23.95	↑+0.46	15/12	29.55	23.75	↓-5.80
		21/10	25.3	24/10	25.4	24.4	↓-1.00	04/11	29.13	24	↓-5.13	20/12	24.99	24.7	↑+0.29
		22/10	25.45	25/10	26.77	24.4	↓-2.37	05/11	23.99	23.9	↑+0.09	21/12	25.78	24.7	↓-1.08
	Round Robin	01/09	27.75	04/09	27.04	28.25	↓-1.21	15/09	29.21	26.85	↓-2.36	31/10	26.43	23.6	↑+2.83
		16/10	27.2	19/10	26.77	27.1	↑+0.33	30/10	29.85	23.95	↓-5.90	15/12	30.02	23.75	↓-6.27
		21/10	25.3	24/10	25.96	24.4	↓-1.56	04/11	25.19	24	↑+1.19	20/12	24.24	24.7	↑+0.46
		22/10	25.45	25/10	25.18	24.4	↑+0.78	05/11	26.41	23.9	↓-2.51	21/12	23.81	24.7	↑+0.89
	Ensemble	01/09	27.75	04/09	26.94	28.25	↓-1.31	15/09	28.43	26.85	↓-1.58	31/10	28.64	23.6	↓-5.04
		16/10	27.2	19/10	25.99	27.1	↑+1.11	30/10	28.56	23.95	↓-4.61	15/12	27.36	23.75	↓-3.61
		21/10	25.3	24/10	24.64	24.4	↑+0.24	04/11	24.77	24	↑+0.77	20/12	26.38	24.7	↓-1.68
		22/10	25.45	25/10	24.66	24.4	↑+0.26	05/11	22.02	23.9	↑+1.88	21/12	25.4	24.7	↑+0.70

TABLE 12: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (LLM: Llama, LSTM Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Hierarchical	31/08	22.8	03/09	22.22	24	↓-1.78	14/09	23.45	24.05	↑+0.60	30/10	20.52	21.2	↑+0.68
		16/10	22.45	19/10	22.85	22.6	↑+0.25	30/10	21.57	21.2	↑+0.37	15/12	25.38	16.3	↓-9.08
		21/10	20.0	24/10	20.8	20.9	↑+0.10	04/11	19.83	20.15	↓-0.32	20/12	20.44	17.8	↓-2.64
		22/10	20.5	25/10	20.83	20.9	↑+0.07	05/11	19.27	20.1	↑+0.83	21/12	18.23	17.8	↑+0.43
	Round Robin	31/08	22.8	03/09	23.25	24	↑+0.75	14/09	23.7	24.05	↑+0.35	30/10	21.8	21.2	↑+0.60
		16/10	22.45	19/10	21.19	22.6	↓-1.41	30/10	22.59	21.2	↓-1.39	15/12	22.73	16.3	↓-6.43
		21/10	20.0	24/10	19.65	20.9	↓-1.25	04/11	19.16	20.15	↓-0.99	20/12	20.3	17.8	↓-2.50
		22/10	20.5	25/10	19.79	20.9	↓-1.11	05/11	20.37	20.1	↑+0.27	21/12	22.54	17.8	↓-4.74
	Ensemble	31/08	22.8	03/09	22.36	24	↓-1.64	14/09	21.75	24.05	↓-2.30	30/10	23.88	21.2	↓-2.68
		16/10	22.45	19/10	23.39	22.6	↑+0.79	30/10	21.54	21.2	↑+0.34	15/12	22.39	16.3	↑+6.09
		21/10	20.0	24/10	20.62	20.9	↑+0.28	04/11	20.38	20.15	↑+0.23	20/12	23.04	17.8	↓-5.24
		22/10	20.5	25/10	19.74	20.9	↓-1.16	05/11	19.36	20.1	↑+0.74	21/12	20.04	17.8	↑+2.24
FPT	Hierarchical	01/09	101.6	04/09	103.64	105	↑+1.36	15/09	100.74	101.9	↓-1.16	31/10	99.52	103.9	↓-4.38
		16/10	89.8	19/10	90.67	88.1	↓-2.57	30/10	95.39	102.7	↑+7.31	15/12	91.25	93.8	↑+2.55
		21/10	93	24/10	92.28	97.7	↓-5.42	04/11	91.49	103.3	↓-11.81	20/12	91.93	93.9	↓-1.97
		22/10	97	25/10	100.01	97.7	↑+2.31	05/11	93.0	100.9	↓-7.90	21/12	99.45	93.9	↓-5.55
	Round Robin	01/09	101.6	04/09	101.11	105	↓-3.89	15/09	101.69	101.9	↑+0.21	31/10	99.16	103.9	↓-4.74
		16/10	89.8	19/10	89.17	88.1	↑+1.07	30/10	92.83	102.7	↑+9.87	15/12	88.84	93.8	↓-4.96
		21/10	93	24/10	92.29	97.7	↓-5.41	04/11	86.98	103.3	↓-16.32	20/12	93.11	93.9	↑+0.79
		22/10	97	25/10	95.73	97.7	↓-1.97	05/11	98.09	100.9	↑+2.81	21/12	100.16	93.9	↓-6.26
	Ensemble	01/09	101.6	04/09	100.9	105	↓-4.10	15/09	100.37	101.9	↓-1.53	31/10	102.29	103.9	↑+1.61
		16/10	89.8	19/10	87.61	88.1	↑+0.49	30/10	101.52	102.7	↑+1.18	15/12	87.69	93.8	↓-6.11
		21/10	93	24/10	91.74	97.7	↓-5.96	04/11	83.49	103.3	↓-19.81	20/12	94.86	93.9	↑+0.96
		22/10	97	25/10	96.38	97.7	↓-1.32	05/11	95.53	100.9	↓-5.37	21/11	97.69	93.9	↓-3.79
VCB	Hierarchical	30/08	68.6	02/09	68.25	68.6	0.35	13/09	67.85	65.8	↑+2.05	29/10	68.02	60.7	↑+7.32
		16/10	62.9	19/10	65.3	61.9	↓-3.40	30/10	60.09	60.6	↑+0.51	15/12	73.78	56.8	↓-16.98
		21/10	59.3	24/10	59.4	59.5	↑+0.10	04/11	60.6	60.1	↑+0.50	20/12	60.64	57.5	↓-3.14
		22/10	59.6	25/10	59.22	59.5	↑+0.28	05/11	59.5	60.8	↓-1.30	21/12	58.0	57.5	↑+0.50
	Round Robin	30/08	68.6	02/09	68.05	68.6	0.55	13/09	69.56	65.8	↓-3.76	29/10	64.52	60.7	↑+3.82
		16/10	62.9	19/10	62.04	61.9	↑+0.14	30/10	58.8	60.6	↑+1.80	15/12	63.9	56.8	↓-7.10
		21/10	59.3	24/10	59.44	59.5	↑+0.06	04/11	57.98	60.1	↓-2.12	20/12	60.96	57.5	↓-3.46
		22/10	59.6	25/10	60.66	59.5	↓-1.16	05/11	58.22	60.8	↓-2.58	21/12	59.7	57.5	↓-2.20
	Ensemble	30/08	68.6	02/09	69.67	68.6	1.07	13/09	71.32	65.8	↓-5.52	29/10	59.23	60.7	↑+1.47
		16/10	62.9	19/10	63.9	61.9	↓-2.00	30/10	63.98	60.6	↓-3.38	15/12	55.02	56.8	↑+1.78
		21/10	59.3	24/10	58.71	59.5	↓-0.79	04/11	57.65	60.1	↓-2.45	20/12	57.91	57.5	↑+0.41
		22/10	59.6	25/10	59.15	59.5	↑+0.35	05/11	59.87	60.8	↑+0.93	21/12	63.2	57.5	↓-5.70
VCS	Hierarchical	30/08	48.8	02/09	47.65	48.8	1.15	13/09	49.17	50	↑+0.83	29/10	49.86	47.6	↓-2.26
		16/10	47.2	19/10	46.45	46.8	↑+0.35	30/10	45.08	47.4	↓-2.32	15/12	46.58	46.5	↑+0.08
		21/10	45.5	24/10	44.01	46.2	↓-2.19	04/11	42.73	47.1	↓-4.37	20/12	44.76	43	↑+1.76
		22/10	45.5	25/10	44.54	46.2	↓-1.66	05/11	44.73	46.9	↓-2.17	21/12	46.2	43	↓-3.20
	Round Robin	30/08	48.8	02/09	49.15	48.8	0.35	13/09	49.82	50	↑+0.18	29/10	49.43	47.6	↓-1.83
		16/10	47.2	19/10	45.8	46.8	↑+1.00	30/10	45.74	47.4	↓-1.66	15/12	49.21	46.5	↓-2.71
		21/10	45.5	24/10	44.65	46.2	↓-1.55	04/11	43.05	47.1	↓-4.05	20/12	44.68	43	↑+1.68
		22/10	45.5	25/10	44.86	46.2	↓-1.34	05/11	44.6	46.9	↓-2.30	21/12	44.05	43	↑+1.05
	Ensemble	30/08	48.8	02/09	48.87	48.8	0.07	13/09	46.8	50	↓-3.20	29/10	46.44	47.6	↑+1.16
		16/10	47.2	19/10	47.13	46.8	↑+0.33	30/10	48.09	47.4	↑+0.69	15/12	48.51	46.5	↓-2.01
		21/10	45.5	24/10	44.87	46.2	↓-1.33	04/11	48.37	47.1	↑+1.27	20/12	45.37	43	↑+2.37
		22/10	45.5	25/10	47.69	46.2	↑+1.49	05/11	47.88	46.9	↑+0.98	21/12	43.33	43	↑+0.33
HPG	Hierarchical	01/09	27.5	04/09	27.2	29.85	↓-2.65	15/09	24.0	30.35	↓-6.35	31/10	26.64	26.7	↑+0.06
		16/10	28.25	19/10	27.85	28	↑+0.15	30/10	28.86	26.9	↓-1.96	15/12	31.1	26.25	↓-4.85
		21/10	26.8	24/10	28.1	26.4	↓-1.70	04/11	26.66	26.75	↑+0.09	20/12	27.1	26.7	↓-0.40
		22/10	26.7	25/10	27.47	26.4	↓-1.07	05/11	26.88	26.3	↓-0.58	21/12	26.0	26.7	0.70
	Round Robin	01/09	27.5	04/09	27.87	29.85	↑+1.98	15/09	28.17	30.35	↑+2.18	31/10	25.52	26.7	↑+1.18
		16/10	28.25	19/10	27.33	28	↑+0.67	30/10	27.92	26.9	↑+1.02	15/12	29.51	26.25	↓-3.26
		21/10	26.8	24/10	25.94	26.4	↑+0.46	04/11	26.69	26.75	↑+0.06	20/12	25.96	26.7	↑+0.74
		22/10	26.7	25/10	27.67	26.4	↓-1.27	05/11	28.31	26.3	↓-2.01	21/12	25.72	26.7	0.98
	Ensemble	01/09	27.5	04/09	27.22	29.85	↓-2.63	15/09	27.71	30.35	↑+2.64	31/10	26.57	26.7	↑+0.13
		16/10	28.25	19/10	26.5	28	↑+1.50	30/10	32.05	26.9	↓-5.15	15/12	26.11	26.25	↑+0.14
		21/10	26.8	24/10	26.66	26.4	↑+0.26	04/11	25.57	26.75	↑+1.18	20/12	26.91	26.7	↓-0.21
		22/10	26.7	25/10	26.5	26.4	↑+0.10	05/11	26.51	26.3	↑+0.21	21/12	26.29	26.7	0.41
	Ensemble	01/09	39.6	04/09	40.52	39.6	0.92	15/09	39.85	39	↓-0.85	31/10	42.56	35.1	↓-7.46
		16/10	41.25	19/10	41.04	40.65	↑+0.39	30/10	35.29	35.7	↑+0.41	15/12	35.59	32	↑+3.59

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TABLE 12 – continued from previous page

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
		21/10	37.65	24/10	38.75	36.1	↓-2.65	04/11	36.46	35	↑+1.46	20/12	39.19	33.6	↓-5.59
		22/10	38	25/10	40.9	36.1	↓-4.80	05/11	38.8	34.1	↓-4.70	21/12	41.41	33.6	↓-7.81
	Hierarchical	01/09	39.6	04/09	38.94	39.6	0.66	15/09	38.79	39	↑+0.21	31/10	41.55	35.1	↓-6.45
		16/10	41.25	19/10	42.52	40.65	↓-1.87	30/10	46.76	35.7	↓-11.06	15/12	37.74	32	↑+5.74
		21/10	37.65	24/10	36.32	36.1	↑+0.22	04/11	38.53	35	↓-3.53	20/12	33.77	33.6	↑+0.17
		22/10	38	25/10	40.89	36.1	↓-4.79	05/11	41.33	34.1	↓-7.23	21/12	29.07	33.6	↑+4.53
	RoundRobin	01/09	39.6	04/09	39.47	39.6	0.13	15/09	38.39	39	↑+0.61	31/10	30.17	35.1	↑+4.93
		16/10	41.25	19/10	41.37	40.65	↓-0.72	30/10	41.63	35.7	↓-5.93	15/12	43.44	32	↓-11.44
		21/10	37.65	24/10	37.16	36.1	↑+1.06	04/11	35.69	35	↑+0.69	20/12	33.84	33.6	↑+0.24
		22/10	38	25/10	35.09	36.1	↑+1.01	05/11	35.07	34.1	↑+0.97	21/12	34.53	33.6	↑+0.93
	Hierarchical	01/09	104.5	04/09	105.61	102.3	↓-3.31	15/09	103.6	104.5	0.90	31/10	99.28	99.2	↑+0.08
		16/10	122	19/10	115.54	116	↑+0.46	30/10	133.52	104	↓-29.52	15/12	111.14	93.9	↑+17.24
		21/10	110.9	24/10	106.67	114.5	↓-7.83	04/11	101.61	100.2	↑+1.41	20/12	116.28	101.5	↓-14.78
		22/10	113	25/10	112.08	114.5	↓-2.42	05/11	121.61	99.6	↓-22.01	21/12	114.3	101.5	↓-12.80
	Round Robin	01/09	104.5	04/09	110.78	102.3	↓-8.48	15/09	104.6	104.5	0.10	31/10	109.81	99.2	↓-10.61
VHM		16/10	122	19/10	127.83	116	↓-11.83	30/10	129.7	104	↓-25.70	15/12	159.5	93.9	↓-65.60
		21/10	110.9	24/10	104.09	114.5	↓-10.41	04/11	113.96	100.2	↓-13.76	20/12	109.0	101.5	↑+7.50
		22/10	113	25/10	115.57	114.5	↑+1.07	05/11	107.31	99.6	↑+7.71	21/12	115.97	101.5	↓-14.47
	Ensemble	01/09	104.5	04/09	104.42	102.3	↑+2.12	15/09	105.57	104.5	1.07	31/10	102.47	99.2	↑+3.27
		16/10	122	19/10	117.56	116	↑+1.56	30/10	146.99	104	↓-42.99	15/12	105.08	93.9	↑+11.18
		21/10	110.9	24/10	113.67	114.5	↑+0.83	04/11	126.98	100.2	↓-26.78	20/12	101.01	101.5	↑+0.49
		22/10	113	25/10	112.04	114.5	↓-2.46	05/11	106.06	99.6	↑+6.46	21/12	114.13	101.5	↓-12.63
	Hierarchical	01/09	78	04/09	77.02	77.5	↑+0.48	15/09	78.69	79.5	↑+0.81	31/10	74.32	82.6	↓-8.28
		16/10	84.5	19/10	85.02	84.5	0.52	30/10	85.2	83.9	↓-1.30	15/12	81.73	77.7	↑+4.03
		21/10	83	24/10	82.3	85.7	↓-3.40	04/11	86.72	81.8	↓-4.92	20/12	81.64	82.9	↑+1.26
		22/10	84.5	25/10	83.65	85.7	↓-2.05	05/11	82.19	80.2	↑+1.99	21/12	86.22	82.9	↓-3.32
	Round Robin	01/09	78	04/09	77.29	77.5	↑+0.21	15/09	77.6	79.5	↓-1.90	31/10	82.25	82.6	↑+0.35
MWG		16/10	84.5	19/10	82.85	84.5	1.65	30/10	83.08	83.9	↑+0.82	15/12	84.71	77.7	↓-7.01
		21/10	83	24/10	84.54	85.7	↑+1.16	04/11	82.54	81.8	↑+0.74	20/12	82.47	82.9	↑+0.43
		22/10	84.5	25/10	83.47	85.7	↓-2.23	05/11	90.21	80.2	↓-10.01	21/12	85.87	82.9	↓-2.97
	Ensemble	01/09	78	04/09	76.62	77.5	↑+0.88	15/09	77.8	79.5	↓-1.70	31/10	78.74	82.6	↑+3.86
		16/10	84.5	19/10	86.34	84.5	1.84	30/10	84.16	83.9	↑+0.26	15/12	89.41	77.7	↓-11.71
		21/10	83	24/10	79.51	85.7	↓-6.19	04/11	81.62	81.8	↑+0.18	20/12	83.22	82.9	↓-0.32
		22/10	84.5	25/10	85.25	85.7	↑+0.45	05/11	82.69	80.2	↑+2.49	21/12	85.48	82.9	↓-2.58
	Hierarchical	01/09	40.3	04/09	44.35	41.85	↑+2.50	15/09	41.55	40.8	↑+0.75	31/10	40.47	41.9	↑+1.43
		16/10	38	19/10	36.28	37.3	↑+1.02	30/10	37.52	41.3	↓-3.78	15/12	38.29	35.1	↓-3.19
		21/10	38.4	24/10	38.52	38.75	↑+0.23	04/11	39.13	39.6	↑+0.47	20/12	39.24	35.2	↓-4.04
		22/10	38.5	25/10	38.39	38.75	↓-0.36	05/11	37.81	39.3	↓-1.49	21/12	37.19	35.2	↑+1.99
	Round Robin	01/09	40.3	04/09	41.31	41.85	↑+0.54	15/09	42.04	40.8	↑+1.24	31/10	39.15	41.9	↓-2.75
CMG		16/10	38	19/10	39.34	37.3	↓-2.04	30/10	39.45	41.3	↑+1.85	15/12	40.82	35.1	↓-5.72
		21/10	38.4	24/10	37.67	38.75	↓-1.08	04/11	36.42	39.6	↓-3.18	20/12	36.43	35.2	↑+1.23
		22/10	38.5	25/10	37.87	38.75	↓-0.88	05/11	39.24	39.3	↑+0.06	21/12	39.08	35.2	↓-3.88
	Ensemble	01/09	40.3	04/09	40.65	41.85	↑+1.20	15/09	40.38	40.8	↑+0.42	31/10	42.34	41.9	↑+0.44
		16/10	38	19/10	37.44	37.3	↑+0.14	30/10	35.86	41.3	↓-5.44	15/12	40.34	35.1	↓-5.24
		21/10	38.4	24/10	39.9	38.75	↑+1.15	04/11	38.51	39.6	↑+1.09	20/12	40.71	35.2	↓-5.51
		22/10	38.5	25/10	37.68	38.75	↓-1.07	05/11	39.97	39.3	↑+0.67	21/12	31.8	35.2	↑+3.40
	Hierarchical	01/09	36.5	04/09	37.7	36.75	↑+0.95	15/09	36.36	34.85	↑+1.51	31/10	37.09	35.85	↓-1.24
		16/10	34.2	19/10	33.25	33.9	↑+0.65	30/10	31.98	35.85	↓-3.87	15/12	38.05	29.6	↓-8.45
		21/10	31.75	24/10	32.24	33.8	↑+1.56	04/11	27.13	35.6	↓-8.47	20/12	32.43	32.65	↑+0.22
		22/10	32.2	25/10	31.55	33.8	↓-2.25	05/11	31.6	34.95	↓-3.35	21/12	32.57	32.65	↑+0.08
	Round Robin	01/09	36.5	04/09	37.8	36.75	↑+1.05	15/09	36.86	34.85	↓-2.01	31/10	34.07	35.85	↑+1.78
KDH		16/10	34.2	19/10	35.28	33.9	↓-1.38	30/10	33.55	35.85	↓-2.30	15/12	27.9	29.6	↑+1.70
		21/10	31.75	24/10	32.17	33.8	↑+1.63	04/11	35.91	35.6	↑+0.31	20/12	32.1	32.65	↑+0.55
		22/10	32.2	25/10	30.05	33.8	↓-3.75	05/11	31.91	34.95	↓-3.04	21/12	32.01	32.65	↓-0.64
	Ensemble	01/09	36.5	04/09	36.65	36.75	↑+0.10	15/09	35.04	34.85	↑+0.19	31/10	34.99	35.85	↑+0.86
		16/10	34.2	19/10	34.65	33.9	↓-0.75	30/10	33.31	35.85	↓-2.54	15/12	35.06	29.6	↓-5.46
		21/10	31.75	24/10	29.22	33.8	↓-4.58	04/11	34.8	35.6	↑+0.80	20/12	30.17	32.65	↓-2.48
		22/10	32.2	25/10	30.93	33.8	↓-2.87	05/11	33.01	34.95	↑+1.94	21/12	32.01	32.65	↓-0.64
	Hierarchical	01/09	98.2	04/09	98.0	99.3	↓-1.30	15/09	97.04	99.5	↓-2.46	31/10	98.34	96	↓-2.34
		16/10	95	19/10	91.81	93.2	↑+1.39	30/10	92.73	93.6	↑+0.87	15/12	91.99	93	↑+1.01
		21/10	89.2	24/10	89.33	92.4	↑+3.07	04/11	83.28	94.6	↓-11.32	20/12	93.05	70.2	↓-22.85
		22/10	91	25/10	90.15	92.4	↓-2.25	05/11	93.49	95	↑+1.51	21/12	69.19	70.2	↑+1.01
	Round Robin	01/09	98.2	04/09	98.94	99.3	↑+0.36	15/09	99.99	99.5	↑+0.49	31/10	98.0	96	↑+2.00
DGC		16/10	95	19/10	92.84	93.2	↑+0.36	30/10	97.48	93.6	↓-3.88	15/12	95.66	93	↓-2.66

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TABLE 12 – continued from previous page

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
		21/10	89.2	24/10	91.04	92.4	↑+1.36	04/11	91.42	94.6	↑+3.18	20/12	103.07	70.2	↓-32.87
		22/10	91	25/10	89.33	92.4	↓-3.07	05/11	93.76	95	↑+1.24	21/12	109.47	70.2	↓-39.27
	Ensemble	01/09	98.2	04/09	99.63	99.3	↑+0.33	15/09	94.65	99.5	↓-4.85	31/10	96.53	96	↑+0.53
		16/10	95	19/10	95.47	93.2	↓-2.27	30/10	93.65	93.6	↑+0.05	15/12	94.31	93	↑+1.31
		21/10	89.2	24/10	85.62	92.4	↓-6.78	04/11	89.57	94.6	↑+5.03	20/12	75.26	70.2	↑+5.06
		22/10	91	25/10	88.55	92.4	↓-3.85	05/11	87.89	95	↓-7.11	21/12	96.04	70.2	↓-25.84
MBB	**Hierarchical**	01/09	27.75	04/09	27.89	28.25	↑+0.36	15/09	28.53	26.85	↓-1.68	31/10	26.07	23.6	↑+2.47
		16/10	27.2	19/10	26.73	27.1	↑+0.37	30/10	27.4	23.95	↓-3.45	15/12	30.05	23.75	↓-6.30
		21/10	25.3	24/10	24.53	24.4	↑+0.13	04/11	24.2	24	↑+0.20	20/12	27.13	24.7	↓-2.43
		22/10	25.45	25/10	26.84	24.4	↓-2.44	05/11	25.75	23.9	↓-1.85	21/12	26.36	24.7	↓-1.66
	Round Robin	01/09	27.75	04/09	30.2	28.25	↑+1.95	15/09	28.28	26.85	↓-1.43	31/10	28.32	23.6	↓-4.72
		16/10	27.2	19/10	28.02	27.1	↓-0.92	30/10	31.26	23.95	↓-7.31	15/12	25.73	23.75	↑+1.98
		21/10	25.3	24/10	23.18	24.4	↑+1.22	04/11	25.65	24	↓-1.65	20/12	24.79	24.7	↑+0.09
		22/10	25.45	25/10	26.52	24.4	↓-2.12	05/11	24.45	23.9	↑+0.55	21/12	23.77	24.7	↑+0.93
	Ensemble	01/09	27.75	04/09	27.9	28.25	↑+0.35	15/09	29.49	26.85	↓-2.64	31/10	35.09	23.6	↓-11.49
		16/10	27.2	19/10	27.98	27.1	↓-0.88	30/10	26.3	23.95	↑+2.35	15/12	21.68	23.75	↑+2.07
		21/10	25.3	24/10	24.25	24.4	↑+0.15	04/11	24.99	24	↑+0.99	20/12	27.35	24.7	↓-2.65
		22/10	25.45	25/10	25.36	24.4	↑+0.96	05/11	24.62	23.9	↑+0.72	21/12	26.54	24.7	↓-1.84

TABLE 13: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (LLM: Llama, Transformer Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	**Hierarchical**	31/08	22.8	03/09	23.64	24	↑+0.36	14/09	21.06	24.05	↓-2.99	30/10	23.76	21.2	↓-2.56
		16/10	22.45	19/10	23.57	22.6	↑+0.97	30/10	24.1	21.2	↓-2.90	15/12	26.53	16.3	↓-10.23
		21/10	20.0	24/10	19.44	20.9	↓-1.46	04/11	21.26	20.15	↑+1.11	20/12	20.78	17.8	↓-2.98
		22/10	20.5	25/10	21.4	20.9	↑+0.50	05/11	21.27	20.1	↓-1.17	21/12	17.66	17.8	↑+0.14
	Round Robin	31/08	22.8	03/09	21.77	24	↓-2.23	14/09	22.45	24.05	↓-1.60	30/10	22.71	21.2	↑+1.51
		16/10	22.45	19/10	22.18	22.6	↓-0.42	30/10	22.12	21.2	↑+0.92	15/12	16.55	16.3	↑+0.25
		21/10	20.0	24/10	20.79	20.9	↑+0.11	04/11	21.63	20.15	↑+1.48	20/12	19.53	17.8	↑+1.73
		22/10	20.5	25/10	19.99	20.9	↓-0.91	05/11	20.04	20.1	↑+0.06	21/12	22.34	17.8	↓-4.54
	Ensemble	31/08	22.8	03/09	25.26	24	↑+1.26	14/09	22.07	24.05	↓-1.98	30/10	24.53	21.2	↓-3.33
		16/10	22.45	19/10	23.47	22.6	↑+0.87	30/10	21.63	21.2	↑+0.43	15/12	16.87	16.3	↑+0.57
		21/10	20.0	24/10	20.34	20.9	↑+0.56	04/11	20.97	20.15	↑+0.82	20/12	20.93	17.8	↓-3.13
		22/10	20.5	25/10	20.43	20.9	↓-0.47	05/11	19.22	20.1	↑+0.88	21/12	20.95	17.8	↓-3.15
FPT	**Hierarchical**	01/09	101.6	04/09	103.42	105	↑+1.58	15/09	102.82	101.9	↑+0.92	31/10	101.14	103.9	↓-2.76
		16/10	89.8	19/10	89.22	88.1	↑+1.12	30/10	91.62	102.7	↑+11.08	15/12	92.18	93.8	↑+1.62
		21/10	93	24/10	90.92	97.7	↓-6.78	04/11	102.65	103.3	↑+0.65	20/12	94.2	93.9	↑+0.30
		22/10	97	25/10	97.44	97.7	↑+0.26	05/11	93.93	100.9	↓-6.97	21/12	100.24	93.9	↓-6.34
	Round Robin	01/09	101.6	04/09	101.65	105	↑+3.35	15/09	101.39	101.9	↓-0.51	31/10	103.54	103.9	↑+0.36
		16/10	89.8	19/10	88.96	88.1	↑+0.86	30/10	83.64	102.7	↓-19.06	15/12	90.52	93.8	↑+3.28
		21/10	93	24/10	90.8	97.7	↓-6.90	04/11	94.23	103.3	↑+9.07	20/12	92.91	93.9	↓-0.99
		22/10	97	25/10	97.9	97.7	↑+0.20	05/11	94.45	100.9	↓-6.45	21/12	96.67	93.9	↑+2.77
	Ensemble	01/09	101.6	04/09	98.49	105	↓-6.51	15/09	100.94	101.9	↓-0.96	31/10	100.2	103.9	↓-3.70
		16/10	89.8	19/10	87.63	88.1	↑+0.47	30/10	95.87	102.7	↑+6.83	15/12	90.66	93.8	↑+3.14
		21/10	93	24/10	91.67	97.7	↓-6.03	04/11	82.46	103.3	↓-20.84	20/12	93.35	93.9	↑+0.55
		22/10	97	25/10	98.2	97.7	↑+0.50	05/11	100.37	100.9	↑+0.53	21/11	99.31	93.9	↓-5.41
VCB	**Hierarchical**	30/08	68.6	02/09	69.61	68.6	1.01	13/09	70.49	65.8	↓-4.69	29/10	76.5	60.7	↓-15.80
		16/10	62.9	19/10	64.6	61.9	↓-2.70	30/10	64.26	60.6	↓-3.66	15/12	56.59	56.8	↑+0.21
		21/10	59.3	24/10	59.13	59.5	↓-0.37	04/11	60.01	60.1	↑+0.09	20/12	56.92	57.5	↑+0.58
		22/10	59.6	25/10	58.67	59.5	↑+0.83	05/11	58.62	60.8	↓-2.18	21/12	59.32	57.5	↑+1.82
	Round Robin	30/08	68.6	02/09	68.25	68.6	0.35	13/09	71.33	65.8	↓-5.53	29/10	74.21	60.7	↓-13.51
		16/10	62.9	19/10	63.06	61.9	↓-1.16	30/10	61.08	60.6	↑+0.48	15/12	65.22	56.8	↓-8.42
		21/10	59.3	24/10	60.34	59.5	↑+0.84	04/11	60.7	60.1	↑+0.60	20/12	58.02	57.5	↑+0.52
		22/10	59.6	25/10	60.44	59.5	↓-0.94	05/11	61.02	60.8	↑+0.22	21/12	61.16	57.5	↓-3.66
	Ensemble	30/08	68.6	02/09	70.09	68.6	1.49	13/09	67.52	65.8	↑+1.72	29/10	64.14	60.7	↑+3.44
		16/10	62.9	19/10	61.82	61.9	↑+0.08	30/10	65.17	60.6	↓-4.57	15/12	65.72	56.8	↓-8.92
		21/10	59.3	24/10	59.78	59.5	↑+0.28	04/11	58.25	60.1	↓-1.85	20/12	55.53	57.5	↑+1.97
		22/10	59.6	25/10	60.22	59.5	↓-0.72	05/11	58.4	60.8	↓-2.40	21/12	59.02	57.5	↑+1.52
VCS	**Hierarchical**	30/08	48.8	02/09	47.69	48.8	1.11	13/09	49.45	50	↑+0.55	29/10	50.47	47.6	↓-2.87
		16/10	47.2	19/10	46.56	46.8	↑+0.24	30/10	45.96	47.4	↓-1.44	15/12	46.36	46.5	↑+0.14
		21/10	45.5	24/10	44.33	46.2	↓-1.87	04/11	44.86	47.1	↓-2.24	20/12	43.56	43	↑+0.56
		22/10	45.5	25/10	47.32	46.2	↑+1.12	05/11	43.82	46.9	↓-3.08	21/12	48.23	43	↓-5.23
	Round Robin	30/08	48.8	02/09	48.21	48.8	0.59	13/09	49.29	50	↑+0.71	29/10	48.72	47.6	↑+1.12
		16/10	47.2	19/10	46.3	46.8	↑+0.50	30/10	46.77	47.4	↓-0.63	15/12	46.73	46.5	↑+0.23
		21/10	45.5	24/10	43.94	46.2	↓-2.26	04/11	45.38	47.1	↓-1.72	20/12	43.26	43	↑+0.26
		22/10	45.5	25/10	45.78	46.2	↑+0.42	05/11	43.01	46.9	↓-3.89	21/12	47.73	43	↓-4.73
	Ensemble	30/08	48.8	02/09	49.98	48.8	1.18	13/09	46.68	50	↓-3.32	29/10	48.22	47.6	↑+0.62
		16/10	47.2	19/10	45.52	46.8	↑+1.28	30/10	48.23	47.4	↑+0.83	15/12	47.75	46.5	↓-1.25
		21/10	45.5	24/10	44.42	46.2	↓-1.78	04/11	47.16	47.1	↑+0.06	20/12	44.45	43	↑+1.45
		22/10	45.5	25/10	46.3	46.2	↑+0.10	05/11	43.9	46.9	↓-3.00	21/12	44.91	43	↑+1.91
HPG	**Hierarchical**	01/09	27.5	04/09	25.39	29.85	↓-4.46	15/09	24.37	30.35	↓-5.98	31/10	26.56	26.7	↑+0.14
		16/10	28.25	19/10	28.12	28	↑+0.12	30/10	28.03	26.9	↑+1.13	15/12	27.91	26.25	↑+1.66
		21/10	26.8	24/10	27.09	26.4	↓-0.69	04/11	26.37	26.75	↑+0.38	20/12	27.25	26.7	↓-0.55
		22/10	26.7	25/10	28.31	26.4	↓-1.91	05/11	24.87	26.3	↑+1.43	21/12	27.39	26.7	0.69
	Round Robin	01/09	27.5	04/09	27.98	29.85	↑+1.87	15/09	28.77	30.35	↑+1.58	31/10	27.74	26.7	↓-1.04
		16/10	28.25	19/10	28.68	28	↓-0.68	30/10	26.28	26.9	↑+0.62	15/12	27.7	26.25	↑+1.45
		21/10	26.8	24/10	28.11	26.4	↓-1.71	04/11	27.01	26.75	↓-0.26	20/12	25.46	26.7	↑+1.24
		22/10	26.7	25/10	27.01	26.4	↓-0.61	05/11	27.11	26.3	↓-0.81	21/12	26.15	26.7	0.55
	Ensemble	01/09	27.5	04/09	28.0	29.85	↑+1.85	15/09	25.13	30.35	↓-5.22	31/10	26.6	26.7	↑+0.10
		16/10	28.25	19/10	27.56	28	↑+0.44	30/10	29.63	26.9	↓-2.73	15/12	28.89	26.25	↓-2.64
		21/10	26.8	24/10	27.38	26.4	↓-0.98	04/11	28.08	26.75	↓-1.33	20/12	27.73	26.7	↓-1.03
		22/10	26.7	25/10	25.55	26.4	↑+0.85	05/11	25.71	26.3	↑+0.59	21/12	26.86	26.7	0.16
	Ensemble	01/09	39.6	04/09	38.3	39.6	1.30	15/09	37.17	39	↑+1.83	31/10	42.3	35.1	↓-7.20
		16/10	41.25	19/10	40.02	40.65	↑+0.63	30/10	41.73	35.7	↓-6.03	15/12	43.06	32	↓-11.06

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TABLE 13 – continued from previous page

Stock	Architecture	Start Date	Start Price	Short-term End Date	Short-term Pred	Short-term Act	Short-term Gap	Medium-term End Date	Medium-term Pred	Medium-term Act	Medium-term Gap	Long-term End Date	Long-term Pred	Long-term Act	Long-term Gap
		21/10	37.65	24/10	38.67	36.1	↓-2.57	04/11	35.72	35	↑+0.72	20/12	36.9	33.6	↑+3.30
		22/10	38	25/10	39.99	36.1	↓-3.89	05/11	33.09	34.1	↑+1.01	21/12	40.75	33.6	↓-7.15
	Hierarchical	01/09	39.6	04/09	38.68	39.6	0.92	15/09	39.72	39	↓-0.72	31/10	42.55	35.1	↓-7.45
		16/10	41.25	19/10	41.13	40.65	↑+0.48	30/10	37.49	35.7	↑+1.79	15/12	50.84	32	↓-18.84
		21/10	37.65	24/10	35.61	36.1	↑+0.49	04/11	36.21	35	↑+1.21	20/12	35.67	33.6	↑+2.07
		22/10	38	25/10	40.05	36.1	↓-3.95	05/11	35.82	34.1	↑+1.72	21/12	42.05	33.6	↓-8.45
	RoundRobin	01/09	39.6	04/09	40.57	39.6	0.97	15/09	40.81	39	↓-1.81	31/10	37.6	35.1	↑+2.50
		16/10	41.25	19/10	41.7	40.65	↓-1.05	30/10	43.55	35.7	↓-7.85	15/12	50.39	32	↓-18.39
		21/10	37.65	24/10	37.52	36.1	↑+1.42	04/11	39.51	35	↓-4.51	20/12	39.1	33.6	↓-5.50
		22/10	38	25/10	37.57	36.1	↑+1.47	05/11	35.11	34.1	↑+1.01	21/12	32.63	33.6	↑+0.97
VHM	**Hierarchical**	01/09	104.5	04/09	102.1	102.3	↑+0.20	15/09	104.26	104.5	0.24	31/10	101.31	99.2	↑+2.11
		16/10	122	19/10	121.05	116	↑+5.05	30/10	133.99	104	↓-29.99	15/12	114.42	93.9	↑+20.52
		21/10	110.9	24/10	110.41	114.5	↓-4.09	04/11	105.07	100.2	↑+4.87	20/12	102.86	101.5	↑+1.36
		22/10	113	25/10	113.74	114.5	↑+0.76	05/11	115.12	99.6	↓-15.52	21/12	116.11	101.5	↓-14.61
	Round Robin	01/09	104.5	04/09	105.86	102.3	↓-3.56	15/09	103.65	104.5	0.85	31/10	107.88	99.2	↓-8.68
		16/10	122	19/10	127.05	116	↓-11.05	30/10	128.94	104	↓-24.94	15/12	106.58	93.9	↑+12.68
		21/10	110.9	24/10	110.18	114.5	↓-4.32	04/11	108.69	100.2	↑+8.49	20/12	108.94	101.5	↑+7.44
		22/10	113	25/10	116.1	114.5	↑+1.60	05/11	122.83	99.6	↓-23.23	21/12	117.16	101.5	↓-15.66
	Ensemble	01/09	104.5	04/09	105.62	102.3	↓-3.32	15/09	105.51	104.5	1.01	31/10	105.58	99.2	↓-6.38
		16/10	122	19/10	123.97	116	↓-7.97	30/10	124.45	104	↓-20.45	15/12	103.94	93.9	↑+10.04
		21/10	110.9	24/10	111.98	114.5	↑+2.52	04/11	106.74	100.2	↑+6.54	20/12	105.24	101.5	↑+3.74
		22/10	113	25/10	113.42	114.5	↑+1.08	05/11	124.21	99.6	↓-24.61	21/12	106.51	101.5	↑+5.01
MWG	**Hierarchical**	01/09	78	04/09	78.93	77.5	↓-1.43	15/09	76.03	79.5	↓-3.47	31/10	74.05	82.6	↓-8.55
		16/10	84.5	19/10	85.45	84.5	0.95	30/10	84.3	83.9	↑+0.40	15/12	83.4	77.7	↑+5.70
		21/10	83	24/10	82.0	85.7	↓-3.70	04/11	83.48	81.8	↓-1.68	20/12	82.16	82.9	↑+0.74
		22/10	84.5	25/10	85.8	85.7	↑+0.10	05/11	80.59	80.2	↑+0.39	21/12	84.24	82.9	↑+1.34
	Round Robin	01/09	78	04/09	77.86	77.5	↑+0.36	15/09	77.22	79.5	↓-2.28	31/10	72.59	82.6	↓-10.01
		16/10	84.5	19/10	83.83	84.5	0.67	30/10	83.17	83.9	↑+0.73	15/12	83.46	77.7	↑+5.76
		21/10	83	24/10	84.77	85.7	↑+0.93	04/11	81.9	81.8	↑+0.10	20/12	82.02	82.9	↑+0.88
		22/10	84.5	25/10	83.51	85.7	↓-2.19	05/11	87.2	80.2	↓-7.00	21/12	83.27	82.9	↑+0.37
	Ensemble	01/09	78	04/09	77.26	77.5	↑+0.24	15/09	78.11	79.5	↑+1.39	31/10	77.73	82.6	↓-4.87
		16/10	84.5	19/10	85.28	84.5	0.78	30/10	84.1	83.9	↑+0.20	15/12	79.13	77.7	↑+1.43
		21/10	83	24/10	83.67	85.7	↑+2.03	04/11	81.47	81.8	↑+0.33	20/12	83.38	82.9	↓-0.48
		22/10	84.5	25/10	82.73	85.7	↓-2.97	05/11	84.57	80.2	↓-4.37	21/12	85.74	82.9	↓-2.84
CMG	**Hierarchical**	01/09	40.3	04/09	38.59	41.85	↓-3.26	15/09	39.86	40.8	↓-0.94	31/10	41.22	41.9	↑+0.68
		16/10	38	19/10	37.58	37.3	↑+0.28	30/10	39.28	41.3	↑+2.02	15/12	37.73	35.1	↑+2.63
		21/10	38.4	24/10	38.98	38.75	↑+0.23	04/11	40.15	39.6	↑+0.55	20/12	38.59	35.2	↓-3.39
		22/10	38.5	25/10	37.35	38.75	↓-1.40	05/11	37.2	39.3	↓-2.10	21/12	38.32	35.2	↑+3.12
	Round Robin	01/09	40.3	04/09	42.68	41.85	↑+0.83	15/09	39.96	40.8	↓-0.84	31/10	40.79	41.9	↑+1.11
		16/10	38	19/10	39.04	37.3	↓-1.74	30/10	35.47	41.3	↓-5.83	15/12	40.43	35.1	↓-5.33
		21/10	38.4	24/10	39.08	38.75	↑+0.33	04/11	38.21	39.6	↓-1.39	20/12	36.25	35.2	↑+1.05
		22/10	38.5	25/10	37.67	38.75	↓-1.08	05/11	38.59	39.3	↑+0.71	21/12	37.97	35.2	↑+2.77
	Ensemble	01/09	40.3	04/09	40.72	41.85	↑+1.13	15/09	41.63	40.8	↑+0.83	31/10	41.34	41.9	↑+0.56
		16/10	38	19/10	39.45	37.3	↓-2.15	30/10	38.96	41.3	↑+2.34	15/12	39.36	35.1	↓-4.26
		21/10	38.4	24/10	38.63	38.75	↑+0.12	04/11	37.09	39.6	↓-2.51	20/12	41.25	35.2	↓-6.05
		22/10	38.5	25/10	39.76	38.75	↑+1.01	05/11	37.6	39.3	↓-1.70	21/12	38.65	35.2	↓-3.45
KDH	**Hierarchical**	01/09	36.5	04/09	36.87	36.75	↑+0.12	15/09	33.9	34.85	↑+0.95	31/10	38.05	35.85	↓-2.20
		16/10	34.2	19/10	34.61	33.9	↓-0.71	30/10	35.8	35.85	↑+0.05	15/12	30.76	29.6	↑+1.16
		21/10	31.75	24/10	30.55	33.8	↓-3.25	04/11	27.85	35.6	↓-7.75	20/12	31.58	32.65	↓-1.07
		22/10	32.2	25/10	30.35	33.8	↓-3.45	05/11	34.53	34.95	↑+0.42	21/12	31.57	32.65	↓-1.08
	Round Robin	01/09	36.5	04/09	35.73	36.75	↓-1.02	15/09	37.4	34.85	↓-2.55	31/10	37.89	35.85	↓-2.04
		16/10	34.2	19/10	33.21	33.9	↑+0.69	30/10	33.87	35.85	↓-1.98	15/12	30.14	29.6	↑+0.54
		21/10	31.75	24/10	30.46	33.8	↓-3.34	04/11	27.9	35.6	↓-7.70	20/12	30.59	32.65	↓-2.06
		22/10	32.2	25/10	30.73	33.8	↓-3.07	05/11	32.66	34.95	↑+2.29	21/12	30.94	32.65	↓-1.71
	Ensemble	01/09	36.5	04/09	36.84	36.75	↑+0.09	15/09	37.45	34.85	↓-2.60	31/10	37.07	35.85	↓-1.22
		16/10	34.2	19/10	35.93	33.9	↓-2.03	30/10	31.26	35.85	↓-4.59	15/12	29.37	29.6	↑+0.23
		21/10	31.75	24/10	34.55	33.8	↑+0.75	04/11	32.0	35.6	↑+3.60	20/12	31.05	32.65	↓-1.60
		22/10	32.2	25/10	32.82	33.8	↑+0.98	05/11	33.4	34.95	↑+1.55	21/12	29.86	32.65	↓-2.79
DGC	**Hierarchical**	01/09	98.2	04/09	98.4	99.3	↑+0.90	15/09	100.23	99.5	↑+0.73	31/10	98.66	96	↓-2.66
		16/10	95	19/10	96.61	93.2	↓-3.41	30/10	93.89	93.6	↑+0.29	15/12	92.87	93	↑+0.13
		21/10	89.2	24/10	92.64	92.4	↑+0.24	04/11	87.83	94.6	↓-6.77	20/12	97.49	70.2	↓-27.29
		22/10	91	25/10	91.59	92.4	↑+0.81	05/11	92.7	95	↑+2.30	21/12	93.66	70.2	↓-23.46
	Round Robin	01/09	98.2	04/09	99.15	99.3	↑+0.15	15/09	97.04	99.5	↓-2.46	31/10	96.98	96	↑+0.98
		16/10	95	19/10	94.26	93.2	↑+1.06	30/10	95.48	93.6	↓-1.88	15/12	97.13	93	↓-4.13

Continued on next page

TABLE 13 – continued from previous page

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
		21/10	89.2	24/10	86.76	92.4	↓-5.64	04/11	90.6	94.6	↑+4.00	20/12	83.38	70.2	↑+13.18
		22/10	91	25/10	90.89	92.4	↓-1.51	05/11	93.18	95	↑+1.82	21/12	101.42	70.2	↓-31.22
	Ensemble	01/09	98.2	04/09	96.96	99.3	↓-2.34	15/09	99.2	99.5	↑+0.30	31/10	100.58	96	↓-4.58
		16/10	95	19/10	97.0	93.2	↓-3.80	30/10	94.51	93.6	↑+0.91	15/12	93.21	93	↑+0.21
		21/10	89.2	24/10	86.98	92.4	↓-5.42	04/11	89.57	94.6	↑+5.03	20/12	99.67	70.2	↓-29.47
		22/10	91	25/10	93.42	92.4	↑+1.02	05/11	89.5	95	↓-5.50	21/12	105.37	70.2	↓-35.17
MBB	**Hierarchical**	01/09	27.75	04/09	28.97	28.25	↑+0.72	15/09	26.97	26.85	↑+0.12	31/10	23.43	23.6	↑+0.17
		16/10	27.2	19/10	26.43	27.1	↑+0.67	30/10	28.56	23.95	↓-4.61	15/12	27.96	23.75	↓-4.21
		21/10	25.3	24/10	23.2	24.4	↑+1.20	04/11	23.24	24	↑+0.76	20/12	24.44	24.7	↑+0.26
		22/10	25.45	25/10	26.63	24.4	↓-2.23	05/11	23.2	23.9	↑+0.70	21/12	26.58	24.7	↓-1.88
	Round Robin	01/09	27.75	04/09	28.56	28.25	↑+0.31	15/09	28.7	26.85	↓-1.85	31/10	28.38	23.6	↓-4.78
		16/10	27.2	19/10	28.23	27.1	↓-1.13	30/10	25.34	23.95	↑+1.39	15/12	28.85	23.75	↓-5.10
		21/10	25.3	24/10	24.12	24.4	↑+0.28	04/11	26.26	24	↓-2.26	20/12	25.87	24.7	↓-1.17
		22/10	25.45	25/10	26.34	24.4	↓-1.94	05/11	25.85	23.9	↓-1.95	21/12	24.59	24.7	↑+0.11
	Ensemble	01/09	27.75	04/09	25.36	28.25	↓-2.89	15/09	27.43	26.85	↑+0.58	31/10	32.17	23.6	↓-8.57
		16/10	27.2	19/10	27.34	27.1	↓-0.24	30/10	30.23	23.95	↓-6.28	15/12	22.97	23.75	↑+0.78
		21/10	25.3	24/10	25.18	24.4	↑+0.78	04/11	25.04	24	↑+1.04	20/12	25.93	24.7	↓-1.23
		22/10	25.45	25/10	27.73	24.4	↓-3.33	05/11	24.1	23.9	↑+0.20	21/12	25.04	24.7	↑+0.34

The following tables provide the comprehensive short-term, medium-term, and long-term forecasts along with the calculated gaps across all evaluated stocks for each underlying foundation model.

TABLE 14: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (Gemini Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Baseline	31/08	22.8	03/09	19.66	24.00	↓-4.34	14/09	26.58	24.05	↑+2.53	30/10	19.44	21.20	↑+1.76
		16/10	22.45	19/10	24.90	22.60	↑+2.30	30/10	17.67	21.20	↑+3.53	15/12	19.59	16.30	↑+3.29
		21/10	20.0	24/10	17.19	20.90	↓-3.71	04/11	23.12	20.15	↑+2.97	20/12	21.13	17.80	↓-3.33
		22/10	20.5	25/10	23.08	20.90	↑+2.18	05/11	16.87	20.10	↑+3.23	21/12	21.22	17.80	↓-3.42
FPT	Baseline	01/09	101.6	04/09	90.02	105.00	↓-14.98	15/09	116.10	101.90	↑+14.20	31/10	118.85	103.90	↑+14.95
		16/10	89.8	19/10	104.49	88.10	↓-16.39	30/10	86.54	102.70	↓-16.16	15/12	113.80	93.80	↑+20.00
		21/10	93	24/10	86.85	97.70	↓-10.85	04/11	89.48	103.30	↓-13.82	20/12	80.35	93.90	↓-13.55
		22/10	97	25/10	105.66	97.70	↑+7.96	05/11	118.73	100.90	↑+17.83	21/12	113.37	93.90	↓-19.47
VCB	Baseline	30/08	68.6	02/09	74.74	68.60	↓-6.14	13/09	80.08	65.80	↓-14.28	29/10	55.04	60.70	↑+5.66
		16/10	62.9	19/10	51.34	61.90	↑+10.56	30/10	65.79	60.60	↓-5.19	15/12	50.75	56.80	↑+6.05
		21/10	59.3	24/10	52.59	59.50	↓-6.91	04/11	71.71	60.10	↑+11.61	20/12	69.44	57.50	↓-11.94
		22/10	59.6	25/10	69.68	59.50	↓-10.18	05/11	52.38	60.80	↓-8.42	21/12	47.77	57.50	↑+9.73
VCS	Baseline	30/08	48.8	02/09	43.43	48.80	↓-5.37	13/09	60.71	50.00	↑+10.71	29/10	40.58	47.60	↑+7.02
		16/10	47.2	19/10	55.76	46.80	↓-8.96	30/10	37.11	47.40	↓-10.29	15/12	55.26	46.50	↓-8.76
		21/10	45.5	24/10	41.34	46.20	↓-4.86	04/11	37.88	47.10	↓-9.22	20/12	37.83	43.00	↑+5.17
		22/10	45.5	25/10	42.48	46.20	↓-3.72	05/11	37.61	46.90	↓-9.29	21/12	47.14	43.00	↓-4.14
HPG	Baseline	01/09	27.5	04/09	25.29	29.85	↓-4.56	15/09	24.52	30.35	↓-5.83	31/10	20.94	26.70	↑+5.76
		16/10	28.25	19/10	25.28	28.00	↑+2.72	30/10	22.43	26.90	↑+4.47	15/12	23.08	26.25	↑+3.17
		21/10	26.8	24/10	21.22	26.40	↑+5.18	04/11	22.53	26.75	↑+4.22	20/12	32.25	26.70	↓-5.55
		22/10	26.7	25/10	23.04	26.40	↑+3.36	05/11	22.65	26.30	↑+3.65	21/12	31.25	26.70	↓-4.55
TCB	Baseline	01/09	39.6	04/09	43.81	39.60	↓-4.21	15/09	44.42	39.00	↓-5.42	31/10	42.56	35.10	↓-7.46
		16/10	41.25	19/10	35.24	40.65	↑+5.41	30/10	39.85	35.70	↑+4.15	15/12	35.34	32.00	↑+3.34
		21/10	37.65	24/10	43.31	36.10	↓-7.21	04/11	31.89	35.00	↑+3.11	20/12	38.54	33.60	↓-4.94
		22/10	38	25/10	30.25	36.10	↑+5.85	05/11	39.18	34.10	↓-5.08	21/12	29.64	33.60	↑+3.96
VHM	Baseline	01/09	104.5	04/09	92.87	102.30	↑+9.43	15/09	113.37	104.50	↓-8.87	31/10	78.59	99.20	↑+20.61
		16/10	122	19/10	140.94	116.00	↓-24.94	30/10	94.20	104.00	↑+9.80	15/12	104.11	93.90	↑+10.21
		21/10	110.9	24/10	94.85	114.50	↓-19.65	04/11	109.41	100.20	↑+9.21	20/12	85.35	101.50	↑+16.15
		22/10	113	25/10	131.42	114.50	↑+16.92	05/11	113.85	99.60	↓-14.25	21/12	91.06	101.50	↑+10.44
MWG	Baseline	01/09	78	04/09	93.56	77.50	↓-16.06	15/09	70.13	79.50	↓-9.37	31/10	64.88	82.60	↓-17.72
		16/10	84.5	19/10	68.91	84.50	↓-15.59	30/10	75.49	83.90	↑+8.41	15/12	66.05	77.70	↑+11.65
		21/10	83	24/10	93.06	85.70	↑+7.36	04/11	71.02	81.80	↑+10.78	20/12	72.98	82.90	↑+9.92
		22/10	84.5	25/10	99.08	85.70	↑+13.38	05/11	72.13	80.20	↑+8.07	21/12	101.11	82.90	↓-18.21
CMG	Baseline	01/09	40.3	04/09	49.70	41.85	↑+7.85	15/09	47.47	40.80	↑+6.67	31/10	51.11	41.90	↑+9.21
		16/10	38	19/10	29.81	37.30	↑+7.49	30/10	46.48	41.30	↑+5.18	15/12	29.06	35.10	↑+6.04
		21/10	38.4	24/10	45.00	38.75	↑+6.25	04/11	44.91	39.60	↑+5.31	20/12	31.60	35.20	↑+3.60
		22/10	38.5	25/10	44.46	38.75	↑+5.71	05/11	43.89	39.30	↑+4.59	21/12	40.77	35.20	↓-5.57
KDH	Baseline	01/09	36.5	04/09	30.68	36.75	↓-6.07	15/09	27.21	34.85	↑+7.64	31/10	29.97	35.85	↑+5.88
		16/10	34.2	19/10	40.84	33.90	↓-6.94	30/10	28.76	35.85	↓-7.09	15/12	32.05	29.60	↑+2.45
		21/10	31.75	24/10	40.46	33.80	↑+6.66	04/11	40.78	35.60	↑+5.18	20/12	35.41	32.65	↑+2.76
		22/10	32.2	25/10	39.60	33.80	↑+5.80	05/11	27.86	34.95	↓-7.09	21/12	28.46	32.65	↓-4.19
DGC	Baseline	01/09	98.2	04/09	116.27	99.30	↑+16.97	15/09	108.20	99.50	↑+8.70	31/10	107.62	96.00	↓-11.62
		16/10	95	19/10	103.39	93.20	↓-10.19	30/10	107.49	93.60	↓-13.89	15/12	78.38	93.00	↑+14.62
		21/10	89.2	24/10	112.49	92.40	↑+20.09	04/11	106.67	94.60	↑+12.07	20/12	61.95	70.20	↑+8.25
		22/10	91	25/10	78.48	92.40	↓-13.92	05/11	103.04	95.00	↑+8.04	21/12	56.27	70.20	↑+13.93
MBB	Baseline	01/09	27.75	04/09	34.20	28.25	↑+5.95	15/09	29.07	26.85	↓-2.22	31/10	26.96	23.60	↑+3.36
		16/10	27.2	19/10	24.05	27.10	↑+3.05	30/10	19.14	23.95	↑+4.81	15/12	28.65	23.75	↓-4.90
		21/10	25.3	24/10	21.04	24.40	↑+3.36	04/11	28.51	24.00	↓-4.51	20/12	21.88	24.70	↑+2.82
		22/10	25.45	25/10	19.96	24.40	↑+4.44	05/11	27.08	23.90	↓-3.18	21/12	27.34	24.70	↓-2.64

TABLE 15: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (GPT Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Baseline	31/08	22.8	03/09	19.29	24.00	↓-4.71	14/09	29.18	24.05	↑+5.13	30/10	23.82	21.20	↓-2.62
		16/10	22.45	19/10	26.41	22.60	↑+3.81	30/10	23.26	21.20	↓-2.06	15/12	18.72	16.30	↑+2.42
		21/10	20.0	24/10	24.59	20.90	↑+3.69	04/11	17.32	20.15	↓-2.83	20/12	20.40	17.80	↓-2.60
		22/10	20.5	25/10	23.42	20.90	↑+2.52	05/11	22.38	20.10	↓-2.28	21/12	15.24	17.80	↑+2.56
FPT	Baseline	01/09	101.6	04/09	116.97	105.00	↑+11.97	15/09	91.29	101.90	↓-10.61	31/10	119.12	103.90	↑+15.22
		16/10	89.8	19/10	68.91	88.10	↑+19.19	30/10	89.33	102.70	↓-13.37	15/12	108.53	93.80	↑+14.73
		21/10	93	24/10	77.82	97.70	↓-19.88	04/11	85.97	103.30	↓-17.33	20/12	105.86	93.90	↑+11.96
		22/10	97	25/10	89.30	97.70	↓-8.40	05/11	115.51	100.90	↑+14.61	21/12	106.04	93.90	↓-12.14
VCB	Baseline	30/08	68.6	02/09	58.17	68.60	↓-10.43	13/09	51.94	65.80	↑+13.86	29/10	72.76	60.70	↓-12.06
		16/10	62.9	19/10	70.09	61.90	↓-8.19	30/10	54.09	60.60	↑+6.51	15/12	48.66	56.80	↑+8.14
		21/10	59.3	24/10	54.08	59.50	↓-5.42	04/11	73.09	60.10	↑+12.99	20/12	51.27	57.50	↑+6.23
		22/10	59.6	25/10	46.70	59.50	↑+12.80	05/11	68.76	60.80	↑+7.96	21/12	68.12	57.50	↓-10.62
VCS	Baseline	30/08	48.8	02/09	54.57	48.80	↓-5.77	13/09	58.43	50.00	↑+8.43	29/10	57.13	47.60	↓-9.53
		16/10	47.2	19/10	41.79	46.80	↑+5.01	30/10	56.66	47.40	↑+9.26	15/12	41.62	46.50	↑+4.88
		21/10	45.5	24/10	55.85	46.20	↑+9.65	04/11	41.91	47.10	↓-5.19	20/12	39.07	43.00	↑+3.93
		22/10	45.5	25/10	49.91	46.20	↑+3.71	05/11	57.06	46.90	↑+10.16	21/12	36.85	43.00	↑+6.15
HPG	Baseline	01/09	27.5	04/09	34.55	29.85	↑+4.70	15/09	25.69	30.35	↓-4.66	31/10	31.71	26.70	↓-5.01
		16/10	28.25	19/10	25.37	28.00	↑+2.63	30/10	22.20	26.90	↑+4.70	15/12	22.23	26.25	↑+4.02
		21/10	26.8	24/10	22.30	26.40	↑+4.10	04/11	32.06	26.75	↓-5.31	20/12	22.00	26.70	↑+4.70
		22/10	26.7	25/10	24.12	26.40	↑+2.28	05/11	31.15	26.30	↓-4.85	21/12	22.57	26.70	↓-4.13
TCB	Baseline	01/09	39.6	04/09	45.80	39.60	↓-6.20	15/09	47.21	39.00	↓-8.21	31/10	29.47	35.10	↑+5.63
		16/10	41.25	19/10	35.60	40.65	↑+5.05	30/10	41.17	35.70	↑+5.47	15/12	27.05	32.00	↑+4.95
		21/10	37.65	24/10	33.06	36.10	↑+3.04	04/11	39.18	35.00	↓-4.18	20/12	39.52	33.60	↓-5.92
		22/10	38	25/10	28.37	36.10	↑+7.73	05/11	29.25	34.10	↑+4.85	21/12	30.04	33.60	↑+3.56
VHM	Baseline	01/09	104.5	04/09	120.53	102.30	↓-18.23	15/09	114.68	104.50	↓-10.18	31/10	80.81	99.20	↑+18.39
		16/10	122	19/10	91.24	116.00	↑+24.76	30/10	83.56	104.00	↑+20.44	15/12	77.11	93.90	↑+16.79
		21/10	110.9	24/10	128.80	114.50	↑+14.30	04/11	90.66	100.20	↑+9.54	20/12	120.13	101.50	↓-18.63
		22/10	113	25/10	103.02	114.50	↓-11.48	05/11	78.55	99.60	↑+21.05	21/12	90.46	101.50	↑+11.04
MWG	Baseline	01/09	78	04/09	87.88	77.50	↓-10.38	15/09	71.23	79.50	↓-8.27	31/10	72.05	82.60	↓-10.55
		16/10	84.5	19/10	96.72	84.50	↓-12.22	30/10	65.57	83.90	↑+18.33	15/12	67.47	77.70	↑+10.23
		21/10	83	24/10	76.82	85.70	↓-8.88	04/11	96.95	81.80	↓-15.15	20/12	65.07	82.90	↑+17.83
		22/10	84.5	25/10	68.27	85.70	↓-17.43	05/11	64.34	80.20	↑+15.86	21/12	98.44	82.90	↓-15.54
CMG	Baseline	01/09	40.3	04/09	33.16	41.85	↓-8.69	15/09	33.83	40.80	↓-6.97	31/10	47.44	41.90	↑+5.54
		16/10	38	19/10	40.52	37.30	↓-3.22	30/10	34.76	41.30	↓-6.54	15/12	39.71	35.10	↓-4.61
		21/10	38.4	24/10	47.09	38.75	↑+8.34	04/11	33.77	39.60	↓-5.83	20/12	32.34	35.20	↑+2.86
		22/10	38.5	25/10	46.62	38.75	↑+7.87	05/11	31.79	39.30	↓-7.51	21/12	42.32	35.20	↓-7.12
KDH	Baseline	01/09	36.5	04/09	41.13	36.75	↑+4.38	15/09	30.09	34.85	↑+4.76	31/10	28.27	35.85	↑+7.58
		16/10	34.2	19/10	30.27	33.90	↑+3.63	30/10	40.67	35.85	↑+4.82	15/12	27.14	29.60	↑+2.46
		21/10	31.75	24/10	30.69	33.80	↓-3.11	04/11	30.23	35.60	↓-5.37	20/12	39.17	32.65	↑+6.52
		22/10	32.2	25/10	29.76	33.80	↓-4.04	05/11	40.61	34.95	↑+5.66	21/12	38.43	32.65	↑+5.78
DGC	Baseline	01/09	98.2	04/09	108.27	99.30	↑+8.97	15/09	86.99	99.50	↓-12.51	31/10	78.39	96.00	↑+17.61
		16/10	95	19/10	111.11	93.20	↓-17.91	30/10	79.46	93.60	↑+14.14	15/12	76.73	93.00	↑+16.27
		21/10	89.2	24/10	108.26	92.40	↑+15.86	04/11	114.74	94.60	↑+20.14	20/12	78.92	70.20	↑+8.72
		22/10	91	25/10	102.55	92.40	↑+10.15	05/11	81.37	95.00	↓-13.63	21/12	58.74	70.20	↑+11.46
MBB	Baseline	01/09	27.75	04/09	24.46	28.25	↓-3.79	15/09	30.14	26.85	↓-3.29	31/10	26.50	23.60	↑+2.90
		16/10	27.2	19/10	22.66	27.10	↑+4.44	30/10	27.95	23.95	↓-4.00	15/12	19.62	23.75	↑+4.13
		21/10	25.3	24/10	20.68	24.40	↑+3.72	04/11	26.36	24.00	↓-2.36	20/12	20.14	24.70	↑+4.56
		22/10	25.45	25/10	21.46	24.40	↑+2.94	05/11	25.91	23.90	↓-2.01	21/12	21.25	24.70	↑+3.45

TABLE 16: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (Llama Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Baseline	31/08	22.8	03/09	26.15	24.00	↑+2.15	14/09	26.29	24.05	↑+2.24	30/10	23.18	21.20	↓-1.98
		16/10	22.45	19/10	25.46	22.60	↑+2.86	30/10	17.16	21.20	↑+4.04	15/12	19.20	16.30	↑+2.90
		21/10	20.0	24/10	23.92	20.90	↑+3.02	04/11	16.24	20.15	↓-3.91	20/12	20.57	17.80	↓-2.77
		22/10	20.5	25/10	23.00	20.90	↑+2.10	05/11	22.66	20.10	↓-2.56	21/12	15.03	17.80	↑+2.77
FPT	Baseline	01/09	101.6	04/09	126.89	105.00	↑+21.89	15/09	122.02	101.90	↑+20.12	31/10	117.23	103.90	↑+13.33
		16/10	89.8	19/10	98.60	88.10	↓-10.50	30/10	89.78	102.70	↓-12.92	15/12	110.53	93.80	↑+16.73
		21/10	93	24/10	84.76	97.70	↓-12.94	04/11	86.35	103.30	↓-16.95	20/12	107.69	93.90	↑+13.79
		22/10	97	25/10	80.00	97.70	↓-17.70	05/11	116.35	100.90	↑+15.45	21/12	75.51	93.90	↑+18.39
VCB	Baseline	30/08	68.6	02/09	76.01	68.60	↓-7.41	13/09	58.83	65.80	↑+6.97	29/10	49.57	60.70	↑+11.13
		16/10	62.9	19/10	50.23	61.90	↑+11.67	30/10	49.25	60.60	↑+11.35	15/12	68.09	56.80	↓-11.29
		21/10	59.3	24/10	54.73	59.50	↓-4.77	04/11	66.97	60.10	↑+6.87	20/12	62.23	57.50	↓-4.73
		22/10	59.6	25/10	72.15	59.50	↓-12.65	05/11	67.16	60.80	↑+6.36	21/12	50.32	57.50	↑+7.18
VCS	Baseline	30/08	48.8	02/09	44.27	48.80	↓-4.53	13/09	40.06	50.00	↓-9.94	29/10	42.76	47.60	↑+4.84
		16/10	47.2	19/10	38.76	46.80	↑+8.04	30/10	39.55	47.40	↓-7.85	15/12	38.47	46.50	↑+8.03
		21/10	45.5	24/10	55.73	46.20	↑+9.53	04/11	53.37	47.10	↑+6.27	20/12	38.93	43.00	↑+4.07
		22/10	45.5	25/10	37.76	46.20	↓-8.44	05/11	41.42	46.90	↓-5.48	21/12	51.06	43.00	↓-8.06
HPG	Baseline	01/09	27.5	04/09	26.06	29.85	↓-3.79	15/09	33.41	30.35	↑+3.06	31/10	23.13	26.70	↑+3.57
		16/10	28.25	19/10	33.74	28.00	↓-5.74	30/10	23.97	26.90	↑+2.93	15/12	23.87	26.25	↑+2.38
		21/10	26.8	24/10	28.96	26.40	↓-2.56	04/11	22.32	26.75	↑+4.43	20/12	28.96	26.70	↓-2.26
		22/10	26.7	25/10	30.63	26.40	↓-4.23	05/11	23.06	26.30	↑+3.24	21/12	23.97	26.70	↓-2.73
TCB	Baseline	01/09	39.6	04/09	33.00	39.60	↓-6.60	15/09	43.37	39.00	↓-4.37	31/10	40.12	35.10	↓-5.02
		16/10	41.25	19/10	45.79	40.65	↓-5.14	30/10	29.41	35.70	↑+6.29	15/12	27.77	32.00	↑+4.23
		21/10	37.65	24/10	30.10	36.10	↑+6.00	04/11	32.04	35.00	↑+2.96	20/12	28.46	33.60	↑+5.14
		22/10	38	25/10	43.98	36.10	↓-7.88	05/11	28.83	34.10	↑+5.27	21/12	36.97	33.60	↑+3.37
VHM	Baseline	01/09	104.5	04/09	91.68	102.30	↑+10.62	15/09	84.19	104.50	↓-20.31	31/10	81.88	99.20	↑+17.32
		16/10	122	19/10	139.47	116.00	↓-23.47	30/10	126.86	104.00	↓-22.86	15/12	79.54	93.90	↑+14.36
		21/10	110.9	24/10	126.81	114.50	↑+12.31	04/11	112.76	100.20	↓-12.56	20/12	122.64	101.50	↓-21.14
		22/10	113	25/10	100.27	114.50	↓-14.23	05/11	117.18	99.60	↓-17.58	21/12	84.24	101.50	↑+17.26
MWG	Baseline	01/09	78	04/09	85.91	77.50	↓-8.41	15/09	65.18	79.50	↓-14.32	31/10	93.52	82.60	↑+10.92
		16/10	84.5	19/10	92.42	84.50	↓-7.92	30/10	93.88	83.90	↓-9.98	15/12	61.11	77.70	↑+16.59
		21/10	83	24/10	98.17	85.70	↑+12.47	04/11	68.92	81.80	↑+12.88	20/12	98.93	82.90	↓-16.03
		22/10	84.5	25/10	101.28	85.70	↑+15.58	05/11	63.47	80.20	↑+16.73	21/12	76.02	82.90	↑+6.88
CMG	Baseline	01/09	40.3	04/09	36.88	41.85	↓-4.97	15/09	34.69	40.80	↓-6.11	31/10	47.66	41.90	↑+5.76
		16/10	38	19/10	43.08	37.30	↓-5.78	30/10	32.37	41.30	↓-8.93	15/12	39.06	35.10	↓-3.96
		21/10	38.4	24/10	31.94	38.75	↓-6.81	04/11	33.71	39.60	↓-5.89	20/12	28.73	35.20	↑+6.47
		22/10	38.5	25/10	31.86	38.75	↓-6.89	05/11	32.52	39.30	↓-6.78	21/12	41.80	35.20	↓-6.60
KDH	Baseline	01/09	36.5	04/09	39.81	36.75	↑+3.06	15/09	27.89	34.85	↑+6.96	31/10	29.75	35.85	↑+6.10
		16/10	34.2	19/10	37.96	33.90	↓-4.06	30/10	28.78	35.85	↓-7.07	15/12	24.60	29.60	↑+5.00
		21/10	31.75	24/10	38.43	33.80	↑+4.63	04/11	40.66	35.60	↑+5.06	20/12	26.45	32.65	↓-6.20
		22/10	32.2	25/10	39.32	33.80	↑+5.52	05/11	42.51	34.95	↑+7.56	21/12	36.14	32.65	↑+3.49
DGC	Baseline	01/09	98.2	04/09	110.36	99.30	↑+11.06	15/09	83.82	99.50	↓-15.68	31/10	83.85	96.00	↑+12.15
		16/10	95	19/10	84.31	93.20	↑+8.89	30/10	108.71	93.60	↓-15.11	15/12	103.99	93.00	↓-10.99
		21/10	89.2	24/10	107.23	92.40	↑+14.83	04/11	111.82	94.60	↑+17.22	20/12	55.14	70.20	↑+15.06
		22/10	91	25/10	72.54	92.40	↓-19.86	05/11	113.73	95.00	↑+18.73	21/12	62.33	70.20	↑+7.87
MBB	Baseline	01/09	27.75	04/09	31.56	28.25	↑+3.31	15/09	29.20	26.85	↓-2.35	31/10	21.64	23.60	↑+1.96
		16/10	27.2	19/10	32.07	27.10	↓-4.97	30/10	27.95	23.95	↓-4.00	15/12	21.27	23.75	↑+2.48
		21/10	25.3	24/10	29.18	24.40	↓-4.78	04/11	28.75	24.00	↓-4.75	20/12	22.48	24.70	↑+2.22
		22/10	25.45	25/10	27.61	24.40	↓-3.21	05/11	19.07	23.90	↑+4.83	21/12	20.33	24.70	↑+4.37

TABLE 17: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (LSTM Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Baseline	31/08	22.8	03/09	19.22	24.00	↓-4.78	14/09	26.86	24.05	↑+2.81	30/10	18.72	21.20	↑+2.48
		16/10	22.45	19/10	26.77	22.60	↑+4.17	30/10	17.48	21.20	↑+3.72	15/12	18.41	16.30	↑+2.11
		21/10	20.0	24/10	23.26	20.90	↑+2.36	04/11	15.94	20.15	↓-4.21	20/12	19.71	17.80	↑+1.91
		22/10	20.5	25/10	19.20	20.90	↓-1.70	05/11	17.47	20.10	↑+2.63	21/12	14.90	17.80	↑+2.90
FPT	Baseline	01/09	101.6	04/09	122.95	105.00	↑+17.95	15/09	82.20	101.90	↓-19.70	31/10	85.50	103.90	↓-18.40
		16/10	89.8	19/10	105.94	88.10	↓-17.84	30/10	123.51	102.70	↑+20.81	15/12	106.43	93.80	↑+12.63
		21/10	93	24/10	87.92	97.70	↓-9.78	04/11	124.96	103.30	↑+21.66	20/12	109.95	93.90	↑+16.05
		22/10	97	25/10	86.66	97.70	↓-11.04	05/11	118.95	100.90	↑+18.05	21/12	107.01	93.90	↓-13.11
VCB	Baseline	30/08	68.6	02/09	80.32	68.60	↓-11.72	13/09	79.34	65.80	↓-13.54	29/10	47.38	60.70	↑+13.32
		16/10	62.9	19/10	69.79	61.90	↓-7.89	30/10	49.08	60.60	↑+11.52	15/12	62.23	56.80	↑+5.43
		21/10	59.3	24/10	47.89	59.50	↓-11.61	04/11	54.82	60.10	↓-5.28	20/12	63.47	57.50	↓-5.97
		22/10	59.6	25/10	65.98	59.50	↓-6.48	05/11	51.18	60.80	↓-9.62	21/12	46.71	57.50	↑+10.79
VCS	Baseline	30/08	48.8	02/09	54.87	48.80	↓-6.07	13/09	45.28	50.00	↓-4.72	29/10	39.19	47.60	↑+8.41
		16/10	47.2	19/10	42.57	46.80	↑+4.23	30/10	53.70	47.40	↑+6.30	15/12	39.28	46.50	↑+7.22
		21/10	45.5	24/10	41.78	46.20	↓-4.42	04/11	50.94	47.10	↑+3.84	20/12	35.82	43.00	↑+7.18
		22/10	45.5	25/10	40.17	46.20	↓-6.03	05/11	39.02	46.90	↓-7.88	21/12	35.48	43.00	↑+7.52
HPG	Baseline	01/09	27.5	04/09	34.92	29.85	↑+5.07	15/09	36.61	30.35	↑+6.26	31/10	21.66	26.70	↑+5.04
		16/10	28.25	19/10	32.15	28.00	↓-4.15	30/10	23.59	26.90	↑+3.31	15/12	30.73	26.25	↓-4.48
		21/10	26.8	24/10	30.39	26.40	↓-3.99	04/11	23.82	26.75	↑+2.93	20/12	29.28	26.70	↓-2.58
		22/10	26.7	25/10	29.71	26.40	↓-3.31	05/11	31.75	26.30	↓-5.45	21/12	29.28	26.70	↓-2.58
TCB	Baseline	01/09	39.6	04/09	46.83	39.60	↓-7.23	15/09	43.10	39.00	↓-4.10	31/10	42.03	35.10	↓-6.93
		16/10	41.25	19/10	45.86	40.65	↓-5.21	30/10	39.16	35.70	↑+3.46	15/12	34.74	32.00	↑+2.74
		21/10	37.65	24/10	29.75	36.10	↑+6.35	04/11	42.14	35.00	↓-7.14	20/12	39.83	33.60	↓-6.23
		22/10	38	25/10	41.36	36.10	↓-5.26	05/11	39.38	34.10	↓-5.28	21/12	30.47	33.60	↑+3.13
VHM	Baseline	01/09	104.5	04/09	114.69	102.30	↓-12.39	15/09	82.13	104.50	↓-22.37	31/10	110.20	99.20	↓-11.00
		16/10	122	19/10	99.38	116.00	↑+16.62	30/10	84.91	104.00	↑+19.09	15/12	103.10	93.90	↑+9.20
		21/10	110.9	24/10	135.00	114.50	↑+20.50	04/11	120.26	100.20	↓-20.06	20/12	112.05	101.50	↓-10.55
		22/10	113	25/10	92.03	114.50	↓-22.47	05/11	111.01	99.60	↑+11.41	21/12	110.69	101.50	↑+9.19
MWG	Baseline	01/09	78	04/09	87.66	77.50	↓-10.16	15/09	70.32	79.50	↓-9.18	31/10	71.47	82.60	↓-11.13
		16/10	84.5	19/10	98.91	84.50	↓-14.41	30/10	92.20	83.90	↓-8.30	15/12	85.92	77.70	↓-8.22
		21/10	83	24/10	78.08	85.70	↓-7.62	04/11	74.72	81.80	↑+7.08	20/12	93.04	82.90	↓-10.14
		22/10	84.5	25/10	100.27	85.70	↑+14.57	05/11	65.36	80.20	↑+14.84	21/12	100.20	82.90	↓-17.30
CMG	Baseline	01/09	40.3	04/09	38.16	41.85	↓-3.69	15/09	34.32	40.80	↓-6.48	31/10	45.80	41.90	↑+3.90
		16/10	38	19/10	41.38	37.30	↓-4.08	30/10	45.32	41.30	↑+4.02	15/12	32.05	35.10	↑+3.05
		21/10	38.4	24/10	46.93	38.75	↑+8.18	04/11	31.39	39.60	↓-8.21	20/12	28.40	35.20	↑+6.80
		22/10	38.5	25/10	43.38	38.75	↑+4.63	05/11	35.86	39.30	↓-3.44	21/12	39.75	35.20	↓-4.55
KDH	Baseline	01/09	36.5	04/09	39.92	36.75	↑+3.17	15/09	29.03	34.85	↑+5.82	31/10	31.28	35.85	↑+4.57
		16/10	34.2	19/10	27.43	33.90	↑+6.47	30/10	42.61	35.85	↑+6.76	15/12	24.06	29.60	↑+5.54
		21/10	31.75	24/10	27.24	33.80	↓-6.56	04/11	27.78	35.60	↓-7.82	20/12	27.56	32.65	↓-5.09
		22/10	32.2	25/10	29.19	33.80	↓-4.61	05/11	41.76	34.95	↑+6.81	21/12	36.39	32.65	↑+3.74
DGC	Baseline	01/09	98.2	04/09	118.39	99.30	↑+19.09	15/09	91.17	99.50	↓-8.33	31/10	106.36	96.00	↓-10.36
		16/10	95	19/10	74.01	93.20	↑+19.19	30/10	109.68	93.60	↓-16.08	15/12	74.21	93.00	↑+18.79
		21/10	89.2	24/10	81.73	92.40	↓-10.67	04/11	80.39	94.60	↓-14.21	20/12	61.11	70.20	↑+9.09
		22/10	91	25/10	99.88	92.40	↑+7.48	05/11	115.75	95.00	↑+20.75	21/12	58.86	70.20	↑+11.34
MBB	Baseline	01/09	27.75	04/09	33.40	28.25	↑+5.15	15/09	23.90	26.85	↑+2.95	31/10	19.84	23.60	↑+3.76
		16/10	27.2	19/10	30.85	27.10	↓-3.75	30/10	21.49	23.95	↑+2.46	15/12	19.39	23.75	↑+4.36
		21/10	25.3	24/10	28.15	24.40	↓-3.75	04/11	21.16	24.00	↑+2.84	20/12	21.02	24.70	↑+3.68
		22/10	25.45	25/10	28.81	24.40	↓-4.41	05/11	20.56	23.90	↑+3.34	21/12	29.27	24.70	↓-4.57

TABLE 18: Comparison of Short-term, Medium-term, and Long-term Forecasts Across Architectures and Stocks (Transformer Baseline)

Stock	Architecture	Start		Short-term				Medium-term				Long-term			
		Date	Price	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap	End Date	Pred	Act	Gap
DXG	Baseline	31/08	22.8	03/09	21.87	24.00	↓-2.13	14/09	19.26	24.05	↓-4.79	30/10	25.47	21.20	↓-4.27
		16/10	22.45	19/10	19.08	22.60	↓-3.52	30/10	25.62	21.20	↓-4.42	15/12	12.87	16.30	↑+3.43
		21/10	20.0	24/10	18.93	20.90	↓-1.97	04/11	23.01	20.15	↑+2.86	20/12	19.89	17.80	↑+2.09
		22/10	20.5	25/10	22.66	20.90	↑+1.76	05/11	23.25	20.10	↓-3.15	21/12	20.05	17.80	↑+2.25
FPT	Baseline	01/09	101.6	04/09	116.43	105.00	↑+11.43	15/09	122.54	101.90	↑+20.64	31/10	86.81	103.90	↓-17.09
		16/10	89.8	19/10	72.77	88.10	↑+15.33	30/10	118.46	102.70	↑+15.76	15/12	75.88	93.80	↓-17.92
		21/10	93	24/10	86.06	97.70	↓-11.64	04/11	91.94	103.30	↓-11.36	20/12	107.59	93.90	↑+13.69
		22/10	97	25/10	118.30	97.70	↑+20.60	05/11	117.20	100.90	↑+16.30	21/12	81.93	93.90	↑+11.97
VCB	Baseline	30/08	68.6	02/09	54.79	68.60	↓-13.81	13/09	54.31	65.80	↑+11.49	29/10	50.52	60.70	↑+10.18
		16/10	62.9	19/10	52.25	61.90	↑+9.65	30/10	69.70	60.60	↓-9.10	15/12	65.06	56.80	↓-8.26
		21/10	59.3	24/10	54.25	59.50	↓-5.25	04/11	53.63	60.10	↓-6.47	20/12	68.11	57.50	↓-10.61
		22/10	59.6	25/10	49.19	59.50	↑+10.31	05/11	50.39	60.80	↓-10.41	21/12	48.09	57.50	↑+9.41
VCS	Baseline	30/08	48.8	02/09	53.04	48.80	↓-4.24	13/09	45.85	50.00	↓-4.15	29/10	37.54	47.60	↑+10.06
		16/10	47.2	19/10	42.83	46.80	↑+3.97	30/10	55.27	47.40	↑+7.87	15/12	52.38	46.50	↓-5.88
		21/10	45.5	24/10	41.92	46.20	↓-4.28	04/11	38.83	47.10	↓-8.27	20/12	37.84	43.00	↑+5.16
		22/10	45.5	25/10	50.74	46.20	↑+4.54	05/11	53.86	46.90	↑+6.96	21/12	46.59	43.00	↓-3.59
HPG	Baseline	01/09	27.5	04/09	27.17	29.85	↓-2.68	15/09	27.37	30.35	↓-2.98	31/10	29.77	26.70	↓-3.07
		16/10	28.25	19/10	22.35	28.00	↑+5.65	30/10	29.54	26.90	↓-2.64	15/12	29.01	26.25	↓-2.76
		21/10	26.8	24/10	24.22	26.40	↑+2.18	04/11	32.33	26.75	↓-5.58	20/12	31.21	26.70	↓-4.51
		22/10	26.7	25/10	28.75	26.40	↓-2.35	05/11	29.72	26.30	↓-3.42	21/12	31.12	26.70	↓-4.42
TCB	Baseline	01/09	39.6	04/09	43.84	39.60	↓-4.24	15/09	45.58	39.00	↓-6.58	31/10	30.36	35.10	↑+4.74
		16/10	41.25	19/10	34.63	40.65	↑+6.02	30/10	43.16	35.70	↓-7.46	15/12	37.39	32.00	↑+5.39
		21/10	37.65	24/10	28.91	36.10	↑+7.19	04/11	39.00	35.00	↓-4.00	20/12	40.42	33.60	↓-6.82
		22/10	38	25/10	29.10	36.10	↑+7.00	05/11	40.56	34.10	↓-6.46	21/12	28.06	33.60	↑+5.54
VHM	Baseline	01/09	104.5	04/09	120.72	102.30	↓-18.42	15/09	126.39	104.50	↓-21.89	31/10	109.88	99.20	↓-10.68
		16/10	122	19/10	138.38	116.00	↓-22.38	30/10	116.97	104.00	↑+12.97	15/12	79.10	93.90	↑+14.80
		21/10	110.9	24/10	133.53	114.50	↑+19.03	04/11	83.78	100.20	↑+16.42	20/12	90.34	101.50	↑+11.16
		22/10	113	25/10	93.74	114.50	↓-20.76	05/11	113.41	99.60	↓-13.81	21/12	90.29	101.50	↑+11.21
MWG	Baseline	01/09	78	04/09	92.82	77.50	↓-15.32	15/09	69.42	79.50	↓-10.08	31/10	95.31	82.60	↑+12.71
		16/10	84.5	19/10	76.16	84.50	↓-8.34	30/10	71.86	83.90	↑+12.04	15/12	68.73	77.70	↑+8.97
		21/10	83	24/10	96.65	85.70	↑+10.95	04/11	73.78	81.80	↑+8.02	20/12	99.09	82.90	↓-16.19
		22/10	84.5	25/10	104.45	85.70	↑+18.75	05/11	68.68	80.20	↑+11.52	21/12	96.75	82.90	↓-13.85
CMG	Baseline	01/09	40.3	04/09	38.24	41.85	↓-3.61	15/09	35.57	40.80	↓-5.23	31/10	32.96	41.90	↓-8.94
		16/10	38	19/10	30.41	37.30	↑+6.89	30/10	50.01	41.30	↑+8.71	15/12	30.22	35.10	↑+4.88
		21/10	38.4	24/10	30.85	38.75	↓-7.90	04/11	44.48	39.60	↑+4.88	20/12	38.78	35.20	↓-3.58
		22/10	38.5	25/10	42.65	38.75	↑+3.90	05/11	33.66	39.30	↓-5.64	21/12	28.41	35.20	↑+6.79
KDH	Baseline	01/09	36.5	04/09	31.46	36.75	↓-5.29	15/09	31.41	34.85	↑+3.44	31/10	29.27	35.85	↑+6.58
		16/10	34.2	19/10	38.16	33.90	↓-4.26	30/10	30.66	35.85	↓-5.19	15/12	25.52	29.60	↑+4.08
		21/10	31.75	24/10	37.83	33.80	↑+4.03	04/11	40.14	35.60	↑+4.54	20/12	39.18	32.65	↑+6.53
		22/10	32.2	25/10	29.12	33.80	↓-4.68	05/11	39.71	34.95	↑+4.76	21/12	26.83	32.65	↓-5.82
DGC	Baseline	01/09	98.2	04/09	117.19	99.30	↑+17.89	15/09	88.37	99.50	↓-11.13	31/10	109.18	96.00	↓-13.18
		16/10	95	19/10	83.47	93.20	↑+9.73	30/10	108.83	93.60	↓-15.23	15/12	101.26	93.00	↓-8.26
		21/10	89.2	24/10	104.68	92.40	↑+12.28	04/11	83.66	94.60	↓-10.94	20/12	55.11	70.20	↑+15.09
		22/10	91	25/10	75.98	92.40	↓-16.42	05/11	105.04	95.00	↑+10.04	21/12	64.49	70.20	↑+5.71
MBB	Baseline	01/09	27.75	04/09	32.74	28.25	↑+4.49	15/09	21.18	26.85	↑+5.67	31/10	19.56	23.60	↑+4.04
		16/10	27.2	19/10	30.40	27.10	↓-3.30	30/10	20.84	23.95	↑+3.11	15/12	19.26	23.75	↑+4.49
		21/10	25.3	24/10	28.01	24.40	↓-3.61	04/11	28.73	24.00	↓-4.73	20/12	28.56	24.70	↓-3.86
		22/10	25.45	25/10	22.28	24.40	↑+2.12	05/11	18.69	23.90	↑+5.21	21/12	29.58	24.70	↓-4.88